

BaseRate - Complete Code

Submission source document. Includes the app, trained model, training data, reproducible training script, and verification checks. Line wrapping and Unicode escapes are for print only; use the source package to run the project.

Educational prototype; extensive AI assistance disclosed. No credentials, patient data, or external API keys are included.

dist/index.html

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
5 <title>BaseRate \u2014 A field guide to screening uncertainty</title>
6 <meta name="description" content="Learn screening uncertainty by counting people, comparing fictional
7 tests and explaining your reasoning to a private on-device AI coach.">
8 <meta name="theme-color" content="#f7f5ef">
9 <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 32
10 32'%3E%3Crect width='32' height='32' fill='%23f7f5ef'/%3E%3Ctext x='6' y='25' font-family='Georgia'
11 font-size='28' fill='%23242c29'%3E%3C/text%3E%3C/svg%3E">
12 <link rel="stylesheet" href="style.css?v=1.1.1">
13 <script src="science.js" defer</script><script src="model.js" defer</script><script
14 src="coach.js?v=1.1.1" defer</script><script src="learning.js?v=1.1.1" defer</script><script
15 src="app.js?v=1.1.1" defer</script><script src="lab-tools.js" defer</script><script
16 src="learning-ui.js?v=1.1.1" defer</script>
17 </head>
18 <body>
19 <a class="skip" href="#lab">Skip to the experiment</a>
20 <header class="site-header"><a class="brand" href="#lab">BaseRate<span>A field guide to screening
21 uncertainty</span></a><nav aria-label="Primary"><a href="#lab">The lab</a><a
22 href="#practice">Practice</a><a href="#coach">AI coach</a><a href="#evidence">Methods</a></nav><a
23 class="header-source" href="https://github.com/Nicholas-afk/baserate" target="_blank"
24 rel="noopener">Source \u2017</a></header>
25 <main>
26 <section class="opening" aria-labelledby="lab-heading"><div><p class="eyebrow">A LEARNING EXPERIMENT /
27 UNIVABIO 2026</p><h1 id="lab-heading">A positive result needs<br><em>a denominator.</em></h1><p
28 class="intro">A test can detect most cases and still produce more false alarms than true positives.
29 Count the people. Change one assumption. See why.</p></div><aside class="opening-note"><span
30 class="small-label">START HERE</span><p>Explore a fictional population below, or predict the answer
31 before you see it.</p><a class="primary-link" href="#practice">Try the three-case lesson <span
32 aria-hidden="true">\u2017</span></a><p class="micro">Fictional rates \xb7 No health details \xb7 No
33 account</p></aside></section>
34 <section id="lab" class="lab" aria-label="Fictional screening experiment">
35 <div class="section-bar"><h2><span class="section-index">01</span> The experiment</h2><div
36 class="bar-actions"><button id="share-experiment" class="text-button">Copy settings
37 link</button><button id="print" class="text-button">Print worksheet</button></div></div>
38 <p class="scope-note">For learning only. These invented rates do not estimate anyone\u2019s health
39 risk or recommend a screening test.</p><p id="link-message" role="status" class="micro"></p>
40 <div class="workspace">
41 <aside class="controls" aria-label="Fictional test settings"><p class="small-label">THREE
42 ASSUMPTIONS</p><label class="scenario-label" for="scenario">Start with</label><select
43 id="scenario"><option value="rare">A rare condition</option><option value="common">A more common
44 condition</option><option value="specific">Fewer false positives</option><option value="custom">Custom
45 experiment</option></select><p id="scenario-note" class="muted">A fictional condition affects 1 in
46 100 people.</p>
47 <div class="control"><label for="prevalence">People with the condition <output for="prevalence"
48 id="prevalence-label">1%</output></label><div class="range-row"><input id="prevalence" type="range"
49 min="0.1" max="50" step="0.1" value="1"><input id="prevalence-number" type="number"
50 aria-label="Prevalence percentage" min="0.1" max="50" step="0.1" value="1"></div><p>Prevalence \xb7
51 the starting group with the condition.</p></div>
52 <div class="control"><label for="sensitivity">Cases the test detects <output for="sensitivity"
53 id="sensitivity-label">90%</output></label><div class="range-row"><input id="sensitivity" type="range"
54 min="1" max="100" step="1" value="90"><input id="sensitivity-number" type="number"
55 aria-label="Sensitivity percentage" min="1" max="100" step="1" value="90"></div><p>Sensitivity \xb7
56 positives among people with the condition.</p></div>
57 <div class="control"><label for="specificity">Non-cases it clears <output for="specificity"
58 id="specificity-label">95%</output></label><div class="range-row"><input id="specificity" type="range"
59 min="1" max="100" step="0.1" value="95"><input id="specificity-number" type="number"
60 aria-label="Specificity percentage" min="1" max="100" step="0.1" value="95"></div><p>Specificity \xb7
61 negatives among people without it.</p></div>
62 <p id="settings-error" role="status" class="error"></p><button class="secondary" id="reset">Reset to
63 the rare case</button><p class="margin-note">Hold the test fixed.<br>Try changing only prevalence.</p>
64 </aside>
65 <div class="population-panel"><div class="panel-top"><div><h3>One thousand people.</h3><p>Four
66 possible outcomes.</p></div><div class="segmented" role="group" aria-label="Population view"><button
67 id="all-view" class="active" aria-pressed="true">All</button><button id="positive-view"
68 aria-pressed="false">Positive</button><button id="tree-view"
69 aria-pressed="false">Tree</button></div></div>
70 <div id="population-wrap"><div id="population" class="population" role="img" aria-label="Population of
71 1000 people"></div><div class="legend"><span><i class="tp"></i>True positive</span><span><i
72 class="fp"></i>False positive</span><span><i class="fn"></i>False negative</span><span><i
73 class="tn"></i>True negative</span></div></div>
74 <div id="frequency-tree" class="frequency-tree" hidden aria-label="Natural frequency tree"></div>
75 <div class="counts"><div><span class="count-tag tp-text">Condition + positive</span><strong
76 id="tp-count">9</strong><small>True positives</small></div><div><span class="count-tag fp-text">No
77 condition + positive</span><strong id="fp-count">49.5</strong><small>False
78 positives</small></div><div><span class="count-tag fn-text">Condition + negative</span><strong
79 id="fn-count">1</strong><small>False negatives</small></div><div><span class="count-tag">No condition
80 + negative</span><strong id="tn-count">940.5</strong><small>True negatives</small></div></div></div><p>
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class="rounding-note">Expected counts may be fractional. Squares are rounded; calculations use the
exact counts.</p>
31 </div>
32 <aside class="result-panel" aria-labelledby="result-heading"><p class="small-label">CHOOSE THE RIGHT
GROUP</p><h3 id="result-heading">Among the people<br>who test positive\&#2026</h3><div
class="result-number"><span id="ppv">15.4</span><span>%</span></div><p id="result-explanation"
aria-live="polite">About 9 of 58.5 expected positive results are true positives.</p><div
class="ratio-bar" aria-hidden="true"><span id="ratio-fill"></span></div><div
class="result-key"><span>True</span><span>False</span></div><div class="formula"><span>True positives
\&#xf7 all positives</span><code id="formula">9 \&#xf7 (9 + 49.5)</code><small>Multiply by 100 for the
percentage.</small></div><p class="insight" id="insight">A small error rate applied to a large group
can outweigh a high detection rate.</p><a class="text-link" href="#coach">Explain this result
\&#2197</a></aside>
33 </div></section>
34 <section id="compare" class="compare-section" aria-labelledby="compare-heading"><div
class="section-bar"><h2 id="compare-heading"><span class="section-index">02</span> Change the starting
population</h2><span class="micro">A counterfactual, not a clinical comparison</span></div><div
class="compare-layout"><div><p class="section-intro">Pin one experiment, then change the lab above.
The test can stay exactly the same while the meaning of a positive result changes.</p><div
class="form-actions"><button id="pin-experiment" class="secondary">Pin current settings as
A</button><button id="compare-common" class="secondary">Compare 1% with 20%</button><p
id="pin-status" role="status" class="micro">A is pinned to the rare-condition example. B follows the
lab.</p><div class="table-scroll"><table class="comparison-table"><caption>Expected outcomes in 1,000
people</caption><thead><tr><th scope="col">Measure</th><th scope="col">A \&#xb7 pinned</th><th
scope="col">B \&#xb7 current</th></tr></thead><tbody id="compare-rows"></tbody></table></div><p
id="compare-insight" class="comparison-insight" aria-live="polite"></p></div><div
class="curve-panel"><div class="panel-top"><div><h3>Prevalence changes the answer.</h3><p
id="curve-test-label">The same test, across different populations.</p></div></div><svg
id="prevalence-curve" viewBox="0 0 640 320" role="img" aria-labelledby="curve-title
curve-description"><title id="curve-title">Positive predictive value as prevalence
changes</title><desc id="curve-description">A curve and a current-experiment marker. Equivalent values
are in the table below.</desc></svg><label for="curve-prevalence">Move the current experiment <output
id="curve-value">1%</output></label><input id="curve-prevalence" type="range" min="0.1" max="50"
step="0.1" value="1"><details><summary>Read the curve as a table</summary><table
class="small-table"><thead><tr><th scope="col">Prevalence</th><th scope="col">True among
positive</th></tr></thead><tbody id="curve-table"></tbody></table></details><p class="micro">The line
uses B\&#2019s test settings. It starts at 0% prevalence; an undefined value is left
blank.</p></div></div>
35 <details id="sequence" class="sequence-details"><summary>Go further: does a second positive test
settle it?</summary><div class="sequence-body"><div><p class="small-label">THE ASSUMPTION
MATTERS</p><h3>Two positives are not automatically independent evidence.</h3><p>Only people positive
on the first test go through this second step. These rates describe that selected group, not the whole
population.</p><label for="sequence-mode">Choose an assumption</label><select
id="sequence-mode"><option value="independent">Independent, identical test</option><option
value="repeat">Perfectly repeated result</option><option value="custom">Custom conditional
rates</option></select><p id="sequence-assumption" class="assumption-note">Independence is an invented
assumption here, not evidence about a real test.</p><div class="sequence-controls"><div><label
for="conditional-detection">Second-positive among true first-positives <output
id="conditional-detection-label">90%</output></label><input id="conditional-detection" type="range"
min="0" max="100" step="1" value="90"></div><div><label for="conditional-false">Second-positive among
false first-positives <output id="conditional-false-label">5%</output></label><input
id="conditional-false" type="range" min="0" max="100" step="0.1" value="5"></div></div><div><div
class="sequence-result"><p class="small-label">AMONG TWO POSITIVE RESULTS</p><strong
id="sequence-ppv">76.6%</strong><p id="sequence-counts"></p><div id="sequence-steps"
class="sequence-steps"></div><p class="micro">A second step can lose true positives as well as remove
false positives. This is a mathematical demonstration, not a recommendation to repeat a real
test.</p></div></div></details></section>
36 <section id="practice" class="practice-section" aria-labelledby="practice-heading"><div
class="section-bar"><h2 id="practice-heading"><span class="section-index">03</span> Predict. Count.
Reconsider.</h2><span id="practice-progress" class="micro">Case 1 of 3</span></div><p
class="section-intro">Three short experiments. Make an estimate before revealing the arithmetic, then
name the group in the denominator. This is practice, not a test of your health knowledge.</p><div
class="practice-layout"><ol id="practice-steps" class="practice-steps" aria-label="Lesson
progress"></ol><div><div id="practice-case"><p id="case-settings" class="small-label"></p><h3
id="case-title" tabindex="-1"></h3><p id="case-prompt"></p><form id="prediction-form"
novalidate><label for="prediction">Your estimate: true positives among positive results</label><div
class="prediction-input"><input id="prediction" type="number" min="0" max="100" step="any" required
placeholder="0\&#2013100" inputmode="decimal"><span>%</span><button id="predict-button"
class="primary">Reveal the counts</button></div><p id="prediction-error" role="status"
class="error"></p></form><div id="case-reveal" class="case-reveal" tabindex="-1" hidden><h4
id="prediction-result"></h4><p id="prediction-difference"></p><code id="case-equation"
class="case-equation"></code><p id="case-explanation"></p><form
id="denominator-form"><fieldset><legend>Which group goes in the denominator?</legend><label><input
type="radio" name="denominator" value="cases"> Everyone with the condition</label><label><input
type="radio" name="denominator" value="positives"> Everyone who tests positive</label><label><input
type="radio" name="denominator" value="population"> All 1,000 people</label></fieldset></form><p
id="denominator-feedback" role="status"></p><button id="practice-next" class="primary" disabled>Next
case</button></div></div><div id="practice-complete" tabindex="-1" hidden><p class="small-label">YOUR
LEARNING NOTEBOOK</p><h3>What changed in your estimates?</h3><p>Your first guesses are kept for this
tab only. A smaller gap on these cases is not proof of learning or a clinical result. Try explaining
the change in your own words to the coach.</p><div class="table-scroll"><table
class="small-table"><thead><tr><th scope="col">Case</th><th scope="col">Guess</th><th
scope="col">Actual</th><th scope="col">Gap</th><th scope="col">Denominator
choices</th></tr></thead><tbody id="notebook-rows"></tbody></table></div><div>

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class="form-actions"><button id="download-notebook" class="secondary">Export my
notebook</button><button id="restart-practice" class="secondary">Start again</button><a
href="#coach">Explain the lesson \u2197</a></div></div></div></div><p class="micro">Answers stay in
this tab. Export saves numeric guesses and selected options to a file; coach explanations are
excluded.</p></section>
37 <section id="coach" class="coach-section" aria-labelledby="coach-heading"><div class="section-bar"><h2
id="coach-heading"><span class="section-index">04</span> Put the reasoning into words</h2><span
class="micro">AI runs on this device</span></div><p class="section-intro">Explain the current lab
result. A small trained classifier suggests a review topic. Confirm or change the focus, check a
factual answer, then revise your explanation. A topic suggestion cannot verify your reasoning.</p><div
class="coach-layout"><form id="coach-form"><label for="explanation">Your explanation of this
experiment</label><textarea id="explanation" maxlength="1800" rows="5" aria-describedby="coach-help"
placeholder="A positive result doesn't prove the condition because\u2026"></textarea><p
class="muted" id="coach-help">Fictional examples only. English in this prototype. Your words stay in
this browser and disappear on reload.</p><div class="form-actions"><button type="submit"
class="primary">Find a review focus</button><button id="example" type="button" class="secondary">Use
an example</button></div></form><div id="coach-response" class="coach-response" aria-live="polite"><p
class="small-label">A SECOND LOOK AT YOUR REASONING</p><h3>Start with the positive group.</h3><p>How
can a rare condition and a small false-positive rate combine to produce this result?</p><p
class="muted">Model feedback is a suggestion. The arithmetic is calculated separately and does not
depend on AI.</p></div></div></section>
38 <section id="evidence" class="evidence-section" aria-labelledby="evidence-heading"><div
class="section-bar"><h2 id="evidence-heading"><span class="section-index">05</span> Methods, evidence
& honest limits</h2><a href="project.html">The project file \u2197</a></div><div
class="evidence-grid"><article><p class="small-label">REPRESENTATION</p><h3>Show the people in the
percentage.</h3><p>Natural frequencies make the relevant groups visible. The square grid and frequency
tree are two views of the same expected counts. The result uses Bayes's theorem; population size
changes counts, not proportions.</p><a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC4604268/"
target="_blank" rel="noopener">Hoffrage et al., 2015's original study \u2197</a><p
class="micro">Research informed the representation. BaseRate itself has not had a learner
study.</p></article><article><p class="small-label">SCREENING</p><h3>Every test has a
context.</h3><p>False positives and missed cases matter. These three rates cannot establish the
benefits, harms or suitability of a real screening program. There is no personal diagnosis, risk
estimate or treatment advice here.</p><a
href="https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq" target="_blank"
rel="noopener">National Cancer Institute's overview \u2197</a></article><article><p
class="small-label">MODEL CARD</p><h3>A small model you can inspect.</h3><p
id="model-summary"></p><details><summary>Read training, evaluation and failure modes</summary><div
id="model-details"></div></details><p class="micro">Extensive AI assistance was used for code,
synthetic examples and educational copy. The app does not send your explanation to a generative model.
Static files are downloaded from GitHub Pages.</p></article></div><div><div
class="methods-details"><summary>Read the equations and assumptions</summary><div
class="methods-content"><p>For population N, prevalence p, sensitivity s and specificity c, expressed
as proportions:</p><code>TP = N * p * s<br>FP = N * (1 - p) * c<br>FN = N *
(1 - p) * s<br>TN = N * (1 - p) * (1 - c)<br>PPV = TP / (TP + FP)</code><p>For a
second test among first-positives: both-positive true cases = TP * conditional detection;
both-positive non-cases = FP * conditional false-positive rate. Reusing unconditional rates
requires a conditional-independence assumption. A zero denominator has no defined
proportion.</p><p>Only visual squares are rounded. Expected counts, comparisons, curves and practice
answers retain the unrounded values.</p></div></div></div></section>
39 </main><footer><a class="brand" href="#lab">BaseRate</a><p>UnivaBio 2026's Educational
prototype<br>Fictional populations. Transparent assumptions.</p><div><a href="project.html">Project
materials</a><a href="https://github.com/Nicholas-afk/baserate" target="_blank" rel="noopener">Code &
datasets \u2197</a></div></div></div></div>
40 </body></html>

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dist/project.html

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1 <!doctype html><html lang="en"><head><meta charset="utf-8"><meta name="viewport"
content="width=device-width,initial-scale=1"><title>BaseRate \xb7 The project file</title><link
rel="stylesheet" href="style.css?v=1.1.1"><style>.materials{max-width:980px;margin:50px auto;padding:0
24px}.materials h1{font-size:clamp(36px,6vw,64px);margin:12px 0 25px}.materials h2{font:28px
Georgia;margin:35px 0 16px}.materials p{line-height:1.65}.materials
video{display:block;width:100%;margin:24px
0;background:#242c29}.material-links{display:flex;flex-wrap:wrap;gap:12px;margin:24px
0}.material-links a{padding:10px 14px;border:1px solid
#81877c;font-size:13px;text-decoration:none}.materials li{margin:12px
0;font-size:14px}</style></head><body><header class="site-header"><a class="brand"
href="index.html">BaseRate</a><nav aria-label="Project navigation"><a
href="index.html">Open the learning lab \u2197</a><a
href="https://github.com/Nicholas-afk/baserate">Source \u2197</a></nav></header><main
class="materials"><p class="eyebrow">UNIVABIO 2026 / AI FOR HUMAN HEALTH</p><h1>Understand the
denominator</h1><p>An interactive field guide to screening uncertainty. Explore fictional
populations, compare experiments, inspect the assumptions behind repeat testing, predict before
revealing the counts, and explain the result to an on-device AI coach.</p><div
class="material-links"><a href="index.html">Try the lab</a><a
href="downloads/BaseRate-Project-Description.pdf">One-page description</a><a
href="downloads/BaseRate-Complete-Code.pdf">Complete code PDF</a><a
href="downloads/BaseRate-Source.zip">Runnable source ZIP</a><a
href="https://github.com/Nicholas-afk/baserate">Repository, data & tests \u2197</a></div><h2>Working
prototype demonstration</h2><video controls preload="metadata" poster="media/demo-thumbnail.png"
aria-label="BaseRate demonstration narrated by the entrant"><source
src="downloads/BaseRate-Demo.mp4?v=final-20261006" type="video/mp4"></video><p>112 seconds of the
final application, with the entrant\u2019s unchanged supplied narration. Actual browser interactions
are shown as edited screen-state captures, with readable holds and cuts rather than a continuous
recording. Staged estimates are demonstration inputs, not learner-study data. The model results,
privacy scope and extensive AI assistance are qualified on screen. No synthesized voice or
reconstructed words are used.</p><p><a href="downloads/BaseRate-Demo.mp4?v=final-20261006">Open or
download the finished MP4 \u2197</a> \xb7 <a
href="https://github.com/Nicholas-afk/baserate/blob/main/docs/demo.md">Demo methods and claim
qualifications \u2197</a></p><h2>What you can investigate</h2><ul><li>Keep detection and clearance
fixed while changing prevalence. Pin experiment A, change B, and compare their positive groups and a
continuous prevalence curve.</li><li>Switch from population squares to a natural-frequency tree. The
table and tree retain raw expected counts; only squares are rounded.</li><li>Explore a second test
among first-positive people. Choose independent tests, perfectly repeated results, or explicit
conditional rates. A repeated result can add no information.</li><li>Complete three
predict/reveal/denominator cases and export a numeric learning notebook. Correct a wrong denominator
before proceeding. The notebook retains first/final choices and first numeric guesses for this tab
only; learner explanations are never exported.</li><li>Inspect influential model features, label
alternatives and uncalibrated votes. Override the suggested topic, answer a factual check, and revise
your explanation. Votes are model diagnostics, not health probabilities or proof of reasoning
quality.</li></ul><h2>Verification and limitations</h2><ul><li>336 arithmetic and population-rounding
combinations; 2,525 prevalence-curve points; 405 conditional repeat-test combinations; five test
suites, practice/export checks and a deterministic model audit.</li><li>100 synthetic English training
explanations. On a separate 30-example synthetic holdout: 29 correct raw classifications; uncertainty
checks accepted 28 labels, of which 27 were correct.</li><li>Known error: one correct distinction
between sensitivity and predictive value was misread. A 24-case development stress audit found further
semantic failures. Feedback is now provisional; classifications and original accuracy are unchanged.
Negation, mixed ideas, unfamiliar wording and other languages can fail. Votes are
uncalibrated.</li><li>No independent learner study, clinical validation or measured health benefit.
All scenarios are fictional; no personal diagnosis, risk estimate or treatment advice.</li></ul><p><a
href="https://github.com/Nicholas-afk/baserate/blob/main/docs/model-card.md">Read the model card and
audit \u2197</a> \xb7 <a
href="https://github.com/Nicholas-afk/baserate/blob/main/docs/research.md">Competition and originality
assessment \u2197</a></p><h2>Privacy and authorship</h2><p>No patient data, analytics, remote
inference, external scripts or fonts. Coach text stays in tab memory, without persistent app storage
or server transmission. Exports save numeric answers and selected choices to a file and exclude
explanations. Loading app/model files and this page\u2019s video metadata/media still contacts GitHub
Pages; hosting providers may retain ordinary request logs. Optional connected browser assistants have
separate data handling. The code, synthetic dataset, educational copy and submission materials were
created with extensive AI assistance at the entrant\u2019s direction. The entrant supplied the
narration recording and must understand and explain the work to judges.</p><p><a
href="https://github.com/Nicholas-afk/baserate/blob/main/docs/architecture.md">How the app and
classifier work \u2197</a> \xb7 <a
href="https://github.com/Nicholas-afk/baserate/blob/main/docs/privacy.md">Data handling \u2197</a>
\b7 <a href="https://github.com/Nicholas-afk/baserate/blob/main/docs/attribution.md">Attribution
\u2197</a></p><h2>Reproduce the work</h2><p>The repository includes the complete static app, synthetic
data, standard-library Python trainer, evaluation details and Node verification checks. The source
ZIP runs locally without app dependencies. The one-page description states what was built, what was
verified and what remains unvalidated.</p></main><footer><a class="brand"
href="index.html">BaseRate</a><p>Educational prototype \xb7 Fictional screening
scenarios</p></footer></body></html>
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dist/style.css

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1 :root{--paper:#f7f5ef;--white:#fffefa;--ink:#242c29;--muted:#626760;--line:#d7d7ca;--strong:#81877c;--
green:#245b4a;--rust:#944a2d;--amber:#b67925;--rose:#9e5065;--wash:#eeeeee4;--sans:ui-sans-serif,-apple
-system,BlinkMacSystemFont,"Segoe UI",sans-serif;--serif:Georgia,"Times New Roman",serif}*{box-sizing:
border-box}html{scroll-behavior:smooth;scroll-padding-top:24px}body{margin:0;background:var(--paper);c
olor:var(--ink);font:16px/1.6 var(--sans)}button,input,select,textarea{font:inherit;a{color:var(--gree
n);text-decoration:underline;text-decoration-thickness:1px}a:active{text-decoration-thickness:2px}butto
n,a,input,select,textarea{touch-action:manipulation}button{cursor:pointer}button:disabled{cursor:default;
opacity:.5}button:focus-visible,a:focus-visible,input:focus-visible,textarea:focus-visible,select:f
ocus-visible,summary:focus-visible,[tabindex="-1"]:focus-visible{outline:3px solid var(--rust);outline
-offset:4px}[hidden]{display:none!important}.skip{position:absolute;top:-100px;left:20px;background:va
r(--white);z-index:10;padding:12px}.skip:focus{top:10px}.site-header{display:flex;align-items:center;g
ap:36px;max-width:1560px;margin:auto;padding:24px 4%;border-bottom:1px solid
var(--line)}.brand{font:700 30px/1.1
var(--serif)};letter-spacing:-1.2px;color:var(--ink);text-decoration:none;white-space:nowrap}.brand
span{display:block;font:11px/1.6 var(--sans)};letter-spacing:.2px;color:var(--muted);margin-top:7px}nav
{display:flex;gap:27px;margin-left:auto}nav
a,.header-source{font-size:14px;text-decoration:none;color:var(--ink)}nav
a:active,.header-source:active{text-decoration:underline}.header-source{border-left:1px solid
var(--line);padding-left:27px}main{max-width:1560px;margin:auto;padding:0
4%}.opening{display:grid;grid-template-columns:1fr 300px;gap:50px;align-items:center;padding:45px 0
38px}.eyebrow,.small-label{font:600 11px/1.5 var(--sans)};letter-spacing:1.2px;margin:0 0
14px;color:var(--muted)}h1{font:400 40px,4.2vw,64px}/1.05
var(--serif)};letter-spacing:-1.8px;margin:0}h1
em{font-weight:400;color:var(--green)}.intro{max-width:680px;color:var(--muted);margin:19px 0
0;font-size:16px;line-height:1.65}.opening-note{border-left:1px solid
var(--line);padding-left:26px}.opening-note p{font-size:14px;margin:0 0
16px}.primary-link{display:flex;justify-content:space-between;border-bottom:1px solid
var(--green);padding-bottom:8px;text-decoration:none;font-size:14px;font-weight:600}.opening-note .mic
ro{font-size:11px;margin-top:14px}.section-bar{display:flex;justify-content:space-between;align-items:
center;gap:24px;border-top:2px solid var(--ink);padding:17px 0 0}.section-bar h2{font:400 25px/1.3
var(--serif)};letter-spacing:-.4px;margin:0}.section-index{font:500 13px var(--sans);color:var(--muted)
;padding-right:12px}.bar-actions{display:flex;gap:20px}.text-button{border:0;background:none;color:var
(--green);font-size:13px;padding:5px 0;text-decoration:underline;text-decoration:underline-offset:4px}.scope-note
{font-size:12px;color:var(--muted);margin:8px 0 18px}.micro,.rounding-note{font-size:12px;color:var(--
muted)};line-height:1.6}.micro:empty,.error:empty{display:none}.workspace{display:grid;grid-template-co
lums:245px minmax(300px,1fr) 260px;border:1px solid
var(--line);background:var(--white)}.controls{padding:25px 22px;border-right:1px solid var(--line)}.sc
enario-label{display:block;font-size:13px;font-weight:600;margin-bottom:7px}select{width:100%;padding:
10px 9px;border:1px solid var(--strong);border-radius:2px;background:var(--white);color:var(--ink);fon
t-size:14px}.muted{font-size:13px;color:var(--muted)}.control{margin:25px 0}.control label{display:fle
x;justify-content:space-between;align-items:center;gap:8px;font-size:13px;line-height:1.5;font-weight:
600}.control output{color:var(--green);font-size:17px;font-variant-numeric:tabular-nums}.range-row{dis
play:flex;align-items:center;gap:10px;margin-top:9px}input[type=range]{accent-color:var(--green);min-h
eight:30px;width:100%;margin:0}input[type=number]{font-variant-numeric:tabular-nums;border:1px solid
var(--strong);background:var(--white);color:var(--ink);padding:7px
6px;border-radius:2px;min-height:38px}.range-row
input[type=number]{width:65px;flex-shrink:0;font-size:13px}.control
p{font-size:12px;color:var(--muted)};line-height:1.5;margin:6px 0 0}.secondary,.primary{padding:10px
15px;min-height:42px;border-radius:2px;font-size:13px;font-weight:600}.secondary{border:1px solid var(
--strong);background:transparent;color:var(--ink)}.secondary:active{background:var(--wash)}.primary{bor
der:1px solid var(--green);background:var(--green)};color:white}.primary:active{background:#173e32}.marg
in-note{border-top:1px solid var(--line);padding-top:16px;margin:22px 0 0;font:italic 17px/1.5
var(--serif)};color:var(--muted)}.error{font-size:13px;color:#9c3529}.population-panel{padding:25px 24p
x;min-width:0}.panel-top{display:flex;justify-content:space-between;align-items:flex-start;gap:18px;ma
rgin-bottom:20px}h3{font:400 24px/1.3 var(--serif)};letter-spacing:-.4px;margin:0}.panel-top
h3{font-size:22px}.panel-top p{font-size:12px;color:var(--muted);margin:5px 0
0}.segmented{display:flex;align-items:center;border:1px solid
var(--strong);border-radius:2px}.segmented button{font-size:12px;padding:8px
10px;background:transparent;border:0;border-right:1px solid var(--line);color:var(--ink)}.segmented
button:last-child{border-right:0}.segmented button.active{background:var(--ink);color:var(--white)}.po
pulation{display:grid;grid-template-columns:repeat(40,1fr);gap:3px;align-content:start;min-height:280p
x}.person{width:100%;aspect-ratio:1;border-radius:0}.person.tp,.legend
.tp{background:var(--green)}.person.fp,.legend
.fp{background:repeating-linear-gradient(135deg,var(--amber) 0 3px,#d8b47f 3px 4px)}.person.fn,.legend
.fn{background:var(--rose);box-shadow:inset 0 0 2px #d5a7b2}.person.tn,.legend
.tn{background:#e3e3d9;border:1px solid
#c6cabc}.person.hidden{opacity:.12}.legend{display:flex;flex-wrap:wrap;gap:8px
14px;color:var(--muted)};font-size:11px;margin-top:16px}.legend
span{display:flex;align-items:center;gap:6px}.legend i{width:10px;height:10px}.counts{display:grid;gri
d-template-columns:repeat(4,1fr);gap:10px;border-top:1px solid var(--line);padding-top:17px;margin-top:
22px}.counts>div{display:flex;flex-direction:column}.count-tag{font-size:10px;font-weight:500;line-he
ight:1.4}.counts strong{font-size:27px;font-weight:500;letter-spacing:-1px;font-variant-numeric:tabula
r-nums;margin-top:4px}.counts small{font-size:11px;color:var(--muted)}.tp-text{color:var(--green)}.fp-
text{color:#8b5417}.fn-text{color:#944359}.rounding-note{font-size:11px;margin:18px 0
0}.result-panel{border-left:1px solid var(--line);padding:25px 23px;background:#eeeeee4}.result-panel
h3{font-size:24px;line-height:1.3}.result-number{display:flex;align-items:baseline;margin:23px 0
6px;gap:2px;color:var(--green)};font-variant-numeric:tabular-nums}.result-number
span:first-child{font:400 70px/1 var(--serif)};letter-spacing:-3px}.result-number
span:last-child{font:400 29px var(--serif)}.result-panel>p:not(.small-label){font-size:13px}.ratio-bar
{height:11px;background:var(--amber);margin-top:18px;overflow:hidden}.ratio-bar span{display:block;hei
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ght:100%;background:var(--green)}.result-key{display:flex;justify-content:space-between;font-size:10px
;color:var(--muted);margin-top:4px}.formula{border-top:1px solid var(--line);border-bottom:1px solid
var(--line);padding:16px 0;margin:20px 0}.formula span,.formula
small{display:block;font-size:11px;color:var(--muted)}.formula
code{display:block;font-size:17px;margin:4px 0;font-variant-numeric:tabular-nums}.text-link{display:bl
ock;font-size:13px;font-weight:600;margin-top:23px}.frequency-tree{min-height:310px}.tree-root{border-
bottom:1px solid var(--ink);font:21px var(--serif);padding:5px 0
16px;margin-bottom:20px}.tree-branches{display:grid;grid-template-columns:1fr
1fr;gap:22px}.tree-branch{border-left:1px solid
var(--strong);padding-left:16px}.tree-parent{font-size:12px;margin-bottom:20px}.tree-parent strong{fon
t-size:23px;display:block;font-weight:500}.tree-leaf{display:flex;justify-content:space-between;align-
items:center;gap:10px;font-size:12px;padding:12px 0;border-top:1px solid var(--line)}.tree-leaf b{font
-size:18px;font-weight:500}.compare-section,.practice-section,.coach-section,.evidence-section{padding
:48px 0 0}.section-intro{font-size:15px;color:var(--muted);max-width:780px;margin:18px 0
22px}.compare-layout{display:grid;grid-template-columns:1fr
1fr;gap:48px;margin-top:20px}.compare-layout .section-intro{margin-top:0}.form-actions{display:flex;al
ign-items:center;gap:10px;flex-wrap:wrap;margin-top:16px}.table-scroll{overflow-x:auto}table{width:100
%;border-collapse:collapse;font-variant-numeric:tabular-nums;font-size:13px;text-align:left}caption{te
xt-align:left;color:var(--muted);font-size:12px;margin-bottom:8px}th{font-weight:600}th,td{border-bott
om:1px solid var(--line);padding:10px
8px}th:first-child,td:first-child{padding-left:0}thead{border-top:1px solid
var(--strong)}.comparison-table td:not(:first-child){font-weight:500}.comparison-table tr:last-child
td{font-size:23px;color:var(--green);font-family:var(--serif)}.comparison-insight{border-left:2px
solid var(--green);padding-left:14px;font-size:14px}.curve-panel{border-left:1px solid
var(--line);padding-left:35px}.curve-panel svg{width:100%;height:auto;display:block}.curve-panel label
{display:flex;justify-content:space-between;font-size:13px;font-weight:600;margin-bottom:6px}.curve-pa
nel output{color:var(--green)}.curve-grid{stroke:var(--line);stroke-width:1}.curve-axis-label{fill:var
(--muted);font:13px var(--sans)}.curve-line{stroke:var(--green);stroke-width:3;fill:none}.curve-guide{
stroke:var(--strong);stroke-width:1;stroke-dasharray:3 4}.curve-dot{fill:var(--green);stroke:var(--whi
te);stroke-width:3}.curve-marker-label{fill:var(--green);font:600 15px
var(--sans)}summary{cursor:pointer;color:var(--green);font-size:13px;padding:8px 0;text-underline-offs
et:4px}details{margin-top:13px}details[open]>summary{margin-bottom:12px}.sequence-details{border-top:1
px solid var(--strong);border-bottom:1px solid var(--line);margin-top:28px;padding:7px
0}.sequence-details>summary{font:400 22px var(--serif);color:var(--ink);padding:13px
0}.sequence-body{display:grid;grid-template-columns:1.15fr 1fr;gap:48px;padding:10px 0
25px}.sequence-body p{font-size:14px}.sequence-body
label{font-size:13px;font-weight:600;display:block}.sequence-body
select{max-width:360px;margin-top:7px}.assumption-note{color:var(--rust);border-left:2px solid
var(--rust);padding-left:12px}.sequence-controls{display:grid;grid-template-columns:1fr
1fr;gap:23px}.sequence-controls label{font-size:12px;line-height:1.5}.sequence-controls output{display
:block;color:var(--green);font-size:17px;margin-top:5px}.sequence-result{background:var(--wash);padding:
25px}.sequence-result>strong{font:400 58px
var(--serif);color:var(--green)}.sequence-steps{border-top:1px solid
var(--line);padding-top:12px}.sequence-steps p{margin:6px
0;font-size:13px}.practice-layout{display:grid;grid-template-columns:270px 1fr;gap:46px;padding:22px 0
25px}.practice-steps{list-style:none;margin:0;padding:0;border-top:1px solid
var(--line)}.practice-steps li{border-bottom:1px solid var(--line);padding:18px
0;font-size:13px;color:var(--muted)}.practice-steps
li.current{color:var(--green);font-weight:600}.practice-steps li.done:after{content:'
\u2713';color:var(--green)}#practice-case,#practice-complete{max-width:820px}.practice-layout
h3{font-size:31px}.practice-layout label{font-size:13px}.prediction-input{display:flex;align-items:cen
ter;gap:10px;margin-top:8px}.prediction-input input{width:110px;font-size:20px;padding:8px
12px}.prediction-input .primary{margin-left:10px}.case-reveal{border-top:1px solid
var(--line);padding-top:20px;margin-top:25px}.case-reveal h4{font:400 27px/1.3
var(--serif);color:var(--green);margin:0}.case-reveal
p{font-size:14px}.case-equation{display:block;padding:13px
15px;background:var(--wash);font-size:16px;line-height:1.6;overflow-wrap:anywhere}.case-reveal
fieldset{margin:20px 0 0;padding:0;border:0}.case-reveal
legend{font-size:14px;font-weight:600;margin-bottom:10px}.case-reveal fieldset
label{display:block;margin:9px 0}input[type=radio]{accent-color:var(--green);width:17px;height:17px;ve
rtical-align:middle;margin-right:6px}.coach-layout{display:grid;grid-template-columns:1fr
1fr;gap:46px}.coach-layout
label{display:block;font-size:14px;font-weight:600;margin-bottom:12px}textarea{width:100%;border:1px
solid var(--strong);background:var(--white);min-height:160px;resize:vertical;border-radius:2px;padding
:15px;line-height:1.6;color:var(--ink);font-size:16px}.coach-response{background:var(--wash);border-le
ft:2px solid var(--green);padding:25px 28px}.coach-response h3{font-size:28px}.coach-response
p{font-size:14px}.coach-icon{display:none}.coach-response
.small-label{font-size:11px;color:var(--green)}.coach-response
.muted{font-size:12px}.feedback-source{border-top:1px solid var(--line);padding-top:13px;margin-top:20
px;font-size:12px}.feedback-tags{display:flex;gap:8px;flex-wrap:wrap}.feedback-tags span{padding:3px
7px;border:1px solid var(--strong);font-size:11px}.coach-response details{border-top:1px solid var(--l
ine);padding-top:5px}.diagnostic-bars{margin-top:13px}.diagnostic-row{display:grid;grid-template-colum
ns:105px 1fr 48px;gap:10px;align-items:center;font-size:11px;margin:7px
0}.diagnostic-meter{height:5px;background:#d8dacf}.diagnostic-meter span{display:block;height:100%;bac
kground:var(--green)}.evidence-section{padding-bottom:45px}.evidence-grid{display:grid;grid-template-c
olumns:1fr 1fr 1fr;gap:40px;padding-top:25px}.evidence-grid article{border-right:1px solid
var(--line);padding-right:30px}.evidence-grid
article:last-child{border-right:0;padding-right:0}.evidence-grid h3{font-size:25px}.evidence-grid
p{font-size:14px;color:var(--muted)}.evidence-grid .small-label,.evidence-grid
.micro{font-size:11px}.evidence-grid a{font-size:12px}.model-metric{color:var(--ink)!important;font-we
ight:600}.methods-details{border-top:1px solid var(--line);margin-top:30px;padding-top:8px}.methods-co
ntent{max-width:850px;font-size:14px}.methods-content
code{display:block;background:var(--wash);padding:18px}.methods-content
p{color:var(--muted)}footer{border-top:1px solid var(--line);padding:27px 4% 35px;max-width:1560px;mar

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gin:auto;display:flex;justify-content:space-between;align-items:center;gap:30px}footer
.brand{font-size:27px}footer
p{font-size:12px;color:var(--muted);margin:0}footer>div{display:flex;gap:22px;font-size:12px}
2 @media(min-width:1450px){.workspace{grid-template-columns:270px minmax(400px,1fr)
285px}.population{min-height:350px;gap:4px}.result-panel{padding:27px}.result-number
span:first-child{font-size:80px}}
3 @media(max-width:1150px){.workspace{grid-template-columns:230px
minmax(300px,1fr)}.result-panel{grid-column:1/-1;border-left:0;border-top:1px solid
var(--line);display:grid;grid-template-columns:1fr 1fr 1fr;gap:0
28px}.result-panel>.small-label{grid-column:1}.result-panel h3{grid-column:1;grid-row:2}.result-number
{grid-column:2;grid-row:1/3;margin:0;align-self:center}.result-panel>#result-explanation{grid-column:3
;grid-row:1/3}.result-panel .ratio-bar,.result-panel
.result-key{display:none}.formula{grid-column:1;grid-row:3;margin:15px 0
0;border-bottom:0}.insight{grid-column:2;grid-row:3}.result-panel
.text-link{grid-column:3;grid-row:3;align-self:center}.opening{grid-template-columns:1fr 240px;gap:30p
x}.compare-layout{gap:25px}.curve-panel{padding-left:25px}.header-source{display:none}.practice-layout
{grid-template-columns:220px 1fr;gap:30px}.sequence-body{gap:30px}}
4 @media(max-width:780px){.site-header{gap:20px;flex-wrap:wrap;padding:22px 5% 15px}.site-header
nav{margin-left:0;width:100%;gap:0;justify-content:space-between}.site-header nav
a{font-size:13px}.brand{font-size:29px}.brand span{font-size:11px}main{padding:0
5%}.opening{grid-template-columns:1fr;padding:30px 0;gap:25px}h1{font-size:43px;letter-spacing:-1.3px}
.intro{font-size:15px}.opening-note{border-left:0;border-top:1px solid var(--line);padding:15px 0
0;max-width:460px}.opening-note .small-label{display:none}.opening-note
p{font-size:13px;margin-bottom:12px}.opening-note .micro{font-size:11px;margin-bottom:0}.section-bar{f
lex-wrap:wrap;gap:8px;padding-top:13px}.section-bar h2{font-size:24px}.bar-actions{gap:18px}.scope-not
e{font-size:11px}.workspace{grid-template-columns:1fr}.controls{border-right:0;border-bottom:1px solid
var(--line);padding:22px}.controls>.small-label{margin-bottom:9px}.control{margin:20px 0}.control
label{font-size:14px}.control p{font-size:12px}.range-row
input[type=number]{width:75px;font-size:15px}.margin-note{display:none}.population-panel{padding:22px
17px}.panel-top{flex-wrap:wrap;gap:10px}.population{grid-template-columns:repeat(25,1fr);gap:3px;min-h
eight:420px}.counts{grid-template-columns:repeat(2,1fr);gap:14px}.count-tag{font-size:12px}.counts
small{font-size:12px}.counts strong{font-size:31px}.legend{font-size:11px}.result-panel{grid-column:au
to;display:block;padding:25px;border-top:1px solid var(--line)}.result-panel h3
br{display:none}.result-number{margin:18px 0}.result-panel .ratio-bar,.result-panel
.result-key{display:block}.result-panel .result-key{display:flex}.formula{margin:20px
0;border-bottom:1px solid var(--line)}.result-panel .text-link{margin-top:15px}.frequency-tree{min-hei
ght:300px}.tree-branches{gap:15px}.tree-branch{padding-left:10px}.compare-section,.practice-section,.c
oach-section,.evidence-section{padding-top:35px}.compare-layout,.sequence-body,.coach-layout,.evidence
-grid{grid-template-columns:1fr;gap:25px}.curve-panel{border-left:0;border-top:1px solid
var(--line);padding:23px 0 0}.sequence-controls{gap:20px}.sequence-details>summary{font-size:21px;line
-height:1.4}.sequence-result{padding:22px}.practice-layout{grid-template-columns:1fr;gap:23px;padding-
top:5px}.practice-steps{display:flex;gap:12px;border-top:0}.practice-steps
li{flex:1;font-size:11px;padding:12px 0;border-top:1px solid
var(--line);border-bottom:0}.practice-layout
h3{font-size:28px}.prediction-input{flex-wrap:wrap}.prediction-input
.primary{margin-left:0}.coach-response{padding:22px}.evidence-grid{padding-top:22px}.evidence-grid
article{border-right:0;border-bottom:1px solid var(--line);padding:0 0 20px}.evidence-grid article:las
t-child{border-bottom:0}.section-intro{font-size:14px}.case-equation{font-size:14px}footer{align-items
:flex-start;flex-wrap:wrap;padding:25px
5%}footer>div{width:100%}table{font-size:12px}th,td{padding:9px 6px}}
5 @media(prefers-reduced-motion:reduce){html{scroll-behavior:auto}}
6 @media print{body{background:white;color:black;font-size:11px}.site-header nav,.header-source,.opening
-note,.bar-actions,.compare-section,.practice-section,.coach-section,.evidence-section,footer,button{d
isplay:none}.opening{padding:15px 0}.opening h1{font-size:34px}.workspace{grid-template-columns:180px
1fr 180px}.controls,.population-panel,.result-panel{padding:12px}.result-panel{display:block;grid-colu
mn:auto;border-left:1px solid #aaa}.population{min-height:180px;gap:2px}.person{print-color-adjust:exa
ct;-webkit-print-color-adjust:exact}.result-number span:first-child{font-size:48px}.counts
strong{font-size:20px}input[type=range]{display:none}.range-row
input{border:0}.scope-note{font-size:10px}.result-panel .ratio-bar,.result-panel
.result-key{display:none}}
7 .coach-check{margin:1.4rem 0;padding:1.25rem 0;border-top:1px solid var(--line);border-bottom:1px
solid var(--line)}
8 .coach-check h4{font-family:Georgia,serif;font-size:1.3rem;margin:0 0 .5rem}.coach-check
select{width:100%;margin:.5rem 0 1rem}.coach-check fieldset{border:0;padding:0;margin:0 0
1rem}.coach-check legend{font-weight:600;margin-bottom:.7rem}.coach-check fieldset
label{display:block;font-weight:400;margin:.65rem 0}.coach-check
input{accent-color:#245b4a}.coach-check-feedback{font-size:.95rem;line-height:1.6}

```

dist/science.js

```
1  (function(root){
2    'use strict';
3    function calculate(prevalence,sensitivity,specificity,population=1000){
4      for(const [value,label] of
5        [[prevalence,'prevalence'],[sensitivity,'sensitivity'],[specificity,'specificity']])
6        if(!Number.isFinite(value)||value<0||value>100) throw new RangeError(label+' must be a percentage from
7          0 to 100.');
```

```
8      if(!Number.isFinite(population)||population<=0)throw new RangeError('Population must be
9        positive.');
```

```
10     const p=prevalence/100,se=sensitivity/100,sp=specificity/100;
11     const tp=population*p*se,fn=population*p*(1-se),fp=population*(1-p)*(1-sp),tn=population*(1-p)*sp;
12     return {tp,fp,fn,tn,positive:tp+fp,negative:tn+fn,ppv:tp+fp>0?tp/(tp+fp)*100:null,npv:tn+fn>0?tn/(
13       tn+fn)*100:null};
14   }
15   function roundedCounts(counts,total=1000){
16     const names=['tp','fn','fp','tn'],out={},remainders=[];
17     let used=0;
18     for(const name of names){out[name]=Math.floor(counts[name]);used+=out[name];remainders.push([name,
19       counts[name]-out[name]]);}
20     remainders.sort((a,b)=>b[1]-a[1]);
21     for(let i=0;i<total-used;i++)out[remainders[i%names.length][0]]++;
22     return out;
23   }
24   function fromSettings(settings,population=1000){
25     if(!settings||typeof settings!=='object')throw new RangeError('An experiment needs three fictional
26       percentage settings.');
```

```
27     return calculate(settings.prevalence,settings.sensitivity,settings.specificity,population);
28   }
29   function compare(settingsA,settingsB){
30     const a=fromSettings(settingsA),b=fromSettings(settingsB);
31     return {a,b,delta:{tp:b.tp-a.tp,fp:b.fp-a.fp,fn:b.fn-a.fn,tn:b.tn-a.tn,ppv:a.ppv===null||b.ppv===n
32       ull?null:b.ppv-a.ppv}};
33   }
34   function prevalenceCurve(sensitivity,specificity){
35     return
36     Array.from({length:101},(_,i)=>({prevalence:i/2,ppv:calculate(i/2,sensitivity,specificity).ppv}));
37   }
38   // These are conditional rates AMONG first-positive people. Independence is
39   // only one possible way to choose them; repeated results need different rates.
40   function sequential(settings,conditionalDetection,conditionalFalsePositive,population=1000){
41     const first=fromSettings(settings,population);
42     for(const value of
43       [conditionalDetection,conditionalFalsePositive])if(!Number.isFinite(value)||value<0||value>100)throw
44       new RangeError('Conditional rates must be percentages from 0 to 100.');
```

```
45     const tp=first.tp*conditionalDetection/100,fp=first.fp*conditionalFalsePositive/100;
46     return {first,tp,fp,positive:tp+fp,ppv:tp+fp>0?tp/(tp+fp)*100:null,rejectedTrue:first.tp-tp,reject
47       edFalse:first.fp-fp};
48   }
49   function formatNumber(value){
50     if(!Number.isFinite(value))throw new RangeError('Only finite values can be displayed.');
```

```
51     const absolute=Math.abs(value);
52     if(absolute>0&&absolute<1e-7)return value.toExponential(2);
53     const digits=absolute>0&&absolute<1?Math.min(7,Math.ceil(-Math.log10(absolute))+1):1;
54     return value.toLocaleString('en-US',{maximumFractionDigits:digits});
55   }
56   root.BaseRateScience={calculate,roundedCounts,compare,prevalenceCurve,sequential,formatNumber};
57   if(typeof module!=='undefined')module.exports=root.BaseRateScience;
58 }(typeof window!=='undefined'?window:globalThis));
```

dist/coach.js

```
1 (function(root){
2   'use strict';
3   const science=typeof module!=='undefined'?require('./science.js'):root.BaseRateScience;
4   const content={
5     unavailable:{title:'The AI model did not load.',focus:'Model unavailable',body:'The AI model is
unavailable, so the coach cannot interpret your wording. Your explanation has not been judged. The
mathematical simulator, comparisons and practice remain available.',question:'Reload to try loading
the model, or use the three-case lesson to inspect the arithmetic.',source:'bmj'},
6     balanced:{title:'Review the positive group.',focus:'Suggested angle: positives',body:'Compare true
positives with everyone who tests positive. This review topic is a model suggestion, not confirmation
that your explanation or arithmetic is correct.',question:'Which group belongs in the
denominator?',source:'bmj'},
7     certainty:{title:'Review what the experiment can establish.',focus:'Suggested angle:
uncertainty',body:'False positives and missed cases can occur. A fictional model cannot establish
certainty about a real person. This topic may fit your wording even if you were rejecting
certainty.',question:'What can these invented rates tell us about a real personal
result?',source:'nci'},
8     sensitivity:{title:'Review the two starting groups.',focus:'Suggested angle: test
rates',body:'Sensitivity starts with people who have the condition. Predictive value starts with
people who test positive. This topic may fit both a correct distinction and a misconception; the model
cannot tell them apart reliably.',question:'For sensitivity, which group is the
denominator?',source:'bmj'},
9     base_rate:{title:'Review the starting population.',focus:'Suggested angle: prevalence',body:'A
large group without the condition can produce many false positives. Compare the same test across
populations; this suggestion does not mean you ignored prevalence.',question:'Keep the test fixed.
Does PPV change between 1% and 20% prevalence?',source:'bmj'},
10    dismissal:{title:'Review benefits and harms.',focus:'Suggested angle: evidence',body:'Errors
matter, but these counts alone cannot establish a real screening program\u2019s benefits, harms or
suitability. The model may also suggest this topic when your explanation already makes that
point.',question:'What evidence is needed beyond these three rates?',source:'nci'},
11    uncertain:{title:'Choose an angle to check.',focus:'No reliable topic suggestion',body:'The model
has too little evidence to suggest a topic reliably. Choose a review focus below. This is not a
judgement of your understanding.',question:'Use a factual self-check, then revise your explanation if
useful.',source:'bmj'},
12    personal:{title:'Keep this experiment fictional.',focus:'Personal results need clinical
context',body:'This lab cannot interpret your personal symptoms, test result, or health risk. Its
numbers describe an invented population. Discuss personal results with an appropriate healthcare
professional.',question:'For this exercise, explain the fictional population shown above
instead.',source:'nci'},
13    empty:{title:'Start with one or two sentences.',focus:'Your reasoning matters',body:'Write an
explanation in your own words, then the coach can look for a reasoning pattern.',question:'Why might a
positive result include a person without the condition?',source:'bmj'}
14  };
15  const sources={bmj:{title:'Gigerenzer \xb7 What are natural frequencies? (BMJ,
2011)',url:'https://www.bmj.com/content/343/bmj.d6386'},nci:{title:'National Cancer Institute \xb7
Cancer Screening
Overview',url:'https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq'}};
16  const indexes=new WeakMap();
17  function tokenize(text,model){const stop=new
Set(model.stopWords),words=(text.toLowerCase().replace(/\u2019/g,"").match(/[a-z]+/g)||[]);return
[...words.filter(w=>!stop.has(w)),...words.slice(0,-1).map((w,i)=>w+'_'+words[i+1])];}
18  function classify(text,model){
19    if(typeof text!='string'||text.length>1800)throw new RangeError('Explanation must be text of at
most 1800 characters.');
```

```
20    if(!text.trim())return {label:'empty',method:'no inference',patterns:[]};
21    if(/(?:\bmys+(?:test|result|symptom|diagnosis|blood|scan)|\bils+(?:have|had|got)\s+(?:as+)?(?:po
sitive|negative|symptoms|cancer|disease|pain)|\b(?:diagnose me|should i(?:get|take|stop|start)|do i
have|am i(?:sick|ill)|my chance of))/i.test(text))return {label:'personal',method:'scope
guard',patterns:[]};
22    if(!model)return {label:'unavailable',method:'model unavailable',patterns:[]};
23    if(!indexes.has(model))indexes.set(model,new Map(model.vocabulary.map((v,i)=>[v,i])));
24    const index=indexes.get(model),feats=tokenize(text,model),known=feats.filter(w=>index.has(w));
25    const scores=model.labels.map(label=>({label,score:model.logPriors[label]+known.reduce((sum,w)=>su
m+model.weights[label][index.get(w)],0)}));
26    const
max=Math.max(...scores.map(s=>s.score)),den=scores.reduce((sum,s)=>sum+Math.exp(s.score-max),0);
27    for(const s of scores)s.vote=Math.exp(s.score-max)/den;
28    scores.sort((a,b)=>b.vote-a.vote);
29    const coverage=known.length/Math.max(1,feats.length),a=model.abstention;
30    const abstain=known.length<a.minKnownFeatures||coverage<a.minCoverage||scores[0].vote<a.minVote||s
cores[0].vote-scores[1].vote<a.minMargin;
31    const first=scores[0],second=scores[1];
32    const patterns=[...new Set(known)].map(word=>({word,diff:model.weights[first.label][index.get(word
)]-model.weights[second.label][index.get(word)]}).filter(x=>x.diff>0).sort((a,b)=>b.diff-a.diff).slic
e(0,4).map(x=>x.word.replace(/_/g,' ')));
33    return {label:abstain?'uncertain':first.label,rawLabel:first.label,method:'Multinomial Naive Bayes
',patterns,coverage,vote:first.vote,scores:scores.map(s=>({label:s.label,vote:s.vote})),abstained:abst
ain};
34  }
35  function feedback(result,counts){const
```

```

    item=content[result.label]||content.uncertain,format=root.BaseRateScience?.formatNumber||String;return
    {...item,provisional:![ 'personal','empty','unavailable'].includes(result.label),source:sources[item.s
    ource],counts:'In the current fictional experiment: ${format(counts.tp)} true positives and
    ${format(counts.fp)} false positives; ${counts.ppv===null?'the positive predictive value is
    undefined':format(counts.ppv)}% of positive results are expected to be true positives'}.}};
36   function exercise(topic,settings){
37     if(!settings)throw new RangeError('Fictional settings are required.');
```

38 const r=science.calculate(settings.prevalence,settings.sensitivity,settings.specificity),fmt=science.formatNumber;

39 const groups=[['positives','Everyone who tests positive'],['cases','Everyone with the condition'],['population','All 1,000 people']];

40 if(topic==='positives')return {question:'For true positives among positive results, which group is the denominator?',choices:groups,correctChoice:'positives',body:'The positive group contains \${fmt(r.tp)} true + \${fmt(r.fp)} false positives = \${fmt(r.positive)}. \${r.ppv===null?'It is empty, so PPV is undefined.':`\${fmt(r.tp)} \xf7 \${fmt(r.positive)} \xd7 100 = \${fmt(r.ppv)}%.`} The condition group is the denominator for sensitivity.'};

41 if(topic==='detection'){const cases=r.tp+r.fn;return {question:'For sensitivity, which group is the denominator?',choices:groups,correctChoice:'cases',body:'Start with all \${fmt(cases)} people with the condition: \${fmt(r.tp)} detected + \${fmt(r.fn)} missed. \${cases===0?'The group is empty, so this observed proportion is undefined.':`\${fmt(r.tp)} \xf7 \${fmt(cases)} \xd7 100 = \${fmt(settings.sensitivity)}%.`} This is a different group from the \${fmt(r.positive)} positive results.'};

42 if(topic==='prevalence'){

43 const a=science.calculate(1,settings.sensitivity,settings.specificity),b=science.calculate(20,settings.sensitivity,settings.specificity),pct=n=>n===null?'undefined':fmt(n)+'%';

44 const correctChoice=a.ppv===null||b.ppv===null?'undefined':Math.abs(b.ppv-a.ppv)<1e-10?'same':b.ppv>a.ppv?'higher':'lower';

45 return {question:'With this test fixed, how does PPV at 20% prevalence compare with PPV at 1%?',choices:[['higher','Higher'],['lower','Lower'],['same','The same'],['undefined','No defined comparison']],correctChoice,body:'The same \${fmt(settings.sensitivity)}% detection and \${fmt(settings.specificity)}% clearance give PPV \${pct(a.ppv)} at 1% and \${pct(b.ppv)} at 20%. \${correctChoice==='same'?`For these settings the proportions are equal; higher prevalence does not always change PPV.`:correctChoice==='undefined'?`A missing positive group prevents a defined comparison.`:'The test is unchanged; the starting groups change the positive denominator.'}};

46 }

47 if(topic==='uncertainty')return {question:'Can these invented rates establish a real person\u2019s diagnosis?',choices:[['context','No; real results need clinical context'],['diagnosis','Yes; this simulated percentage establishes it']],correctChoice:'context',body:'Even 100% in a fictional experiment is conditional on its invented assumptions. This simulator cannot interpret a real person\u2019s result or establish a diagnosis.'};

48 if(topic==='evidence')return {question:'What is needed to judge a real screening program\u2019s overall value?',choices:[['benefits','Evidence about benefits, harms and the population'],['errors','Only whether any false result occurs']],correctChoice:'benefits',body:'The counts demonstrate errors and uncertainty. Benefits, harms, downstream actions and the population need separate evidence. An error alone does not establish that every program is worthless.'};

49 throw new RangeError('Unknown review focus.');

50 }

51 function checkAnswer(topic,choice,settings){

52 const item=exercise(topic,settings);

53 if(!item.choices.some([value]>=>value===choice))throw new RangeError('Choose one of the displayed answers.');

54 return {correct:choice===item.correctChoice,body:item.body};

55 }

56 root.BaseRateCoach={classify,feedback,tokenize,exercise,checkAnswer};

57 if(typeof module!=='undefined')module.exports=root.BaseRateCoach;

58 }(typeof window!=='undefined'?window:globalThis);

dist/app.js

```
1  'use strict';
2  const $=id=>document.getElementById(id);
3  const settings=[{prevalence:1,'sensitivity':1,'specificity':1};
4  const presets={rare:[1,90,95],common:[20,90,95],specific:[1,90,99]};
5  let positiveOnly=false,current;
6  let lastExplanation=null;
7  let coachReviewTopic=null;
8  function number(n){return BaseRateScience.formatNumber(n);}
9  function update(preserveInput=null){
10   const values=settings.map(id=>Number($ (id).value));
11   current=BaseRateScience.calculate(...values);
12   settings.forEach((id,i)=>{$(id+'-label').textContent=number(values[i]+'%');$(id).setAttribute('aria-
valueText',number(values[i]+' percent'))});
13   settings.forEach((id,i)=>{const input=$(id+'-number');if(id!==(preserveInput))input.value=values[i];in
put.removeAttribute('aria-invalid')});
14   $('settings-error').textContent='';
15   for(const id of ['tp','fp','fn','tn'])$(id+'-count').textContent=number(current[id]);
16   $('ppv').textContent=current.ppv===null?'\u2014':number(current.ppv);
17   $('result-explanation').textContent=current.ppv===null?'There are no expected positive results, so
this percentage is undefined.':`About ${number(current.tp)} of ${number(current.positive)} expected
positive results are true positives.`;
18   $('ratio-fill').style.width=(current.ppv??0)+'%';
19   $('formula').textContent=`${number(current.tp)} \xf7 (${number(current.tp)} +
${number(current.fp)})`;
20   $('insight').textContent=current.ppv===null?'A proportion needs a denominator. With no positive
results, there is no positive group to compare.':`current.fp>current.tp?Even a capable test can
produce many false positives when a condition is rare.`As prevalence rises or false positives fall,
a larger share of positive results can be true positives.`;
21   const counts=BaseRateScience.roundedCounts(current),fragment=document.createDocumentFragment();
22   for(const key of ['tp','fn','fp','tn'])for(let i=0;i<counts[key];i++){const
square=document.createElement('span');square.className='person
'+key+(positiveOnly&&['fn','tn'].includes(key)?'
hidden':'');square.setAttribute('aria-hidden','true');fragment.append(square);}
23   $('population').replaceChildren(fragment);
24   $('population').setAttribute('aria-label','Rounded illustration of 1000 people: ${counts.tp} true
positives, ${counts.fp} false positives, ${counts.fn} false negatives, ${counts.tn} true negatives.
${positiveOnly?'Negative results are dimmed.':''}');
25   if(lastExplanation!=null)showFeedback(lastExplanation);
26   window.dispatchEvent(new CustomEvent('baserate:experiment',{detail:{settings:Object.fromEntries(sett
ings.map((id,i)=>[id,values[i]])),counts:{...current}}});
27 }
28 function setExperiment(values,preserveInput=null){
29   for(const id of settings)if(!Number.isFinite(values[id])||values[id]<Number($ (id).min)||values[id]
>Number($ (id).max))throw new RangeError('Fictional '+id+' is outside the available range.');
```

```

51   const label=node('label','Review
focus'),select=node('select');select.id='coach-topic';label.htmlFor=select.id;
52   for(const [value,text] of [['','Choose a focus'],['positives','The positive
group'],['detection','The detection rate'],['prevalence','Changing
prevalence'],['uncertainty','Uncertainty and context'],['evidence','Benefits and harms']]){const
option=node('option',text);option.value=value;select.append(option);}
53   select.value=coachReviewTopic??suggested;
54   const form=node('form'),question=node('fieldset'),status=node('p',undefined,'coach-check-feedback'),
button=node('button','Check this answer','secondary'),revise=node('button','Revise my
explanation','text-button');
55   form.noValidate=true;status.setAttribute('role','status');status.id='coach-check-feedback';button.ty
pe='submit';button.id='coach-check-submit';
56   function renderQuestion(){
57     status.textContent='';question.replaceChildren();button.disabled=!select.value;
58     if(!select.value){question.append(node('p','Choose a focus to start the self-check.'));return;}
59     const item=BaseRateCoach.exercise(select.value,BaseRateApp.getExperiment().settings);question.appe
nd(node('legend',item.question));
60     for(const [value,text] of item.choices){const row=node('label'),input=node('input');input.type='ra
dio';input.name='coach-check-answer';input.value=value;row.append(input,document.createTextNode('
'+text));question.append(row);}
61   }
62   select.addEventListener('change',()=>{coachReviewTopic=select.value;renderQuestion();});
63   form.addEventListener('change',()=>{status.textContent='';});
64   form.addEventListener('submit',event=>{
65     event.preventDefault();const answer=question.querySelector('input:checked');
66     if(!answer){status.textContent='Choose one of the displayed
answers.';question.querySelector('input')?.focus();return;}
67     const
check=BaseRateCoach.checkAnswer(select.value,answer.value,BaseRateApp.getExperiment().settings);
68     status.textContent=(check.correct?'Correct for this question. ':'Revisit the selected answer.
')+check.body+' This checks your selected answer, not the quality of your explanation.';
69   });
70   revise.type='button';revise.addEventListener('click',()=>$('explanation').focus());
71   form.append(question,button,status,revise);section.append(heading,node('p',result.label==='unavailab
le'? 'This factual check works without the AI model.':suggested?'The model suggests a review topic. You
can change it; a suggestion is not a verdict.': 'Choose a topic yourself; the model has no reliable
suggestion.','micro'),label,select,form);panel.append(section);renderQuestion();
72 }
73 function showFeedback(text){
74   const result=BaseRateCoach.classify(text>window.BaseRateModel),f=BaseRateCoach.feedback(result,curre
nt),panel=$( 'coach-response' );
75   const icon=node('span','\u2733','coach-icon');icon.setAttribute('aria-hidden','true');
76   panel.replaceChildren(icon,node('p',f.focus.toUpperCase(),'small-label'),node('h3',f.title),node('p'
,f.body));
77   if(!['personal','empty'].includes(result.label))panel.append(node('p',f.counts,'current-counts'));
78   panel.append(node('p',f.question,'coach-question'));
79   if(result.patterns.length&&!result.abstained){const tags=node('div','','feedback-tags');for(const
pattern of result.patterns)tags.append(node('span',pattern));panel.append(node('p','Patterns that
influenced the model:','muted'),tags);}
80   const source=node('div','Read the evidence: ','feedback-source'),link=node('a',f.source.title);link.
href=f.source.url;link.target='_blank';link.rel='noopener';source.append(link);panel.append(source);
81   panel.append(node('p','A model-selected topic does not establish a misconception or correct
reasoning. Negation, quotations and mixed ideas can be misread.','muted'));
82   addCoachCheck(panel,result);
83   if(result.scores){
84     const diagnostics=node('details'),summary=node('summary','Inspect the model\u2019s topic
suggestion');diagnostics.append(summary);
85     diagnostics.append(node('p','Feature coverage: ${number(result.coverage*100)}%.
${result.abstained?'The uncertainty gate withheld a topic suggestion.': 'The leading pattern passed the
prototype\u2019s uncertainty gate; that does not verify its meaning.'},'muted'));
86     const bars=node('div','','diagnostic-bars'),names={balanced:'Contextual',certainty:'Certainty',sen
sitivity:'Test rates',base_rate:'Prevalence',dismissal:'Dismissal'};
87     for(const score of result.scores){const row=node('div','','diagnostic-row'),meter=node('span','','
diagnostic-meter'),fill=node('span');fill.style.width=score.vote*100+'%';meter.append(fill);row.append
(node('span',names[score.label]),meter,node('span',score.vote>0&&score.vote*100<.1?'<0.1%':number(scor
e.vote*100)+'%'));bars.append(row);}
88     diagnostics.append(bars,node('p','These are relative, uncalibrated model votes. They are not
probabilities of correctness or health outcomes. The feature snippets above are associations, not a
complete explanation.','muted'));panel.append(diagnostics);
89   }
90   if(!['personal','empty'].includes(result.label)){const
next=node('a',result.label==='base_rate'? 'Next: compare the starting populations
\u2019: result.label==='sensitivity'? 'Next: practice choosing the denominator \u2019: 'Next: try a
prediction before revealing the counts
\u2019: 'text-link');next.href=result.label==='base_rate'? '#compare': '#practice';panel.append(next);}
91   return {label:result.label,focus:f.focus,counts:{...current},abstained:!!result.abstained};
92 }
93 $('coach-form').addEventListener('submit',e=>{e.preventDefault();coachReviewTopic=null;lastExplanation
=$('explanation').value;showFeedback(lastExplanation)});
94 let exampleIndex=0;
95 const examples=['The test detects 90 percent of cases, so 90 percent of positive results must be real
cases.','A positive result does not prove someone has the condition because false positives can
outnumber true positives when it is rare.','How common the condition is does not matter to a positive
result.'];

```

```

96 $('example').addEventListener('click',()=>{ $('explanation').value=examples[exampleIndex++%examples.length];lastExplanation=null;coachReviewTopic=null; $('coach-response').replaceChildren(node('p','EXAMPLE LOADED','small-label'),node('h3','Review this example.'),node('p','Select Find a review focus to review this example with the current experiment. The previous self-check has been cleared.')); $('explanation').focus();});
97 if(window.BaseRateModel){const m=window.BaseRateModel,e=m.evaluation;
98   $('model-summary').textContent='A ${m.algorithm} classifier trained on ${m.trainingCount} synthetic English explanations suggests a review topic. You can override it and check a factual answer separately. Your text is processed on this device.`';
99   const details= $('model-details');details.replaceChildren(node('p','Model ${m.version}; feedback policy 1.1. Word and adjacent-word features; additive smoothing; five original reasoning labels mapped to review topics. Training code and dataset are included in the source package.`),node('p','Reproduced byte for byte: ${e.correctCount}/${e.evaluatedCount} raw classifications matched the predefined synthetic labels; ${e.acceptedCount}/${e.evaluatedCount} passed uncertainty checks, ${e.acceptedCorrect} correct. The training and holdout sets have no exact duplicate sentences, but share authorship, task and familiar claim patterns.`),node('p','A 24-probe author-created stress audit found negation, ambiguous and mixed explanations can receive strong votes for the wrong pattern. Two of ten flawed or mixed probes previously received contextual praise. The revised feedback gives provisional review topics and factual self-checks, never correctness endorsements based on text. These probes informed the change; they are development regressions, not a fresh benchmark.`),node('p','The original weights, data and 30-example holdout are unchanged, including the known correct sensitivity distinction misread as confusion. No improved classification accuracy is claimed. See model/audit.json and docs/model-card.md in the repository for reproduction, lexical split checks and all probes.`),node('p','Uncertainty checks use feature coverage, model vote and separation between labels. Votes are uncalibrated and cannot detect every semantic mistake. A correct selected self-check answer does not verify the explanation or demonstrate learning.`),node('p','No real learner study, clinical validation, or measured health benefit. The AI never computes the simulator\u0019s numbers or generates medical facts. Explanations stay in tab memory; the app does not persist them or send them to a server. Reload clears them. Loading downloads static files from GitHub Pages; a connected browser assistant has separate data handling.`),node('p','Model and feedback were authored with AI assistance. The synthetic dataset contains no patient data.`));
100 }else{
101   $('model-summary').textContent='The AI model is unavailable. The simulator, comparisons and mathematical practice still work.`';
102   $('model-details').replaceChildren(node('p','Model data did not load. Reload to try again, or inspect the source package. No AI feedback about your wording can be given until the model is available.`));
103 }
104 if(document.modelContext?.registerTool){
105   const lifecycle=new AbortController();
106   for(const tool of [
107     {name:'configure_fictional_experiment',title:'Configure fictional experiment',description:'Change the visible educational simulation. No diagnosis or personal risk estimate.',inputSchema:{type:'object',properties:{prevalence:{type:'number',minimum:0.1,maximum:50},sensitivity:{type:'number',minimum:1,maximum:100},specificity:{type:'number',minimum:1,maximum:100},required:settings,additionalProperties:false},annotations:{readOnlyHint:false},execute(input){for(const id of settings){if(!Number.isFinite(input?.[id])||input[id]<Number($id).min||input[id]>Number($id).max))throw new Error('Invalid fictional percentage: '+id);}for(const id of settings){$id.value=input[id];$('scenario').value='custom';$('scenario-note').textContent='Explore your own fictional assumptions.`';update();return {settings:Object.fromEntries(settings.map(id=>[id,Number($id).value])}),expectedCounts:{...current}};}},
108     {name:'check_learning_explanation',title:'Check learning explanation',description:'Put an English explanation into the visible local AI coach and display educational feedback. Inference runs locally; a connected assistant has its own conversation/data handling.',inputSchema:{type:'object',properties:{explanation:{type:'string',maxLength:1800},required:['explanation'],additionalProperties:false},annotations:{readOnlyHint:false,untrustedContentHint:true},execute(input){if(typeof input?.explanation!='string'||input.explanation.length>1800)throw new Error('Explanation must be text, at most 1800 characters.`'); $('explanation').value=input.explanation;coachReviewTopic=null;lastExplanation=input.explanation;return showFeedback(lastExplanation);}}
109   ]){try{Promise.resolve(document.modelContext.registerTool(tool,{signal:lifecycle.signal})).catch(()=>>{});}catch{}}
110   window.addEventListener('pagehide',()=>lifecycle.abort(),{once:true});
111 }
112 update();
113 // Fixed, fictional demonstration links; no arbitrary learner text is accepted in URLs.
114 const demo=new URLSearchParams(window.location.search).get('demo');
115 if(demo==='common'){$('scenario').value='common';preset('common');}
116 if(demo==='coach'){$('explanation').value=examples[0];lastExplanation=examples[0];showFeedback(lastExplanation);}
117 if(demo==='model')document.querySelector('#evidence details').open=true;

```

dist/learning.js

```
1 (function(root){
2   'use strict';
3   const science=typeof module!=='undefined'?require('./science.js'):root.BaseRateScience;
4   const cases=Object.freeze([
5     {id:'rare',title:'The rare-condition
6     puzzle',settings:{prevalence:1,sensitivity:90,specificity:95},prompt:'The test detects 90% of cases.
7     What percentage of positive results do you expect to be true positives?',explanation:'The group
8     without the condition is much larger. Five percent of that group produces 49.5 false positives, beside
9     9 true positives. Sensitivity starts with cases; predictive value starts with positives.'},
10    {id:'common',title:'Same test, different
11    population',settings:{prevalence:20,sensitivity:90,specificity:95},prompt:'Only prevalence has
12    changed, from 1% to 20%. Predict the percentage of positive results that are true
13    positives.',explanation:'The test has not improved. The starting population has changed: 180 true
14    positives and 40 false positives. A percentage describing the test alone does not tell the whole
15    story.'},
16    {id:'specific',title:'Fewer false
17    alarms',settings:{prevalence:1,sensitivity:90,specificity:99},prompt:'Prevalence is back to 1%, but
18    specificity is now 99%. Predict the percentage of positive results that are true
19    positives.',explanation:'The smaller false-positive rate gives 9.9 false positives, alongside the same
20    9 true positives. That raises predictive value, but still does not make a positive result a
21    certainty.'}
22  ].map(item=>Object.freeze({...item,settings:Object.freeze(item.settings)}));
23  function gradePrediction(caseId,guess){
24    const item=cases.find(c=>c.id===caseId);
25    if(!item)throw new RangeError('Unknown learning case.');
```

11 if(!Number.isFinite(guess)||guess<0||guess>100)throw new RangeError('Enter a percentage from 0 to 100.');

13 const counts=science.calculate(item.settings.prevalence,item.settings.sensitivity,item.settings.specificity);

14 const error=Math.abs(counts.ppv-guess);

15 return {actual:counts.ppv,error,close:error<=2,count};

16 }

17 function notebook(answers){

18 return {version:2,purpose:'Fictional screening practice; not a validated learning assessment',storage:'This export was created on request. The app retains answers in tab memory, with no persistent app storage; this downloaded file persists independently.',attempts:answers.map(answer=>{

19 const

20 item=cases.find(c=>c.id===answer.caseId),result=gradePrediction(answer.caseId,answer.prediction);

21 const valid=value=>['positives','cases','population'].includes(value),denominator=valid(answer.denominator)?answer.denominator:null;

22 return {caseId:item.id,settings:{...item.settings},prediction:answer.prediction,actual:result.actual,absoluteErrorPercentagePoints:result.error,denominator,firstDenominator:valid(answer.firstDenominator)?answer.firstDenominator:denominator,denominatorAttempts:Number.isSafeInteger(answer.denominatorAttempts)&&answer.denominatorAttempts>=0?answer.denominatorAttempts:denominator?1:0};

23 }}}

24 }

25 function checkDenominator(caseId,choice){

26 if(!['positives','cases','population'].includes(choice))throw new RangeError('Choose a displayed denominator.');

27 const r=gradePrediction(caseId,0).counts,fmt=science.formatNumber,cases=r.tp+r.fn;

28 if(choice==='positives')return {correct:true,body:`Yes. Start with \${fmt(r.positive)} positive results: \${fmt(r.tp)} true + \${fmt(r.fp)} false. \${fmt(r.tp)} \xf7 \${fmt(r.positive)} \xd7 100 = \${fmt(r.ppv)}%`};

29 if(choice==='cases')return {correct:false,body:`That group contains \${fmt(cases)} people with the condition. \${fmt(r.tp)} \xf7 \${fmt(cases)} \xd7 100 = \${fmt(r.tp/cases\*100)}% is sensitivity. This question starts with everyone who tested positive. Try that group before continuing.`};

30 return {correct:false,body:`\${fmt(r.tp)} \xf7 1,000 \xd7 100 = \${fmt(r.tp/10)}% counts true positives in the whole population. This question asks about the smaller positive group. Try that group before continuing.`};

31 }

32 root.BaseRateLearning={cases,gradePrediction,notebook,checkDenominator};

33 if(typeof module!=='undefined')module.exports=root.BaseRateLearning;

34 })(typeof window!=='undefined'?window:globalThis);

dist/learning-ui.js

```
1 (function(){
2   'use strict';
3   const get=id=>document.getElementById(id),cases=BaseRateLearning.cases;
4   let index=0,answers=[],revealed=false,denominatorDone=false;
5   const fmt=value=>BaseRateScience.formatNumber(value);
6   function display(){
7     const item=cases[index];revealed=false;denominatorDone=false;
8     get('practice-progress').textContent=`Case ${index+1} of ${cases.length}`;
9     get('case-title').textContent=item.title;get('case-prompt').textContent=item.prompt;
10    get('case-settings').textContent=`Prevalence ${item.settings.prevalence}% \xb7 Detection
    ${item.settings.sensitivity}% \xb7 Clearance ${item.settings.specificity}%`;
11    get('prediction').value='';get('prediction').disabled=false;
12    get('predict-button').disabled=false;get('prediction-error').textContent='';
13    get('case-reveal').hidden=true;get('practice-next').disabled=true;
14    get('denominator-feedback').textContent='';
15    for(const input of
    document.querySelectorAll('[name="denominator"]')){input.checked=false;input.disabled=false;}
16    get('practice-next').textContent=index===cases.length-1?'See my notebook':'Next case';
17    get('practice-complete').hidden=true;get('practice-case').hidden=false;
18    get('practice-steps').replaceChildren(...cases.map((c,i)=>{
19      const li=document.createElement('li');li.textContent=`0${i+1}
    ${c.title}`;li.className=i===index?'current':i<index?'done':'';
20      if(i===index)li.setAttribute('aria-current','step');return li;
21    }));
22  }
23  get('prediction-form').addEventListener('submit',event=>{
24    event.preventDefault();if(revealed)return;
25    const prediction=get('prediction').valueAsNumber;
26    try{
27      const item=cases[index],result=BaseRateLearning.gradePrediction(item.id,prediction);
28      answers[index]={caseId:item.id,prediction,denominator:null,firstDenominator:null,denominatorAttempts:0};revealed=true;
29      get('prediction-error').textContent='';
30      get('prediction').disabled=true;get('predict-button').disabled=true;
31      get('case-reveal').hidden=false;
32      get('prediction-result').textContent=`${fmt(result.actual)}% of positive results are true
    positives.`;
33      get('prediction-difference').textContent=`Your first estimate: ${fmt(prediction)}%. Gap:
    ${fmt(result.error)} percentage points. ${result.close?'You were within 2 points.': 'Use the
    denominator below to see the difference.'}`;
34      get('case-equation').textContent=`${fmt(result.counts.tp)} \xf7 (${fmt(result.counts.tp)} +
    ${fmt(result.counts.fp)}) \xd7 100 = ${fmt(result.actual)}%`;
35      get('case-explanation').textContent=item.explanation;
36      get('case-reveal').focus({preventScroll:true});
37    }catch{get('prediction-error').textContent='Enter a number from 0 to 100.';}
38  });
39  get('denominator-form').addEventListener('change',event=>{
40    if(!revealed||denominatorDone||event.target.name!=='denominator')return;
41    const answer=answers[index],choice=event.target.value,result=BaseRateLearning.checkDenominator(cases[index].id,choice);
42    answer.firstDenominator??=choice;answer.denominator=choice;answer.denominatorAttempts++;
43    denominatorDone=result.correct;
44    get('denominator-feedback').textContent=result.body;
45    get('practice-next').disabled=!denominatorDone;
46    if(denominatorDone){for(const input of document.querySelectorAll('[name="denominator"]'))input.disabled=true;get('practice-next').focus({preventScroll:true});}
47  });
48  get('practice-next').addEventListener('click',()=>{
49    if(!revealed||!denominatorDone)return;
50    if(index<cases.length-1){index++;display();get('case-title').focus({preventScroll:true});}
51    else{
52      get('practice-case').hidden=true;get('practice-complete').hidden=false;
53      get('notebook-rows').replaceChildren(...answers.map(answer=>{
54        const item=cases.find(c=>c.id===answer.caseId),r=BaseRateLearning.gradePrediction(item.id,answer.prediction),tr=document.createElement('tr');
55        const names={cases:'Condition group',population:'Whole population',positives:'Positive group'},choices=answer.firstDenominator===answer.denominator?names[answer.denominator]:names[answer.firstDenominator]+' \u2192 '+names[answer.denominator];
56        for(const value of [item.title,fmt(answer.prediction)+'%',fmt(r.actual)+'%',fmt(r.error)+'%',choices+' ('+answer.denominatorAttempts+'
    choice'+(answer.denominatorAttempts===1?'':'s')+')']){const
    cell=document.createElement('td');cell.textContent=value;tr.append(cell);}return tr;
57      }));
58      get('practice-complete').focus({preventScroll:true});
59      for(const li of
    get('practice-steps').children){li.className='done';li.removeAttribute('aria-current');}
60    }
61  });
62  function restart(){index=0;answers=[];display();get('case-title').focus({preventScroll:true});}
63  get('restart-practice').addEventListener('click',restart);
```

```

64     get('download-notebook').addEventListener('click',()=>{
65         const url=URL.createObjectURL(new
66         Blob([JSON.stringify(BaseRateLearning.notebook(answers),null,2)],{type:'application/json'}));
67         const link=document.createElement('a');link.href=url;link.download='BaseRate-Learning-Notebook.js
68         on';link.click();setTimeout(()=>URL.revokeObjectURL(url),1000);
69     });
70     window.BaseRateLesson={notebook:()=>BaseRateLearning.notebook(answers),restart};
71     display();
72     if(new URLSearchParams(window.location.search).get('demo')==='practice'){
73         get('prediction').value=90;get('prediction-form').requestSubmit();
74         get('prediction-difference').textContent='Demonstration estimate: 90%. The expected answer is
75         15.4%. This is a fictional example, not a learner result.';
76     }
77 }
78 })();

```

dist/lab-tools.js

```
1  (function(){
2    'use strict';
3    const get-id=>document.getElementById(id), science=BaseRateScience;
4    const fmt=n=>n===null?'undefined':science.formatNumber(n);
5    const pct=n=>n===null?'Undefined':fmt(n)+'%';
6    const element=(tag, text, css)=>{const
e=document.createElement(tag);if(text!==undefined)e.textContent=text;if(css)e.className=css;return
e;};
7    let pinned={prevalence:1, sensitivity:90, specificity:95};
8    function tree(settings, counts){
9      const root=element('div', '1,000
people', 'tree-root'), branches=element('div', undefined, 'tree-branches');
10     for(const [heading, total, positive, negative, positiveClass, negativeClass] of [
11       ['With the condition', 1000*settings.prevalence/100, counts.tp, counts.fn, 'tp-text', 'fn-text'],
12       ['Without the condition', 1000*(1-settings.prevalence/100), counts.fp, counts.tn, 'fp-text', '']
13     ]){
14       const branch=element('div', undefined, 'tree-branch'), parent=element('div', heading, 'tree-parent');
15       parent.append(element('strong', fmt(total))); branch.append(parent);
16       for(const [label, n, css] of
[[['Positive', positive, positiveClass], ['Negative', negative, negativeClass]]){const
leaf=element('div', undefined, 'tree-leaf '+css); leaf.append(element('span', label), element('b', fmt(n)));
branch.append(leaf);}branches.append(branch);
17     }
18     get('frequency-tree').replaceChildren(root, branches);
19   }
20   function comparison(settings){
21     const pair=science.compare(pinned, settings), rows=[['Prevalence', pct(pinned.prevalence), pct(setting
s.prevalence)], ['Detection / clearance', pct(pinned.sensitivity)+' /
'+pct(pinned.specificity), pct(settings.sensitivity)+' / '+pct(settings.specificity)], ['True
positives', fmt(pair.a.tp), fmt(pair.b.tp)], ['False positives', fmt(pair.a.fp), fmt(pair.b.fp)], ['All
positive results', fmt(pair.a.positive), fmt(pair.b.positive)], ['True among
positives', pct(pair.a.ppv), pct(pair.b.ppv)]];
22     get('compare-rows').replaceChildren(...rows.map(row=>{const
tr=element('tr'); row.forEach((value, i)=>{const
cell=element(i===0?'th':'td', value); if(i===0) cell.scope='row'; tr.append(cell);}); return tr;}));
23     const
sameTest=pinned.sensitivity===settings.sensitivity&& pinned.specificity===settings.specificity;
24     get('compare-insight').textContent=pair.delta.ppv===null?'One experiment has no positive group, so
a percentage-point difference is undefined.':'B differs by
${pair.delta.ppv}>0?'+':'-'}${fmt(pair.delta.ppv)} percentage points.
${sameTest?(pinned.prevalence===settings.prevalence?'Both experiments have identical settings.':'The
test settings are identical; only prevalence differs.'):'Prevalence or test settings differ. This
comparison does not isolate a single cause.'}`;
25   }
26   const ns='http://www.w3.org/2000/svg';
27   function svgNode(tag, attrs, text){const n=document.createElementNS(ns, tag);for(const [key, value] of
Object.entries(attrs))n.setAttribute(key, value);if(text!==undefined)n.textContent=text;return n;}
28   function curve(settings, counts){
29     const svg=get('prevalence-curve'), parts=[svgNode('title', {id:'curve-title'}, 'How prevalence
changes positive predictive value'), svgNode('desc', {id:'curve-description'}, 'Detection
${fmt(settings.sensitivity)} percent, clearance ${fmt(settings.specificity)} percent. Current
prevalence ${fmt(settings.prevalence)} percent gives ${pct(counts.ppv)} true positives among
positives. Equivalent values are in the table.')]
30     const x=p=>60+p/50*540, y=p=>245-p/100*210;
31     for(const value of [0, 25, 50, 75, 100])parts.push(svgNode('line', {x1:60, x2:600, y1:y(value), y2:y(value)
}, class:'curve-grid'), svgNode('text', {x:47, y:y(value)+4, 'text-anchor':'end', class:'curve-axis-label'
}, value+'%'));
32     for(const value of [0, 10, 20, 30, 40, 50])parts.push(svgNode('text', {x:x(value), y:269, 'text-anchor':'m
iddle', class:'curve-axis-label', value+'%'}));
33     parts.push(svgNode('text', {x:60, y:17, class:'curve-axis-label'}, 'True among positive
results'), svgNode('text', {x:330, y:305, 'text-anchor':'middle', class:'curve-axis-label'}, 'Prevalence in
the starting population'));
34     const points=science.prevalenceCurve(settings.sensitivity, settings.specificity); let
path='', connected=false;
35     for(const p of points){if(p.ppv===null){connected=false; continue;}path+=(connected?'L':'M')+x(p.pr
evalence).toFixed(2)+' '+y(p.ppv).toFixed(2)+' '+connected=true;}
36     parts.push(svgNode('path', {d:path, class:'curve-line'}));
37     if(counts.ppv!==null){const px=x(settings.prevalence), py=y(counts.ppv); parts.push(svgNode('line', {
x1:px, x2:px, y1:py, y2:245, class:'curve-guide'}, svgNode('circle', {cx:px, cy:py, r:6, class:'curve-dot'}), s
vgNode('text', {x:Math.min(530, Math.max(80, px+13)), y:Math.max(33, py-12), class:'curve-marker-label'}, pct
(counts.ppv)}));
38     svg.replaceChildren(...parts);
39     get('curve-test-label').textContent='Detection ${fmt(settings.sensitivity)}% \xb7 Clearance
${fmt(settings.specificity)}%';
40     get('curve-prevalence').value=settings.prevalence; get('curve-value').textContent=pct(settings.prev
alence);
41     get('curve-table').replaceChildren(...[0, .1, 1, 5, 10, 20, 50].map(p=>{const tr=element('tr'), th=elemen
t('th', pct(p)); th.scope='row'; tr.append(th, element('td', pct(science.calculate(p, settings.sensitivity, s
ettings.specificity).ppv)); return tr;}));
42   }
```

```

42  function sequence(settings){
43    const mode=get('sequence-mode').value;
44    if(mode==='independent'){get('conditional-detection').value=settings.sensitivity;get('conditional-
false').value=Number((100-settings.specificity).toFixed(1));}
45    if(mode==='repeat'){get('conditional-detection').value=100;get('conditional-false').value=100;}
46    const d=Number(get('conditional-detection').value),f=Number(get('conditional-false').value),r=science.sequential(settings,d,f);
47    get('conditional-detection-label').textContent=pct(d);get('conditional-false-label').textContent=pct(f);
48    get('sequence-assumption').textContent=[independent:'Assumption: the two results are independent within each condition group. Real tests can share errors. Independence is invented here, not evidence about a real test.',repeat:'Assumption: every first-positive result is repeated, including every false positive. The second result adds no new information.',custom:'These conditional rates apply only among people positive on the first test. Their values need separate evidence in a real setting.'][mode];
49    get('sequence-ppv').textContent=pct(r.ppv);
50    get('sequence-counts').textContent=r.ppv===null?'No one is expected to be positive twice. The proportion is undefined.':`${fmt(r.tp)} true and ${fmt(r.fp)} false positives remain: ${fmt(r.tp)}\xf7 ${fmt(r.positive)} \xd7 100.`;
51    get('sequence-steps').replaceChildren(element('p','First positives: ${fmt(r.first.tp)} true + ${fmt(r.first.fp)} false'),element('p','Both positive: ${fmt(r.tp)} true + ${fmt(r.fp)} false'),element('p','Second step rejects: ${fmt(r.rejectedTrue)} true + ${fmt(r.rejectedFalse)} false first-positives'));
52  }
53  function render(detail){tree(detail.settings,detail.counts);comparison(detail.settings);curve(detail.settings,detail.counts);sequence(detail.settings);}
54  get('pin-experiment').addEventListener('click',()=>{pinned={...BaseRateApp.getExperiment().settings};get('pin-status').textContent=`A pinned: ${pct(pinned.prevalence)} prevalence, ${pct(pinned.sensitivity)} detection, ${pct(pinned.specificity)} clearance. Changes now affect B only.`;render(BaseRateApp.getExperiment());});
55  get('compare-common').addEventListener('click',()=>{pinned={prevalence:1,sensitivity:90,specificity:95};get('pin-status').textContent=`A: rare condition at 1%. B: same test at 20% prevalence.`;BaseRateApp.setExperiment({prevalence:20,sensitivity:90,specificity:95});});
56  get('curve-prevalence').addEventListener('input',()=>BaseRateApp.setExperiment({...BaseRateApp.getExperiment().settings,prevalence:Number(get('curve-prevalence').value)}));
57  get('sequence-mode').addEventListener('change',()=>sequence(BaseRateApp.getExperiment().settings));
58  for(const id of ['conditional-detection','conditional-false'])get(id).addEventListener('input',()=>{get('sequence-mode').value='custom';sequence(BaseRateApp.getExperiment().settings)});
59  get('share-experiment').addEventListener('click',async()=>{
60    const url=new URL(window.location.href);url.search='';url.hash='lab';
61    const s=BaseRateApp.getExperiment().settings;url.searchParams.set('p',s.prevalence);url.searchParams.set('se',s.sensitivity);url.searchParams.set('sp',s.specificity);
62    try{await navigator.clipboard.writeText(url.href);get('link-message').textContent='Copied. This link contains only fictional rate settings.';}catch{get('link-message').textContent='Copy this settings link: ${url.href}`;}}
63  });
64  window.addEventListener('baserate:experiment',event=>render(event.detail));
65  const params=new URLSearchParams(window.location.search),keys=['p','se','sp'];
66  if(keys.some(key=>params.has(key))){
67    const valid=keys.every(key=>params.getAll(key).length===1&&/^(?:\d+(?:\.\d+)?|\.\d+)$/.test(params.get(key)));
68    try{if(!valid)throw new Error('Invalid settings');BaseRateApp.setExperiment({prevalence:Number(params.get('p')),sensitivity:Number(params.get('se')),specificity:Number(params.get('sp'))});get('link-message').textContent='Loaded fictional settings from this link.';}catch{get('link-message').textContent='This settings link is incomplete or outside the available ranges. The default experiment is shown.';}}
69  }
70  render(BaseRateApp.getExperiment());
71  const demo=new URLSearchParams(window.location.search).get('demo');
72  if(demo==='compare')BaseRateApp.setExperiment({prevalence:20,sensitivity:90,specificity:95});
73  if(demo==='sequence')get('sequence').open=true;
74  })();

```

model/train.py

```
1  """Reproducible, dependency-free educational text-model training. No clinical data."""
2  import collections, hashlib, json, math, pathlib, re
3  ROOT = pathlib.Path(__file__).resolve().parents[1]
4  data = json.loads((ROOT / 'model/dataset.json').read_text())
5  STOP = {'a', 'an', 'the', 'is', 'are', 'am', 'be', 'being', 'been', 'of', 'to', 'in', 'and', 'or', 'it', 'its', 'if', 'for', 'as', 'at', 'i', 'we', 'you', 'they', 'their', 'there', 'this', 'that', 'these', 'those', 'with', 'by', 'from', 'on', 'us', 'our', 'my', 'has', 'have', 'had', 'was', 'were', 'can', 'could', 'would', 'should'}
6  def features(text):
7      words = re.findall(r"[a-z]+", text.lower().replace('\u2019', ''))
8      return [w for w in words if w not in STOP] + [words[i]+'_'+words[i+1] for i in
9              range(len(words)-1)]
10     labels = list(data['train'])
11     counts = {label: collections.Counter() for label in labels}
12     vocabulary = set()
13     for label, examples in data['train'].items():
14         for example in examples:
15             counts[label].update(features(example))
16             vocabulary.update(features(example))
17     vocabulary = sorted(vocabulary)
18     alpha = 0.7
19     weights = {label:
20         [math.log((counts[label][word]+alpha)/(sum(counts[label].values()+alpha*len(vocabulary)))) for word in
21             vocabulary] for label in labels}
22     model = {'version': '1.0.0', 'algorithm': 'Multinomial Naive Bayes', 'labels': labels, 'vocabulary': vocabula
23 ry, 'weights': weights, 'logPriors': {l: math.log(len(data['train'][l])/sum(map(len, data['train'].values())))
24 for l in labels}, 'stopWords': sorted(STOP), 'trainingCount': sum(map(len, data['train'].values())), 'hol
25 doutCount': sum(map(len, data['holdout'].values())), 'datasetSha256': hashlib.sha256((ROOT/'model/dataset.
26 json').read_bytes()).hexdigest(), 'abstention': {'minKnownFeatures': 4, 'minCoverage': 0.18, 'minVote': 0.65,
27 'minMargin': 0.2}, 'warning': 'Scores are uncalibrated model votes, not probabilities of correctness.
28 Synthetic English evaluation is not evidence of clinical or learner benefit.'}
29 index = {word: i for i, word in enumerate(vocabulary)}
30 def predict(text):
31     feats = features(text)
32     known = [index[w] for w in feats if w in index]
33     scores = {label: model['logPriors'][label] + sum(weights[label][i] for i in known) for label in labels}
34     maximum = max(scores.values())
35     exp = {k: math.exp(v - maximum) for k, v in scores.items()}
36     votes = {k: v / sum(exp.values()) for k, v in exp.items()}
37     ordered = sorted(votes, key=votes.get, reverse=True)
38     a = model['abstention']
39     abstain = len(known) < a['minKnownFeatures'] or len(known) / max(1, len(feats)) < a['minCoverage'] or
40 votes[ordered[0]] < a['minVote'] or votes[ordered[0]] - votes[ordered[1]] < a['minMargin']
41     return ordered[0], votes[ordered[0]], abstain
42 confusion = {l: {p: 0 for p in labels} for l in labels}
43 rows = []
44 for actual, examples in data['holdout'].items():
45     for text in examples:
46         predicted, vote, abstain = predict(text)
47         confusion[actual][predicted] += 1
48         rows.append({'text': text, 'actual': actual, 'predicted': predicted, 'vote': round(vote, 4), 'abstained
49 ': abstain})
50 correct = sum(r['actual'] == r['predicted'] for r in rows)
51 accepted = [r for r in rows if not r['abstained']]
52 metrics = {'evaluatedCount': len(rows), 'correctCount': correct, 'rawAccuracy': correct / len(rows), 'acceptedCo
53 unt': len(accepted), 'acceptedCorrect': sum(r['actual'] == r['predicted'] for r in
54 accepted), 'coverage': len(accepted) / len(rows), 'confusionMatrix': confusion, 'limits': 'Small,
55 author-created English holdout. Training and holdout share a fictional task; no external,
56 multilingual, clinical, or educational-outcome validation.'}
57 model['evaluation'] = metrics
58 (ROOT/'dist/model.js').write_text('window.BaseRateModel =
59 '+json.dumps(model, separators=(',', ':'))+';\n')
60 (ROOT/'model/evaluation.json').write_text(json.dumps({'metrics': metrics, 'predictions': rows}, indent=2)+
61 '\n')
62 print(json.dumps(metrics, indent=2))
```

model/dataset.json

```
1 {
2   "description": "Author-created fictional English learner explanations. Labels describe educational
   reasoning, never a health status. Held-out sentences were authored separately; no real learners or
   patient data are included.",
3   "train": {
4     "balanced": [
5       "A positive result does not prove the condition because false positives happen.",
6       "The chance after a positive result depends on how common the condition is.",
7       "We need to count true positives and false positives among everyone who tests positive.",
8       "A rare condition means many people without it may test positive.",
9       "Sensitivity is not the same as the chance of the condition after a positive test.",
10      "Prevalence matters because the group without the condition can be much larger.",
11      "A negative result does not completely rule out the condition because cases can be missed.",
12      "Both false positives and false negatives are possible in a screening test.",
13      "The proportion with the condition among positives is true positives divided by all positives.",
14      "A result is uncertain and has to be interpreted in context.",
15      "Most positives could be false when the condition is uncommon.",
16      "Even a good test cannot make a positive result a guaranteed diagnosis.",
17      "The same sensitivity can give different predictive values in different populations.",
18      "The result depends on prevalence and the number of people incorrectly flagged.",
19      "Testing positive is not a certainty; some positive people do not have the condition.",
20      "False positives can outnumber true positives when there are few actual cases.",
21      "A positive is a clue rather than proof, since the test can be wrong.",
22      "Changing how common the condition is changes the meaning of a positive result.",
23      "To understand a positive we compare detected cases with all positive results.",
24      "A negative test can miss cases, so it is not an absolute guarantee."
25    ],
26    "certainty": [
27      "A positive result proves you have the condition.",
28      "Everyone with a positive test definitely has the disease.",
29      "Testing positive means a guaranteed diagnosis.",
30      "The person must be ill because their test is positive.",
31      "A positive result is certain proof of the condition.",
32      "There is no uncertainty after a positive test.",
33      "A negative result proves you cannot have the condition.",
34      "If the result is negative it is impossible to have the disease.",
35      "Everyone who tests negative is definitely healthy.",
36      "A negative test guarantees there are no cases missed.",
37      "The test is positive so there is a hundred percent chance of disease.",
38      "Positive tests always mean actual disease.",
39      "The result is conclusive and there cannot be any mistakes.",
40      "A positive is a definite diagnosis with no need for context.",
41      "You always have the condition if the screening says positive.",
42      "A negative rules out disease completely.",
43      "Screening gives absolute certainty about who is ill.",
44      "The result tells us for sure whether the condition is present.",
45      "No positive person could be free of the disease.",
46      "Nobody with a negative result can have the condition."
47    ],
48    "sensitivity": [
49      "The sensitivity is ninety percent so a positive person has a ninety percent chance of
   disease.",
50      "Sensitivity tells us the chance of being ill after a positive result.",
51      "Ninety percent sensitivity means ninety percent of positive results are true.",
52      "Positive predictive value and sensitivity are the same number.",
53      "The detection rate is the probability a positive person has the condition.",
54      "A test that detects ninety percent of cases means only ten percent of positives are false.",
55      "The sensitivity directly answers how likely disease is after testing positive.",
56      "I use the sensitivity as the probability of disease in the positive group.",
57      "A ninety percent detection rate gives every positive person ninety percent risk.",
58      "Since the sensitivity is high most positive people have the condition regardless of
   prevalence.",
59      "Specificity is the percentage of positive results that are correct.",
60      "The specificity tells how many of the positive group are really ill.",
61      "Ninety five percent specificity means ninety five percent of positives have the disease.",
62      "The test accuracy is simply the chance of disease if positive.",
63      "The sensitivity percentage can be read as the probability of a correct positive.",
64      "True positives over all positives is the definition of sensitivity.",
65      "Sensitivity and the proportion ill among positive tests are identical.",
66      "A detection rate of ninety percent means nine out of ten positive tests are actual cases.",
67      "You can use specificity to find the disease chance among people testing positive.",
68      "The probability after a positive test is equal to the detection rate."
69    ],
70    "base_rate": [
71      "How common the condition is does not matter to the result.",
72      "Prevalence is irrelevant if the test is accurate.",
73      "The number of people with the condition makes no difference.",
74      "A positive result means the same thing in every population.",
75      "We can ignore the base rate because the test performs well.",
76      "The chance after a positive is unchanged when prevalence changes."
```

```

77     "It does not matter whether the condition is rare or common.",
78     "All groups have the same predictive value if they use the same test.",
79     "We only need sensitivity and specificity, not prevalence.",
80     "The group size without disease is irrelevant to the positive results.",
81     "Changing the population cannot change the meaning of a positive result.",
82     "The frequency of the condition has nothing to do with interpreting tests.",
83     "Rare and common conditions produce the same probability after a positive.",
84     "Prevalence should be left out of the calculation.",
85     "The base rate has no effect on positive predictive value.",
86     "We do not need to know how widespread the condition is.",
87     "An accurate test makes the prevalence unimportant.",
88     "Testing a different group will not change the proportion of true positives.",
89     "How many people start with disease is not needed for a positive prediction.",
90     "I would ignore prevalence and focus only on the test performance."
91 ],
92 "dismissal": [
93     "False positives mean screening tests are completely useless.",
94     "If there are false positives we should never use screening.",
95     "Any false negative means the test has no value at all.",
96     "A test with mistakes is worthless and should be thrown away.",
97     "Screening is pointless because it cannot be perfect.",
98     "If most positives are false all screening is a waste of time.",
99     "False positives prove no one should ever be screened.",
100    "An imperfect test cannot help anybody.",
101    "We should reject every screening test that sometimes gives false results.",
102    "All screening tests are bad because false alarms happen.",
103    "A false positive makes the whole test useless.",
104    "There is no benefit to screening if any cases are missed.",
105    "The only worthwhile test is a perfect test with no errors.",
106    "A test that misses even one case should never be used.",
107    "These errors show screening has no purpose.",
108    "A lot of false alarms means screening is always harmful.",
109    "I would dismiss screening altogether because false positives exist.",
110    "Any mistake makes the test totally pointless.",
111    "Tests with false positives cannot ever be useful.",
112    "An imperfect screen is a waste and has zero benefit."
113 ],
114 },
115 "holdout": {
116     "balanced": [
117         "Positive results can include people without the condition, so I compare true positives with all
positives.",
118         "If a condition is rare, the large healthy group can create lots of false positives.",
119         "A positive screening result is not a diagnosis because the test sometimes flags non cases.",
120         "A negative result is not certain because false negatives can occur.",
121         "How common the condition is affects the proportion of real cases among positive results.",
122         "Sensitivity measures detected cases rather than the probability of disease among positives."
123     ],
124     "certainty": [
125         "If someone screens positive they must have the condition for sure.",
126         "A negative result is an absolute guarantee of no disease.",
127         "The screen confirms the disease with complete certainty.",
128         "Positive means definitely ill and negative means definitely well.",
129         "Every positive result is conclusive proof of disease.",
130         "It is impossible for somebody with a negative screen to be ill."
131     ],
132     "sensitivity": [
133         "The test detects ninety percent of cases, therefore ninety percent of positive people are
sick.",
134         "I think sensitivity is the chance a positive test is a real case.",
135         "The ninety percent sensitivity is also the positive predictive value.",
136         "Specificity is how likely somebody is to have the condition given a positive.",
137         "Sensitivity means the proportion with disease in the positive group.",
138         "Ninety percent case detection tells us nine out of ten positives have disease."
139     ],
140     "base_rate": [
141         "The disease frequency is irrelevant to interpreting a positive test.",
142         "I can leave prevalence out because sensitivity and specificity are enough.",
143         "It makes no difference if the disease is uncommon or widespread.",
144         "The same test means the same probability in any group.",
145         "The base rate does not affect the fraction of true positives.",
146         "There is no reason to consider how common the condition is."
147     ],
148     "dismissal": [
149         "Because false positives occur, screening should never be used at all.",
150         "A screening test that misses a case is entirely worthless.",
151         "False alarms mean the test has zero value.",
152         "An imperfect test is always a waste of effort.",
153         "Any false result proves screening is pointless.",
154         "If tests are not perfect there cannot be any screening benefit."
155     ]
156 }
157 }

```

model/evaluation.json

```
1 {
2   "metrics": {
3     "evaluatedCount": 30,
4     "correctCount": 29,
5     "rawAccuracy": 0.9666666666666667,
6     "acceptedCount": 28,
7     "acceptedCorrect": 27,
8     "coverage": 0.9333333333333333,
9     "confusionMatrix": {
10      "balanced": {
11        "balanced": 5,
12        "certainty": 0,
13        "sensitivity": 1,
14        "base_rate": 0,
15        "dismissal": 0
16      },
17      "certainty": {
18        "balanced": 0,
19        "certainty": 6,
20        "sensitivity": 0,
21        "base_rate": 0,
22        "dismissal": 0
23      },
24      "sensitivity": {
25        "balanced": 0,
26        "certainty": 0,
27        "sensitivity": 6,
28        "base_rate": 0,
29        "dismissal": 0
30      },
31      "base_rate": {
32        "balanced": 0,
33        "certainty": 0,
34        "sensitivity": 0,
35        "base_rate": 6,
36        "dismissal": 0
37      },
38      "dismissal": {
39        "balanced": 0,
40        "certainty": 0,
41        "sensitivity": 0,
42        "base_rate": 0,
43        "dismissal": 6
44      }
45    },
46    "limits": "Small, author-created English holdout. Training and holdout share a fictional task; no
external, multilingual, clinical, or educational-outcome validation."
47  },
48  "predictions": [
49    {
50      "text": "Positive results can include people without the condition, so I compare true positives
with all positives.",
51      "actual": "balanced",
52      "predicted": "balanced",
53      "vote": 0.9999,
54      "abstained": false
55    },
56    {
57      "text": "If a condition is rare, the large healthy group can create lots of false positives.",
58      "actual": "balanced",
59      "predicted": "balanced",
60      "vote": 0.9628,
61      "abstained": false
62    },
63    {
64      "text": "A positive screening result is not a diagnosis because the test sometimes flags non
cases.",
65      "actual": "balanced",
66      "predicted": "balanced",
67      "vote": 0.9848,
68      "abstained": false
69    },
70    {
71      "text": "A negative result is not certain because false negatives can occur.",
72      "actual": "balanced",
73      "predicted": "balanced",
74      "vote": 0.9855,
75      "abstained": false
76    },
77  ]
}
```

```

78     "text": "How common the condition is affects the proportion of real cases among positive
results.",
79     "actual": "balanced",
80     "predicted": "balanced",
81     "vote": 0.955,
82     "abstained": false
83 },
84 {
85     "text": "Sensitivity measures detected cases rather than the probability of disease among
positives.",
86     "actual": "balanced",
87     "predicted": "sensitivity",
88     "vote": 0.9808,
89     "abstained": false
90 },
91 {
92     "text": "If someone screens positive they must have the condition for sure.",
93     "actual": "certainty",
94     "predicted": "certainty",
95     "vote": 0.9712,
96     "abstained": false
97 },
98 {
99     "text": "A negative result is an absolute guarantee of no disease.",
100    "actual": "certainty",
101    "predicted": "certainty",
102    "vote": 0.9967,
103    "abstained": false
104 },
105 {
106    "text": "The screen confirms the disease with complete certainty.",
107    "actual": "certainty",
108    "predicted": "certainty",
109    "vote": 0.8101,
110    "abstained": false
111 },
112 {
113    "text": "Positive means definitely ill and negative means definitely well.",
114    "actual": "certainty",
115    "predicted": "certainty",
116    "vote": 0.9847,
117    "abstained": false
118 },
119 {
120    "text": "Every positive result is conclusive proof of disease.",
121    "actual": "certainty",
122    "predicted": "certainty",
123    "vote": 0.996,
124    "abstained": false
125 },
126 {
127    "text": "It is impossible for somebody with a negative screen to be ill.",
128    "actual": "certainty",
129    "predicted": "certainty",
130    "vote": 0.9991,
131    "abstained": false
132 },
133 {
134    "text": "The test detects ninety percent of cases, therefore ninety percent of positive people
are sick.",
135    "actual": "sensitivity",
136    "predicted": "sensitivity",
137    "vote": 1.0,
138    "abstained": false
139 },
140 {
141    "text": "I think sensitivity is the chance a positive test is a real case.",
142    "actual": "sensitivity",
143    "predicted": "sensitivity",
144    "vote": 0.9525,
145    "abstained": false
146 },
147 {
148    "text": "The ninety percent sensitivity is also the positive predictive value.",
149    "actual": "sensitivity",
150    "predicted": "sensitivity",
151    "vote": 1.0,
152    "abstained": false
153 },
154 {
155    "text": "Specificity is how likely somebody is to have the condition given a positive.",
156    "actual": "sensitivity",
157    "predicted": "sensitivity",
158    "vote": 0.6772,

```

```

159     "abstained": false
160 },
161 {
162     "text": "Sensitivity means the proportion with disease in the positive group.",
163     "actual": "sensitivity",
164     "predicted": "sensitivity",
165     "vote": 0.9981,
166     "abstained": false
167 },
168 {
169     "text": "Ninety percent case detection tells us nine out of ten positives have disease.",
170     "actual": "sensitivity",
171     "predicted": "sensitivity",
172     "vote": 1.0,
173     "abstained": false
174 },
175 {
176     "text": "The disease frequency is irrelevant to interpreting a positive test.",
177     "actual": "base_rate",
178     "predicted": "base_rate",
179     "vote": 0.5632,
180     "abstained": true
181 },
182 {
183     "text": "I can leave prevalence out because sensitivity and specificity are enough.",
184     "actual": "base_rate",
185     "predicted": "base_rate",
186     "vote": 0.6504,
187     "abstained": false
188 },
189 {
190     "text": "It makes no difference if the disease is uncommon or widespread.",
191     "actual": "base_rate",
192     "predicted": "base_rate",
193     "vote": 0.869,
194     "abstained": false
195 },
196 {
197     "text": "The same test means the same probability in any group.",
198     "actual": "base_rate",
199     "predicted": "base_rate",
200     "vote": 0.9962,
201     "abstained": false
202 },
203 {
204     "text": "The base rate does not affect the fraction of true positives.",
205     "actual": "base_rate",
206     "predicted": "base_rate",
207     "vote": 0.9767,
208     "abstained": false
209 },
210 {
211     "text": "There is no reason to consider how common the condition is.",
212     "actual": "base_rate",
213     "predicted": "base_rate",
214     "vote": 0.6138,
215     "abstained": true
216 },
217 {
218     "text": "Because false positives occur, screening should never be used at all.",
219     "actual": "dismissal",
220     "predicted": "dismissal",
221     "vote": 1.0,
222     "abstained": false
223 },
224 {
225     "text": "A screening test that misses a case is entirely worthless.",
226     "actual": "dismissal",
227     "predicted": "dismissal",
228     "vote": 0.9985,
229     "abstained": false
230 },
231 {
232     "text": "False alarms mean the test has zero value.",
233     "actual": "dismissal",
234     "predicted": "dismissal",
235     "vote": 0.9997,
236     "abstained": false
237 },
238 {
239     "text": "An imperfect test is always a waste of effort.",
240     "actual": "dismissal",
241     "predicted": "dismissal",
242     "vote": 0.9993,

```

```

243     "abstained": false
244 },
245 {
246     "text": "Any false result proves screening is pointless.",
247     "actual": "dismissal",
248     "predicted": "dismissal",
249     "vote": 0.9869,
250     "abstained": false
251 },
252 {
253     "text": "If tests are not perfect there cannot be any screening benefit.",
254     "actual": "dismissal",
255     "predicted": "dismissal",
256     "vote": 0.991,
257     "abstained": false
258 }
259 ]
260 }

```

tests/verify.cjs

```
1  const assert=require('node:assert/strict');
2  const fs=require('node:fs');
3  const vm=require('node:vm');
4  const path=require('node:path');
5  const root=path.resolve(__dirname, '..');
6  const science=require('../dist/science.js');
7  const coach=require('../dist/coach.js');
8  const context={window:{}};
9  vm.runInNewContext(fs.readFileSync(path.join(root, 'dist/model.js'), 'utf8'), context);
10 const model=context.window.BaseRateModel;
11 function close(a,b){assert.ok(Math.abs(a-b)<1e-9, `${a} != ${b}`);}
12 const rare=science.calculate(1,90,95);
13 close(rare.tp,9);close(rare.fp,49.5);close(rare.fn,1);close(rare.tn,940.5);close(rare.ppv,100*9/58.5);
14 close(science.calculate(20,90,95).ppv,100*180/220);
15 close(science.calculate(1,90,99).ppv,100*9/18.9);
16 assert.equal(science.calculate(0,90,100).ppv,null);
17 assert.equal(science.calculate(100,100,0).npv,null);
18 assert.equal(science.calculate(1,100,100).ppv,100);
19 assert.throws(()=>science.calculate(NaN,90,95),RangeError);
20 assert.throws(()=>science.calculate(101,90,95),RangeError);
21 assert.throws(()=>science.calculate(1,90,95,0),RangeError);
22 let combinations=0;
23 for(const p of [0,.1,1,7.3,20,50,99.9,100])for(const se of [0,1,10,63,90,99.9,100])for(const sp of
[0,1,50,95,99.9,100]){
24   const r=science.calculate(p,se,sp),rounded=science.roundedCounts(r);
25   close(r.tp+r.fp+r.fn+r.tn,1000);
26   assert.equal(Object.values(rounded).reduce((a,b)=>a+b,0),1000);
27   for(const key of
['tp','fn','fp','tn']){assert.ok(r[key]>=0);assert.ok(Math.abs(rounded[key]-r[key])<=1.0000001);}
28   for(const value of [r.ppv,r.npv])assert.ok(value===null||(value>=0&&value<=100));
29   close(r.ppv??0,science.calculate(p,se,sp,10000).ppv??0);
30   combinations++;
31 }
32 assert.equal(coach.classify('',model).label,'empty');
33 assert.equal(coach.classify('My test result came back positive, do I have
cancer?',model).label,'personal');
34 assert.equal(coach.classify('\u4f60\u597d\u4e16\u754c',model).label,'uncertain');
35 assert.equal(coach.classify('banana bicycle orchestra',model).label,'uncertain');
36 assert.throws(()=>coach.classify('x'.repeat(1801),model),RangeError);
37 assert.equal(coach.classify('A positive result does not prove the condition because false positives
happen.',model).label,'balanced');
38 assert.equal(coach.classify('A positive result proves you have the
condition.',model).label,'certainty');
39 const expected=JSON.parse(fs.readFileSync(path.join(root, 'model/evaluation.json'), 'utf8'));
40 for(const row of expected.predictions){const actual=coach.classify(row.text,model);assert.equal(actual
.rawLabel,row.predicted);assert.equal(actual.abstained,row.abstained);}
41 const dataset=JSON.parse(fs.readFileSync(path.join(root, 'model/dataset.json'), 'utf8'));
42 const train=new Set(Object.values(dataset.train).flat());
43 for(const text of Object.values(dataset.holdout).flat())assert.ok(!train.has(text),'Holdout sentence
occurs in training');
44 const sourceFiles=['index.html','style.css','science.js','model.js','coach.js','app.js'];
45 for(const file of sourceFiles)assert.ok(fs.statSync(path.join(root, 'dist', file)).size>0);
46 console.log(JSON.stringify({result:'PASS',scienceCombinations:combinations,modelParityCases:expected.p
redictions.length,checks:['Bayes examples','edge denominators','count
conservation','rounding','invalid inputs','scope guard','out of vocabulary abstention','negation
example','train/holdout disjointness','browser/Python model parity']}));
```

tests/extensions.cjs

```
1  const assert=require('node:assert/strict');
2  const science=require('../dist/science.js');
3  const rare={prevalence:1,sensitivity:90,specificity:95};
4  const common={...rare,prevalence:20};
5  const close=(a,b)=>assert.ok(Math.abs(a-b)<1e-8,`${a} != ${b}`);
6  const pair=science.compare(rare,common);
7  close(pair.a.ppv,9/58.5*100);
8  close(pair.b.ppv,180/220*100);
9  close(pair.delta.ppv,180/220*100-9/58.5*100);
10 close(pair.delta.fp,-9.5);close(pair.delta.tp,171);
11 assert.deepEqual(rare,{prevalence:1,sensitivity:90,specificity:95});
12 assert.equal(science.compare({prevalence:0,sensitivity:90,specificity:100},rare).delta.ppv,null);
13 let curvePoints=0;
14 for(const se of [0,1,50,90,100])for(const sp of [0,1,50,95,100]){
15   const curve=science.prevalenceCurve(se,sp);
16   assert.equal(curve.length,101);assert.equal(curve[0].prevalence,0);assert.equal(curve[100].prevalenc
e,50);
17   let previous=0;
18   for(const point of curve){
19     close(point.prevalence*2,Math.round(point.prevalence*2));
20     assert.equal(point.ppv,science.calculate(point.prevalence,se,sp).ppv);
21     if(point.ppv!==null){assert.ok(point.ppv+1e-8>=previous);previous=point.ppv;}
22     curvePoints++;
23   }
24 }
25 const independent=science.sequential(rare,90,5);
26 close(independent.tp,8.1);close(independent.fp,2.475);
27 close(independent.ppv,8.1/(8.1+2.475)*100);
28 close(independent.rejectedTrue,.9);close(independent.rejectedFalse,47.025);
29 const repeated=science.sequential(rare,100,100);
30 close(repeated.ppv,science.calculate(1,90,95).ppv);
31 close(repeated.tp,9);close(repeated.fp,49.5);
32 assert.equal(science.sequential(rare,0,0).ppv,null);
33 assert.equal(science.sequential(rare,100,0).ppv,100);
34 let sequenceCases=0;
35 for(const p of [0,.1,1,20,100])for(const se of [0,90,100])for(const sp of [0,95,100])for(const d of
[0,90,100])for(const f of [0,5,100]){
36   const r=science.sequential({prevalence:p,sensitivity:se,specificity:sp},d,f);
37   close(r.tp+r.rejectedTrue,r.first.tp);close(r.fp+r.rejectedFalse,r.first.fp);
38   assert.ok(r.tp>=0&&r.fp>=0&&r.rejectedTrue>=-1e-10&&r.rejectedFalse>=-1e-10);
39   assert.ok(r.ppv===null||(r.ppv>=0&&r.ppv<=100));sequenceCases++;
40 }
41 for(const bad of [NaN,Infinity,-1,101]){
42   assert.throws(()=>science.prevalenceCurve(bad,90),RangeError);
43   assert.throws(()=>science.sequential(rare,bad,5),RangeError);
44   assert.throws(()=>science.sequential(rare,90,bad),RangeError);
45   assert.throws(()=>science.compare({...rare,prevalence:bad},common),RangeError);
46 }
47 console.log(JSON.stringify({result:'PASS',curvePoints,sequenceCases,checks:['comparison
units','immutable inputs','monotonic prevalence curve','conditional repeated tests','undefined
proportions','conservation','invalid rates']}));
```

tests/learning.cjs

```
1  const assert=require('node:assert/strict');
2  const learning=require('../dist/learning.js');
3  const close=(a,b)=>assert.ok(Math.abs(a-b)<1e-9);
4  assert.equal(learning.cases.length,3);
5  assert.ok(Object.isFrozen(learning.cases[0].settings));
6  close(learning.gradePrediction('rare',90).actual,9/58.5*100);
7  close(learning.gradePrediction('rare',90).error,90-9/58.5*100);
8  close(learning.gradePrediction('common',80).actual,180/220*100);
9  close(learning.gradePrediction('specific',50).actual,9/18.9*100);
10 assert.equal(learning.gradePrediction('rare',15.4).close,true);
11 assert.equal(learning.gradePrediction('rare',90).close,false);
12 for(const guess of [0,100])assert.ok(Number.isFinite(learning.gradePrediction('rare',guess).error));
13 for(const guess of ['',null,undefined,NaN,Infinity,-.1,100.1,'15'])assert.throws(()=>learning.gradePrediction('rare',guess),RangeError);
14 assert.throws(()=>learning.gradePrediction('missing',50),RangeError);
15 const answers=[{caseId:'rare',prediction:90,denominator:'cases',explanation:'private learner words',arbitrary:'secret'}];
16 const notebook=learning.notebook(answers);
17 assert.equal(notebook.attempts.length,1);assert.equal(notebook.attempts[0].prediction,90);
18 assert.equal(notebook.attempts[0].denominator,'cases');
19 assert.equal(JSON.stringify(notebook).includes('private learner words'),false);
20 assert.equal(JSON.stringify(notebook).includes('secret'),false);
21 assert.equal(notebook.attempts[0].settings.prevalence,1);
22 assert.equal(learning.notebook([]).attempts.length,0);
23 assert.deepEqual(answers,[{caseId:'rare',prediction:90,denominator:'cases',explanation:'private learner words',arbitrary:'secret'}]);
24 console.log(JSON.stringify({result:'PASS',checks:['three case calculations','first guess error','immutable content','invalid guesses','text-free notebook','empty notebook']}));
```

tests/display.cjs

```
1  const assert=require('node:assert/strict');
2  const science=require('../dist/science.js');
3  const coach=require('../dist/coach.js');
4  const result=science.sequential({prevalence:.1,sensitivity:1,specificity:100},1,0);
5  assert.equal(result.ppv,100);
6  assert.equal(science.formatNumber(result.tp),'0.0001');
7  assert.equal(science.formatNumber(result.positive),'0.0001');
8  assert.equal(science.formatNumber(result.first.tp),'0.01');
9  assert.equal(science.formatNumber(49.5),'49.5');
10 assert.equal(science.formatNumber(940.5),'940.5');
11 assert.equal(science.formatNumber(0),'0');
12 for(const value of [.000000000001,.0001,.001,.01,.1,-.0001])assert.notEqual(Number(science.formatNumber
(value).replace(/,/g,'')),0);
13 const unavailable=coach.classify('False positives can outnumber true positives.',null);
14 assert.equal(unavailable.label,'unavailable');
15 assert.match(coach.feedback(unavailable,result.first).body,/model.*unavailable/i);
16 assert.match(coach.feedback(unavailable,result.first).counts,/0\.01 true positives/);
17 assert.match(coach.feedback(unavailable,result).counts,/0\.0001 true positives/);
18 console.log(JSON.stringify({result:'PASS',checks:['nonzero count precision','defined repeat
denominator','very small display values','missing model message']}));
```

tests/coaching.cjs

```
1 // Breaks caught: unsupported model praise; wrong conditional question arithmetic;
2 // invalid choices accepted as correct; lost first denominator after correction.
3 const assert=require('node:assert/strict'),fs=require('node:fs'),vm=require('node:vm');
4 const science=require('../dist/science.js'),coach=require('../dist/coach.js'),learning=require('../dist/learning.js');
5 const context={window:{}};vm.runInNewContext(fs.readFileSync(require('node:path').join(__dirname,'../dist/model.js'),'utf8'),context);
6 const settings={prevalence:1,sensitivity:90,specificity:95};
7 const bad='All positives are actual cases, without exception.';
8 const inferred=coach.classify(bad,context.window.BaseRateModel),feedback=coach.feedback(inferred,science.calculate(1,90,95));
9 assert.doesNotMatch(feedback.title,/you are connecting/i,'An incorrect explanation must not receive a correctness endorsement');
10 assert.equal(feedback.provisional,true,'A model-selected topic must remain a suggestion, even for a high vote');
11 for(const [topic,choice,correct] of [
12   ['positives','positives',true],['positives','cases',false],['detection','cases',true],['detection','positives',false],
13   ['prevalence','higher',true],['prevalence','same',false],['uncertainty','context',true],['uncertainty','diagnosis',false],
14   ['evidence','benefits',true],['evidence','errors',false]
15 ]) assert.equal(coach.checkAnswer(topic,choice,settings).correct,correct,topic+' '+choice);
16 assert.equal(coach.exercise('prevalence',settings).correctChoice,'higher');
17 assert.match(coach.exercise('detection',{prevalence:0,sensitivity:90,specificity:95}).body,/empty.*undefined/);
18 assert.match(coach.exercise('positives',{prevalence:0,sensitivity:0,specificity:100}).body,/empty.*undefined/);
19 assert.equal(coach.exercise('prevalence',{prevalence:1,sensitivity:0,specificity:100}).correctChoice,'undefined');
20 assert.equal(coach.exercise('prevalence',{prevalence:1,sensitivity:0,specificity:95}).correctChoice,'same');
21 assert.equal(coach.exercise('prevalence',{prevalence:1,sensitivity:90,specificity:100}).correctChoice,'same','Perfect clearance keeps PPV100% at both positive prevalences');
22 assert.match(coach.checkAnswer('positives','cases',settings).body,/58\.\.5/);
23 assert.match(coach.checkAnswer('detection','positives',settings).body,/10/);
24 assert.throws(()=>coach.checkAnswer('positives','invented',settings),RangeError);
25 assert.throws(()=>coach.exercise('invented',settings),RangeError);
26 assert.throws(()=>coach.exercise('positives',{...settings,prevalence:NaN}),RangeError);
27 assert.equal(learning.checkDenominator('rare','positives').correct,true);
28 assert.equal(learning.checkDenominator('rare','cases').correct,false);
29 assert.match(learning.checkDenominator('rare','cases').body,/90%/);
30 assert.match(learning.checkDenominator('rare','population').body,/0\.\.9%/);
31 assert.throws(()=>learning.checkDenominator('rare','invented'),RangeError);
32 assert.throws(()=>learning.checkDenominator('missing','positives'),RangeError);
33 const answers=[{caseId:'rare',prediction:90,denominator:'positives',firstDenominator:'cases',denominatorAttempts:2,explanation:'never export this'}];
34 const notebook=learning.notebook(answers);
35 assert.equal(notebook.version,2);assert.equal(notebook.attempts[0].firstDenominator,'cases');
36 assert.equal(notebook.attempts[0].denominator,'positives');assert.equal(notebook.attempts[0].denominatorAttempts,2);
37 assert.ok(!JSON.stringify(notebook).includes('never export this'));
38 console.log(JSON.stringify({result:'PASS',checks:['provisional coach feedback','five factual self-checks','perfect-clearance edge','specific denominator correction','first and corrected choices exported without text']}));
39 // Reproduce a review finding through the actual application event handlers.
40 // An unsubmitted example must not retain an answered check for old lab counts.
41 {
42   const elements=[];
43   class Element {
44     constructor(tag='div'){this.tag=tag;this.children=[];this.listeners={};this.style={};this.classList=[toggle(){}];this.attrs={};this._text='';this.value='';elements.push(this);}
45     set textContent(value){this._text=value??'';this.children=[];this.value=value;}
46     get textContent(){return this._text+this.children.map(e=>e.textContent).join('');}
47     append(...children){this.children.push(...children);for(const child of children)child.parent=this;}
48     replaceChildren(...children){this.children=[];this._text='';this.append(...children);}
49     setAttribute(key,value){this.attrs[key]=value;}
50     removeAttribute(key){delete this.attrs[key];}
51     addEventListener(type,handler){this.listeners[type]=handler;}
52     focus(){}
53     all(){return this.children.flatMap(e=>[...e.all()]);}
54     querySelector(selector){return this.all().find(e=>e.tag==='input'&&(selector!=='input:checked' || e.checked))||null;}
55     emit(type){this.listeners[type]?.({preventDefault(){},target:this});}
56   }
57   const byId={},document={
58     getElementById:id=>elements.findLast(e=>e.id===id)||byId[id]?=>Object.assign(new Element(),{id}),
59     createElement:tag=>new Element(tag),createDocumentFragment:()=>new Element('fragment'),
60     createTextNode:text=>Object.assign(new Element('text'),{textContent:text})
61   };
62   for(const [id,value,min,max] of [['prevalence',1,.1,50],['sensitivity',90,1,100],['specificity',95,1,
```

```

100]])Object.assign(document.getElementById(id), {value,min,max});
63  const appContext={document,URLSearchParams,CustomEvent:class{},location:{search:''},dispatchEvent(){}
,addEventListener(){}},BaseRateScience:science,BaseRateCoach:coach};
64  appContext.window=appContext;
65  vm.runInNewContext(fs.readFileSync(require('node:path').join(__dirname,'../dist/app.js'),'utf8'),appC
ontext);
66  document.getElementById('explanation').value='A fictional
explanation';document.getElementById('coach-form').emit('submit');
67  const select=document.getElementById('coach-topic');select.value='positives';select.emit('change');
68  const form=document.getElementById('coach-check-submit').parent;form.querySelector('input').checked=t
rue;form.emit('submit');
69  assert.match(document.getElementById('coach-response').textContent,/58\.5/);
70  document.getElementById('example').emit('click');
71  appContext.BaseRateApp.setExperiment({prevalence:20,sensitivity:90,specificity:95});
72  assert.match(document.getElementById('result-explanation').textContent,/180 of 220/);
73  assert.doesNotMatch(document.getElementById('coach-response').textContent,/58\.5|Correct for this
question/, 'Loading an unsubmitted example must clear stale answered checks');
74  console.log(JSON.stringify({result:'PASS',check:'example loading clears answered coach state'}));
75  }

```

model/audit.py

```
1  """Reproduce the model, inspect lexical split proximity, and compare feedback policies.
2
3  Python and Node standard libraries only. Probes are development cases used to
4  improve feedback, NOT an independent benchmark or evidence of learning benefit.
5  """
6  import argparse
7  import difflib
8  import hashlib
9  import json
10 from pathlib import Path
11 import re
12 import shutil
13 import subprocess
14 import sys
15 import tempfile
16
17 ROOT = Path(__file__).resolve().parents[1]
18
19
20 def audit():
21     with tempfile.TemporaryDirectory(prefix='baserate-audit-') as directory:
22         temporary = Path(directory)
23         (temporary / 'model').mkdir()
24         (temporary / 'dist').mkdir()
25         for name in ['train.py', 'dataset.json']:
26             shutil.copy2(ROOT / 'model' / name, temporary / 'model' / name)
27             subprocess.run([sys.executable, str(temporary / 'model/train.py')],
28                           check=True, capture_output=True, text=True)
29             reproduction = {name: (ROOT / name).read_bytes() == (temporary / name).read_bytes()
30                             for name in ['dist/model.js', 'model/evaluation.json']}
31         data = json.loads((ROOT / 'model/dataset.json').read_text())
32         training = [(label, text) for label, texts in data['train'].items() for text in texts]
33         holdout = [(label, text) for label, texts in data['holdout'].items() for text in texts]
34
35         def tokens(text):
36             return set(re.findall(r'[a-z]+', text.lower()))
37
38         def similarity(left, right):
39             a, b = tokens(left), tokens(right)
40             return len(a & b) / max(1, len(a | b))
41
42         neighbors = []
43         for label, text in holdout:
44             nearest_label, nearest = max(training, key=lambda pair: similarity(text, pair[1]))
45             neighbors.append({'holdout': text, 'label': label, 'nearestTrain': nearest,
46                             'nearestLabel': nearest_label,
47                             'tokenJaccard': round(similarity(text, nearest), 3),
48                             'sequenceRatio': round(difflib.SequenceMatcher(None, text.lower(),
49                                     nearest.lower()).ratio(), 3)})
49         javascript = r'''
50 const fs=require('node:fs'),path=require('node:path'),vm=require('node:vm');
51 const root=process.argv[1],science=require(path.join(root,'dist/science.js'));
52 const before=require(path.join(root,'model/feedback-v1.cjs')),after=require(path.join(root,'dist/coach
53 .js'));
54 const context={window:{}};vm.runInNewContext(fs.readFileSync(path.join(root,'dist/model.js'),'utf8'),c
55 ontext);
56 const model=context.window.BaseRateModel,counts=science.calculate(1,90,95);
57 const rows=JSON.parse(fs.readFileSync(path.join(root,'model/probes.json'),'utf8')).probes.map(probe=>{
58 const original=before.classify(probe.text,model),current=after.classify(probe.text,model);
59 const oldFeedback=before.feedback(original,counts),feedback=after.feedback(current,counts);
60 return
61 {...probe,rawLabel:current.rawLabel,label:current.label,vote:current.vote,coverage:current.coverage,
62 abstained:current.abstained,classificationUnchanged:JSON.stringify(original)==JSON.stringify(curren
63 t),
64 originalTitle:oldFeedback.title,currentTitle:feedback.title,provisional:feedback.provisional,
65 originalPraise:oldFeedback.title==='You are connecting the
66 pieces.',currentPraise:feedback.title==='You are connecting the pieces.'});
67 });
68 process.stdout.write(JSON.stringify(rows));
69 '''
70     rows = json.loads(subprocess.run(['node', '-e', javascript, str(ROOT)],
71                                     check=True, capture_output=True, text=True).stdout)
72     flawed = [row for row in rows if row['group'] in ['mixed', 'misconception']]
73     correct = [row for row in rows if row['group'] == 'correct']
74     metrics = json.loads((ROOT / 'model/evaluation.json').read_text())['metrics']
75     return {
76         'purpose': 'Reproducible internal audit. Same-author synthetic splits and development probes;
77 no independent learner, clinical or learning-outcome evidence.',
78         'modelReproduction': reproduction,
79         'sha256': {name: hashlib.sha256((ROOT / name).read_bytes()).hexdigest()

```

```

74         for name in ['model/train.py', 'model/dataset.json', 'dist/model.js',
75 'model/evaluation.json', 'model/probes.json']],
76         'reportedMetricsReproduced': metrics,
77         'split': {'trainingCount': len(training), 'holdoutCount': len(holdout),
78 'exactSentenceOverlap': len(set(t for _, t in training) & set(t for _, t in
79 holdout)),
80 'meanNearestTokenJaccard': round(sum(r['tokenJaccard'] for r in neighbors) /
81 len(neighbors), 6),
82 'nearestTokenJaccardAtLeastHalf': sum(r['tokenJaccard'] >= .5 for r in neighbors),
83 'sameLabelNearest': sum(r['label'] == r['nearestLabel'] for r in neighbors),
84 'limitations': 'Token/sequence similarity cannot establish semantic independence.
Both splits reuse familiar claim types and authoring style.',
85 'nearestNeighbors': neighbors},
86 'developmentProbes': {'count': len(rows), 'classificationUnchanged':
all(r['classificationUnchanged'] for r in rows),
87 'flawedOrMixedCount': len(flawed),
88 'originalPraisedFlawedOrMixed': sum(r['originalPraise'] for r in
flawed),
89 'currentPraisedFlawedOrMixed': sum(r['currentPraise'] for r in flawed),
90 'correctProbesAcceptedAsMisconceptionPattern': sum(r['label'] in
['sensitivity', 'certainty', 'base_rate', 'dismissal'] for r in correct),
91 'correctProbeCount': len(correct),
92 'allCurrentResponsesProvisional': all(r['provisional'] for r in rows),
93 'limitations': 'These author-labeled probes informed this feedback
revision. Fewer praising titles measures response wording, not classification accuracy, misconception
detection or learning gain.',
94 'rows': rows},
95 'feedbackPolicy': '1.1: provisional, overridable review topics; separate factual self-checks.
Model/data/holdout unchanged.'
96 }
97
98 if __name__ == '__main__':
99     parser = argparse.ArgumentParser(description=__doc__)
100     parser.add_argument('--write', action='store_true', help='write the deterministic audit artifact')
101     args = parser.parse_args()
102     result = audit()
103     serialized = json.dumps(result, indent=2, ensure_ascii=False) + '\n'
104     artifact = ROOT / 'model/audit.json'
105     if args.write:
106         artifact.write_text(serialized)
107     elif not artifact.exists() or artifact.read_text() != serialized:
108         raise SystemExit('Audit artifact is absent or stale. Inspect changes, then run with --write.')
109     if not all(result['modelReproduction'].values()):
110         raise SystemExit('Distributed model/evaluation did not reproduce byte for byte.')
111     print(json.dumps({'result': 'PASS', 'reproduction': result['modelReproduction'],
'split': {k: v for k, v in result['split'].items() if k != 'nearestNeighbors'},
'probes': {k: v for k, v in result['developmentProbes'].items() if k !=
'rows'}}, indent=2))

```

model/probes.json

```
1 {
2   "purpose": "Author-created development stress probes, used to choose feedback changes; not a
   pristine holdout, independent benchmark or real learner data. Groups are author judgements, not
   clinical labels.",
3   "probes": [
4     {
5       "group": "correct",
6       "text": "Sensitivity measures detected cases rather than the probability of disease among
   positives."
7     },
8     {
9       "group": "correct",
10      "text": "Sensitivity starts with people who have the condition, whereas predictive value starts
   with positive results."
11    },
12    {
13      "group": "correct",
14      "text": "A positive result is not conclusive proof of disease."
15    },
16    {
17      "group": "correct",
18      "text": "Prevalence is not irrelevant: it determines how many non-cases can become false
   positives."
19    },
20    {
21      "group": "correct",
22      "text": "False alarms do not make every screening program useless; benefits and harms need
   evidence."
23    },
24    {
25      "group": "correct",
26      "text": "Nine true cases among 58.5 positives is about fifteen percent, not ninety percent."
27    },
28    {
29      "group": "correct",
30      "text": "Specificity counts negative results among people without the condition, not true cases
   among positives."
31    },
32    {
33      "group": "correct",
34      "text": "It is wrong to say that every positive result is a guaranteed diagnosis."
35    },
36    {
37      "group": "mixed",
38      "text": "Prevalence matters, but ninety percent sensitivity still means ninety percent of
   positives are true."
39    },
40    {
41      "group": "mixed",
42      "text": "False positives can outnumber true positives when the condition is rare, so the test is
   completely useless."
43    },
44    {
45      "group": "mixed",
46      "text": "The result depends on the starting population, although a positive proves the disease
   for sure."
47    },
48    {
49      "group": "mixed",
50      "text": "False negatives can occur, but a negative result guarantees no condition."
51    },
52    {
53      "group": "mixed",
54      "text": "I know sensitivity and predictive value have different denominators, yet both are
   always the same percentage."
55    },
56    {
57      "group": "mixed",
58      "text": "We count all positive results in the denominator; in this case 9 divided by 58.5 is
   ninety percent."
59    },
60    {
61      "group": "ambiguous",
62      "text": "Sensitivity is ninety percent."
63    },
64    {
65      "group": "ambiguous",
66      "text": "The test is pretty accurate."
67    },
68    {
```

```

69     "group": "ambiguous",
70     "text": "There are false positives."
71 },
72 {
73     "group": "ambiguous",
74     "text": "What does prevalence mean?"
75 },
76 {
77     "group": "ambiguous",
78     "text": "My friend says a positive result proves disease. Is that really correct?"
79 },
80 {
81     "group": "ambiguous",
82     "text": "I am unsure whether the denominator is cases or positives."
83 },
84 {
85     "group": "misconception",
86     "text": "All positives are actual cases, without exception."
87 },
88 {
89     "group": "misconception",
90     "text": "Prevalence can be ignored for any test."
91 },
92 {
93     "group": "misconception",
94     "text": "The true-positive rate is exactly the probability that a positive is a case."
95 },
96 {
97     "group": "misconception",
98     "text": "One false alarm makes every screening test pointless."
99 }
100 ]
101 }

```

model/audit.json

```
1 {
2   "purpose": "Reproducible internal audit. Same-author synthetic splits and development probes; no
   independent learner, clinical or learning-outcome evidence.",
3   "modelReproduction": {
4     "dist/model.js": true,
5     "model/evaluation.json": true
6   },
7   "sha256": {
8     "model/train.py": "8e9e3f7435066abde2120fae6f179aceaf38f592baea9742a1b816205b44f3f7",
9     "model/dataset.json": "daa9465efc6e649ff68519e20d4a8ded91d1b4b23e36b017c3586d1cf74086e0",
10    "dist/model.js": "d1d2cd1f4568f8664d0cc2e64bd4132bc358baca6be5b62c88d370bb8eeeb69e",
11    "model/evaluation.json": "84e17dcb3f572579e6d3aa499d4437a13044382066c959b04454e1c4f359842b",
12    "model/probes.json": "22f9b6e69ae95ec2bf8033e014c32b2c0dcd66a0f4e49fbf082dbfd96a3572fd"
13  },
14  "reportedMetricsReproduced": {
15    "evaluatedCount": 30,
16    "correctCount": 29,
17    "rawAccuracy": 0.9666666666666667,
18    "acceptedCount": 28,
19    "acceptedCorrect": 27,
20    "coverage": 0.9333333333333333,
21    "confusionMatrix": {
22      "balanced": {
23        "balanced": 5,
24        "certainty": 0,
25        "sensitivity": 1,
26        "base_rate": 0,
27        "dismissal": 0
28      },
29      "certainty": {
30        "balanced": 0,
31        "certainty": 6,
32        "sensitivity": 0,
33        "base_rate": 0,
34        "dismissal": 0
35      },
36      "sensitivity": {
37        "balanced": 0,
38        "certainty": 0,
39        "sensitivity": 6,
40        "base_rate": 0,
41        "dismissal": 0
42      },
43      "base_rate": {
44        "balanced": 0,
45        "certainty": 0,
46        "sensitivity": 0,
47        "base_rate": 6,
48        "dismissal": 0
49      },
50      "dismissal": {
51        "balanced": 0,
52        "certainty": 0,
53        "sensitivity": 0,
54        "base_rate": 0,
55        "dismissal": 6
56      }
57    },
58    "limits": "Small, author-created English holdout. Training and holdout share a fictional task; no
   external, multilingual, clinical, or educational-outcome validation."
59  },
60  "split": {
61    "trainingCount": 100,
62    "holdoutCount": 30,
63    "exactSentenceOverlap": 0,
64    "meanNearestTokenJaccard": 0.332533,
65    "nearestTokenJaccardAtLeastHalf": 0,
66    "sameLabelNearest": 23,
67    "limitations": "Token/sequence similarity cannot establish semantic independence. Both splits
   reuse familiar claim types and authoring style.",
68    "nearestNeighbors": [
69      {
70        "holdout": "Positive results can include people without the condition, so I compare true
   positives with all positives.",
71        "label": "balanced",
72        "nearestTrain": "The proportion with the condition among positives is true positives divided
   by all positives.",
73        "nearestLabel": "balanced",
74        "tokenJaccard": 0.3,
75        "sequenceRatio": 0.613

```

```

76     },
77     {
78         "holdout": "If a condition is rare, the large healthy group can create lots of false
positives.",
79         "label": "balanced",
80         "nearestTrain": "If most positives are false all screening is a waste of time.",
81         "nearestLabel": "dismissal",
82         "tokenJaccard": 0.286,
83         "sequenceRatio": 0.222
84     },
85     {
86         "holdout": "A positive screening result is not a diagnosis because the test sometimes flags
non cases.",
87         "label": "balanced",
88         "nearestTrain": "A positive result does not prove the condition because false positives
happen.",
89         "nearestLabel": "balanced",
90         "tokenJaccard": 0.3,
91         "sequenceRatio": 0.571
92     },
93     {
94         "holdout": "A negative result is not certain because false negatives can occur.",
95         "label": "balanced",
96         "nearestTrain": "A negative result does not completely rule out the condition because cases
can be missed.",
97         "nearestLabel": "balanced",
98         "tokenJaccard": 0.3,
99         "sequenceRatio": 0.628
100    },
101    {
102        "holdout": "How common the condition is affects the proportion of real cases among positive
results.",
103        "label": "balanced",
104        "nearestTrain": "Changing how common the condition is changes the meaning of a positive
result.",
105        "nearestLabel": "balanced",
106        "tokenJaccard": 0.389,
107        "sequenceRatio": 0.723
108    },
109    {
110        "holdout": "Sensitivity measures detected cases rather than the probability of disease among
positives.",
111        "label": "balanced",
112        "nearestTrain": "I use the sensitivity as the probability of disease in the positive group.",
113        "nearestLabel": "sensitivity",
114        "tokenJaccard": 0.278,
115        "sequenceRatio": 0.642
116    },
117    {
118        "holdout": "If someone screens positive they must have the condition for sure.",
119        "label": "certainty",
120        "nearestTrain": "You always have the condition if the screening says positive.",
121        "nearestLabel": "certainty",
122        "tokenJaccard": 0.333,
123        "sequenceRatio": 0.441
124    },
125    {
126        "holdout": "A negative result is an absolute guarantee of no disease.",
127        "label": "certainty",
128        "nearestTrain": "A negative test can miss cases, so it is not an absolute guarantee.",
129        "nearestLabel": "balanced",
130        "tokenJaccard": 0.353,
131        "sequenceRatio": 0.645
132    },
133    {
134        "holdout": "The screen confirms the disease with complete certainty.",
135        "label": "certainty",
136        "nearestTrain": "Everyone with a positive test definitely has the disease.",
137        "nearestLabel": "certainty",
138        "tokenJaccard": 0.231,
139        "sequenceRatio": 0.389
140    },
141    {
142        "holdout": "Positive means definitely ill and negative means definitely well.",
143        "label": "certainty",
144        "nearestTrain": "Sensitivity and the proportion ill among positive tests are identical.",
145        "nearestLabel": "sensitivity",
146        "tokenJaccard": 0.214,
147        "sequenceRatio": 0.296
148    },
149    {
150        "holdout": "Every positive result is conclusive proof of disease.",
151        "label": "certainty",
152        "nearestTrain": "A positive result is certain proof of the condition.",

```

```

153     "nearestLabel": "certainty",
154     "tokenJaccard": 0.417,
155     "sequenceRatio": 0.667
156 },
157 {
158     "holdout": "It is impossible for somebody with a negative screen to be ill.",
159     "label": "certainty",
160     "nearestTrain": "If the result is negative it is impossible to have the disease.",
161     "nearestLabel": "certainty",
162     "tokenJaccard": 0.294,
163     "sequenceRatio": 0.46
164 },
165 {
166     "holdout": "The test detects ninety percent of cases, therefore ninety percent of positive
people are sick.",
167     "label": "sensitivity",
168     "nearestTrain": "A test that detects ninety percent of cases means only ten percent of
positives are false.",
169     "nearestLabel": "sensitivity",
170     "tokenJaccard": 0.368,
171     "sequenceRatio": 0.735
172 },
173 {
174     "holdout": "I think sensitivity is the chance a positive test is a real case.",
175     "label": "sensitivity",
176     "nearestTrain": "Sensitivity is not the same as the chance of the condition after a positive
test.",
177     "nearestLabel": "balanced",
178     "tokenJaccard": 0.412,
179     "sequenceRatio": 0.575
180 },
181 {
182     "holdout": "The ninety percent sensitivity is also the positive predictive value.",
183     "label": "sensitivity",
184     "nearestTrain": "Positive predictive value and sensitivity are the same number.",
185     "nearestLabel": "sensitivity",
186     "tokenJaccard": 0.385,
187     "sequenceRatio": 0.397
188 },
189 {
190     "holdout": "Specificity is how likely somebody is to have the condition given a positive.",
191     "label": "sensitivity",
192     "nearestTrain": "The chance after a positive result depends on how common the condition is.",
193     "nearestLabel": "balanced",
194     "tokenJaccard": 0.333,
195     "sequenceRatio": 0.397
196 },
197 {
198     "holdout": "Sensitivity means the proportion with disease in the positive group.",
199     "label": "sensitivity",
200     "nearestTrain": "I use the sensitivity as the probability of disease in the positive group.",
201     "nearestLabel": "sensitivity",
202     "tokenJaccard": 0.429,
203     "sequenceRatio": 0.789
204 },
205 {
206     "holdout": "Ninety percent case detection tells us nine out of ten positives have disease.",
207     "label": "sensitivity",
208     "nearestTrain": "Ninety five percent specificity means ninety five percent of positives have
the disease.",
209     "nearestLabel": "sensitivity",
210     "tokenJaccard": 0.353,
211     "sequenceRatio": 0.663
212 },
213 {
214     "holdout": "The disease frequency is irrelevant to interpreting a positive test.",
215     "label": "base_rate",
216     "nearestTrain": "The group size without disease is irrelevant to the positive results.",
217     "nearestLabel": "base_rate",
218     "tokenJaccard": 0.429,
219     "sequenceRatio": 0.657
220 },
221 {
222     "holdout": "I can leave prevalence out because sensitivity and specificity are enough.",
223     "label": "base_rate",
224     "nearestTrain": "We only need sensitivity and specificity, not prevalence.",
225     "nearestLabel": "base_rate",
226     "tokenJaccard": 0.267,
227     "sequenceRatio": 0.611
228 },
229 {
230     "holdout": "It makes no difference if the disease is uncommon or widespread.",
231     "label": "base_rate",
232     "nearestTrain": "If the result is negative it is impossible to have the disease.",

```

```

233     "nearestLabel": "certainty",
234     "tokenJaccard": 0.312,
235     "sequenceRatio": 0.331
236 },
237 {
238     "holdout": "The same test means the same probability in any group.",
239     "label": "base_rate",
240     "nearestTrain": "A positive result means the same thing in every population.",
241     "nearestLabel": "base_rate",
242     "tokenJaccard": 0.286,
243     "sequenceRatio": 0.566
244 },
245 {
246     "holdout": "The base rate does not affect the fraction of true positives.",
247     "label": "base_rate",
248     "nearestTrain": "Testing a different group will not change the proportion of true positives.",
249     "nearestLabel": "base_rate",
250     "tokenJaccard": 0.294,
251     "sequenceRatio": 0.603
252 },
253 {
254     "holdout": "There is no reason to consider how common the condition is.",
255     "label": "base_rate",
256     "nearestTrain": "How common the condition is does not matter to the result.",
257     "nearestLabel": "base_rate",
258     "tokenJaccard": 0.429,
259     "sequenceRatio": 0.479
260 },
261 {
262     "holdout": "Because false positives occur, screening should never be used at all.",
263     "label": "dismissal",
264     "nearestTrain": "If there are false positives we should never use screening.",
265     "nearestLabel": "dismissal",
266     "tokenJaccard": 0.312,
267     "sequenceRatio": 0.625
268 },
269 {
270     "holdout": "A screening test that misses a case is entirely worthless.",
271     "label": "dismissal",
272     "nearestTrain": "A test that misses even one case should never be used.",
273     "nearestLabel": "dismissal",
274     "tokenJaccard": 0.333,
275     "sequenceRatio": 0.589
276 },
277 {
278     "holdout": "False alarms mean the test has zero value.",
279     "label": "dismissal",
280     "nearestTrain": "Any false negative means the test has no value at all.",
281     "nearestLabel": "dismissal",
282     "tokenJaccard": 0.357,
283     "sequenceRatio": 0.708
284 },
285 {
286     "holdout": "An imperfect test is always a waste of effort.",
287     "label": "dismissal",
288     "nearestTrain": "An imperfect screen is a waste and has zero benefit.",
289     "nearestLabel": "dismissal",
290     "tokenJaccard": 0.357,
291     "sequenceRatio": 0.653
292 },
293 {
294     "holdout": "Any false result proves screening is pointless.",
295     "label": "dismissal",
296     "nearestTrain": "Screening is pointless because it cannot be perfect.",
297     "nearestLabel": "dismissal",
298     "tokenJaccard": 0.25,
299     "sequenceRatio": 0.465
300 },
301 {
302     "holdout": "If tests are not perfect there cannot be any screening benefit.",
303     "label": "dismissal",
304     "nearestTrain": "There is no benefit to screening if any cases are missed.",
305     "nearestLabel": "dismissal",
306     "tokenJaccard": 0.375,
307     "sequenceRatio": 0.45
308 }
309 ]
310 },
311 "developmentProbes": {
312     "count": 24,
313     "classificationUnchanged": true,
314     "flawedOrMixedCount": 10,
315     "originalPraisedFlawedOrMixed": 2,
316     "currentPraisedFlawedOrMixed": 0,

```

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317     "correctProbesAcceptedAsMisconceptionPattern": 5,
318     "correctProbeCount": 8,
319     "allCurrentResponsesProvisional": true,
320     "limitations": "These author-labeled probes informed this feedback revision. Fewer praising titles
measures response wording, not classification accuracy, misconception detection or learning gain.",
321     "rows": [
322         {
323             "group": "correct",
324             "text": "Sensitivity measures detected cases rather than the probability of disease among
positives.",
325             "rawLabel": "sensitivity",
326             "label": "sensitivity",
327             "vote": 0.9808012911978988,
328             "coverage": 0.7142857142857143,
329             "abstained": false,
330             "classificationUnchanged": true,
331             "originalTitle": "Check which group you are counting.",
332             "currentTitle": "Review the two starting groups.",
333             "provisional": true,
334             "originalPraise": false,
335             "currentPraise": false
336         },
337         {
338             "group": "correct",
339             "text": "Sensitivity starts with people who have the condition, whereas predictive value
starts with positive results.",
340             "rawLabel": "sensitivity",
341             "label": "uncertain",
342             "vote": 0.49901806319793324,
343             "coverage": 0.48,
344             "abstained": true,
345             "classificationUnchanged": true,
346             "originalTitle": "Let\u2019s use the counts as our anchor.",
347             "currentTitle": "Choose an angle to check.",
348             "provisional": true,
349             "originalPraise": false,
350             "currentPraise": false
351         },
352         {
353             "group": "correct",
354             "text": "A positive result is not conclusive proof of disease.",
355             "rawLabel": "certainty",
356             "label": "certainty",
357             "vote": 0.9167186456519244,
358             "coverage": 0.8571428571428571,
359             "abstained": false,
360             "classificationUnchanged": true,
361             "originalTitle": "Leave room for uncertainty.",
362             "currentTitle": "Review what the experiment can establish.",
363             "provisional": true,
364             "originalPraise": false,
365             "currentPraise": false
366         },
367         {
368             "group": "correct",
369             "text": "Prevalence is not irrelevant: it determines how many non-cases can become false
positives.",
370             "rawLabel": "balanced",
371             "label": "balanced",
372             "vote": 0.9849016249450843,
373             "coverage": 0.5416666666666666,
374             "abstained": false,
375             "classificationUnchanged": true,
376             "originalTitle": "You are connecting the pieces.",
377             "currentTitle": "Review the positive group.",
378             "provisional": true,
379             "originalPraise": true,
380             "currentPraise": false
381         },
382         {
383             "group": "correct",
384             "text": "False alarms do not make every screening program useless; benefits and harms need
evidence.",
385             "rawLabel": "dismissal",
386             "label": "dismissal",
387             "vote": 0.9697847021194123,
388             "coverage": 0.46153846153846156,
389             "abstained": false,
390             "classificationUnchanged": true,
391             "originalTitle": "Errors are part of the assessment.",
392             "currentTitle": "Review benefits and harms.",
393             "provisional": true,
394             "originalPraise": false,
395             "currentPraise": false

```

```

396     },
397     {
398         "group": "correct",
399         "text": "Nine true cases among 58.5 positives is about fifteen percent, not ninety percent.",
400         "rawLabel": "sensitivity",
401         "label": "sensitivity",
402         "vote": 0.9991577229019015,
403         "coverage": 0.5909090909090909,
404         "abstained": false,
405         "classificationUnchanged": true,
406         "originalTitle": "Check which group you are counting.",
407         "currentTitle": "Review the two starting groups.",
408         "provisional": true,
409         "originalPraise": false,
410         "currentPraise": false
411     },
412     {
413         "group": "correct",
414         "text": "Specificity counts negative results among people without the condition, not true
cases among positives.",
415         "rawLabel": "balanced",
416         "label": "balanced",
417         "vote": 0.9998357289682823,
418         "coverage": 0.6538461538461539,
419         "abstained": false,
420         "classificationUnchanged": true,
421         "originalTitle": "You are connecting the pieces.",
422         "currentTitle": "Review the positive group.",
423         "provisional": true,
424         "originalPraise": true,
425         "currentPraise": false
426     },
427     {
428         "group": "correct",
429         "text": "It is wrong to say that every positive result is a guaranteed diagnosis.",
430         "rawLabel": "certainty",
431         "label": "certainty",
432         "vote": 0.8786283424496205,
433         "coverage": 0.6842105263157895,
434         "abstained": false,
435         "classificationUnchanged": true,
436         "originalTitle": "Leave room for uncertainty.",
437         "currentTitle": "Review what the experiment can establish.",
438         "provisional": true,
439         "originalPraise": false,
440         "currentPraise": false
441     },
442     {
443         "group": "mixed",
444         "text": "Prevalence matters, but ninety percent sensitivity still means ninety percent of
positives are true.",
445         "rawLabel": "sensitivity",
446         "label": "sensitivity",
447         "vote": 0.9999999998321463,
448         "coverage": 0.76,
449         "abstained": false,
450         "classificationUnchanged": true,
451         "originalTitle": "Check which group you are counting.",
452         "currentTitle": "Review the two starting groups.",
453         "provisional": true,
454         "originalPraise": false,
455         "currentPraise": false
456     },
457     {
458         "group": "mixed",
459         "text": "False positives can outnumber true positives when the condition is rare, so the test
is completely useless.",
460         "rawLabel": "balanced",
461         "label": "balanced",
462         "vote": 0.9999938391352348,
463         "coverage": 0.8928571428571429,
464         "abstained": false,
465         "classificationUnchanged": true,
466         "originalTitle": "You are connecting the pieces.",
467         "currentTitle": "Review the positive group.",
468         "provisional": true,
469         "originalPraise": true,
470         "currentPraise": false
471     },
472     {
473         "group": "mixed",
474         "text": "The result depends on the starting population, although a positive proves the disease
for sure.",
475         "rawLabel": "certainty",

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476     "label": "certainty",
477     "vote": 0.9930030748440959,
478     "coverage": 0.6086956521739131,
479     "abstained": false,
480     "classificationUnchanged": true,
481     "originalTitle": "Leave room for uncertainty.",
482     "currentTitle": "Review what the experiment can establish.",
483     "provisional": true,
484     "originalPraise": false,
485     "currentPraise": false
486 },
487 {
488     "group": "mixed",
489     "text": "False negatives can occur, but a negative result guarantees no condition.",
490     "rawLabel": "certainty",
491     "label": "certainty",
492     "vote": 0.831716242578965,
493     "coverage": 0.5263157894736842,
494     "abstained": false,
495     "classificationUnchanged": true,
496     "originalTitle": "Leave room for uncertainty.",
497     "currentTitle": "Review what the experiment can establish.",
498     "provisional": true,
499     "originalPraise": false,
500     "currentPraise": false
501 },
502 {
503     "group": "mixed",
504     "text": "I know sensitivity and predictive value have different denominators, yet both are
always the same percentage.",
505     "rawLabel": "base_rate",
506     "label": "base_rate",
507     "vote": 0.9371322579369675,
508     "coverage": 0.46153846153846156,
509     "abstained": false,
510     "classificationUnchanged": true,
511     "originalTitle": "Give prevalence a place in the story.",
512     "currentTitle": "Review the starting population.",
513     "provisional": true,
514     "originalPraise": false,
515     "currentPraise": false
516 },
517 {
518     "group": "mixed",
519     "text": "We count all positive results in the denominator; in this case 9 divided by 58.5 is
ninety percent.",
520     "rawLabel": "sensitivity",
521     "label": "sensitivity",
522     "vote": 0.9992353750157679,
523     "coverage": 0.5833333333333334,
524     "abstained": false,
525     "classificationUnchanged": true,
526     "originalTitle": "Check which group you are counting.",
527     "currentTitle": "Review the two starting groups.",
528     "provisional": true,
529     "originalPraise": false,
530     "currentPraise": false
531 },
532 {
533     "group": "ambiguous",
534     "text": "Sensitivity is ninety percent.",
535     "rawLabel": "sensitivity",
536     "label": "sensitivity",
537     "vote": 0.9999575342095176,
538     "coverage": 1,
539     "abstained": false,
540     "classificationUnchanged": true,
541     "originalTitle": "Check which group you are counting.",
542     "currentTitle": "Review the two starting groups.",
543     "provisional": true,
544     "originalPraise": false,
545     "currentPraise": false
546 },
547 {
548     "group": "ambiguous",
549     "text": "The test is pretty accurate.",
550     "rawLabel": "base_rate",
551     "label": "uncertain",
552     "vote": 0.5854269892047604,
553     "coverage": 0.5714285714285714,
554     "abstained": true,
555     "classificationUnchanged": true,
556     "originalTitle": "Let\u2019s use the counts as our anchor.",
557     "currentTitle": "Choose an angle to check.",

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558     "provisional": true,
559     "originalPraise": false,
560     "currentPraise": false
561 },
562 {
563     "group": "ambiguous",
564     "text": "There are false positives.",
565     "rawLabel": "dismissal",
566     "label": "dismissal",
567     "vote": 0.8960862305265518,
568     "coverage": 1,
569     "abstained": false,
570     "classificationUnchanged": true,
571     "originalTitle": "Errors are part of the assessment.",
572     "currentTitle": "Review benefits and harms.",
573     "provisional": true,
574     "originalPraise": false,
575     "currentPraise": false
576 },
577 {
578     "group": "ambiguous",
579     "text": "What does prevalence mean?",
580     "rawLabel": "base_rate",
581     "label": "uncertain",
582     "vote": 0.655848744431607,
583     "coverage": 0.42857142857142855,
584     "abstained": true,
585     "classificationUnchanged": true,
586     "originalTitle": "Let\u2019s use the counts as our anchor.",
587     "currentTitle": "Choose an angle to check.",
588     "provisional": true,
589     "originalPraise": false,
590     "currentPraise": false
591 },
592 {
593     "group": "ambiguous",
594     "text": "My friend says a positive result proves disease. Is that really correct?",
595     "rawLabel": "certainty",
596     "label": "certainty",
597     "vote": 0.9656013686454958,
598     "coverage": 0.5789473684210527,
599     "abstained": false,
600     "classificationUnchanged": true,
601     "originalTitle": "Leave room for uncertainty.",
602     "currentTitle": "Review what the experiment can establish.",
603     "provisional": true,
604     "originalPraise": false,
605     "currentPraise": false
606 },
607 {
608     "group": "ambiguous",
609     "text": "I am unsure whether the denominator is cases or positives.",
610     "rawLabel": "balanced",
611     "label": "uncertain",
612     "vote": 0.48839184555279735,
613     "coverage": 0.2857142857142857,
614     "abstained": true,
615     "classificationUnchanged": true,
616     "originalTitle": "Let\u2019s use the counts as our anchor.",
617     "currentTitle": "Choose an angle to check.",
618     "provisional": true,
619     "originalPraise": false,
620     "currentPraise": false
621 },
622 {
623     "group": "misconception",
624     "text": "All positives are actual cases, without exception.",
625     "rawLabel": "balanced",
626     "label": "balanced",
627     "vote": 0.7640742316130136,
628     "coverage": 0.75,
629     "abstained": false,
630     "classificationUnchanged": true,
631     "originalTitle": "You are connecting the pieces.",
632     "currentTitle": "Review the positive group.",
633     "provisional": true,
634     "originalPraise": true,
635     "currentPraise": false
636 },
637 {
638     "group": "misconception",
639     "text": "Prevalence can be ignored for any test.",
640     "rawLabel": "balanced",
641     "label": "uncertain",

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642     "vote": 0.40518696542595917,
643     "coverage": 0.4,
644     "abstained": true,
645     "classificationUnchanged": true,
646     "originalTitle": "Let\u2019s use the counts as our anchor.",
647     "currentTitle": "Choose an angle to check.",
648     "provisional": true,
649     "originalPraise": false,
650     "currentPraise": false
651 },
652 {
653     "group": "misconception",
654     "text": "The true-positive rate is exactly the probability that a positive is a case.",
655     "rawLabel": "sensitivity",
656     "label": "sensitivity",
657     "vote": 0.9716179316559089,
658     "coverage": 0.55,
659     "abstained": false,
660     "classificationUnchanged": true,
661     "originalTitle": "Check which group you are counting.",
662     "currentTitle": "Review the two starting groups.",
663     "provisional": true,
664     "originalPraise": false,
665     "currentPraise": false
666 },
667 {
668     "group": "misconception",
669     "text": "One false alarm makes every screening test pointless.",
670     "rawLabel": "dismissal",
671     "label": "dismissal",
672     "vote": 0.999891763027237,
673     "coverage": 0.6,
674     "abstained": false,
675     "classificationUnchanged": true,
676     "originalTitle": "Errors are part of the assessment.",
677     "currentTitle": "Review benefits and harms.",
678     "provisional": true,
679     "originalPraise": false,
680     "currentPraise": false
681 }
682 ]
683 },
684 "feedbackPolicy": "1.1: provisional, overridable review topics; separate factual self-checks.
Model/data/holdout unchanged."
685 }

```

model/feedback-v1.cjs

```
1 (function(root){
2   'use strict';
3   const content={
4     unavailable:{title:'The AI model did not load.',focus:'Model unavailable',body:'The AI model is
unavailable, so the coach cannot interpret your wording. Your explanation has not been judged. The
mathematical simulator, comparisons and practice remain available.',question:'Reload to try loading
the model, or use the three-case lesson to inspect the arithmetic.',source:'bmj'},
5     balanced:{title:'You are connecting the pieces.',focus:'Prevalence + uncertainty',body:'Your
wording resembles the model\u2019s examples of contextual reasoning. Check that your explanation
connects false positives to the size of the group without the condition.',question:'What changes if
the condition becomes more common but the test stays the same?',source:'bmj'},
6     certainty:{title:'Leave room for uncertainty.',focus:'A result is not a guarantee',body:'The model
detected wording that may treat a screening result as certainty. Positive results can include false
positives; negative results can include missed cases. If you were rejecting certainty, the model may
have misread your wording.',question:'In this experiment, which people show why a positive or negative
result is not a guarantee?',source:'nci'},
7     sensitivity:{title:'Check which group you are counting.',focus:'Sensitivity \u2260 predictive
value',body:'The model detected wording about sensitivity or specificity. Sensitivity starts with
people who have the condition. Positive predictive value starts with people who test positive. Those
denominators are different. If you already made that distinction, the model may have misread your
wording.',question:'To interpret a positive result, should your denominator be all cases or all
positive results?',source:'bmj'},
8     base_rate:{title:'Give prevalence a place in the story.',focus:'The starting population
matters',body:'The model detected wording that may leave out how common the condition is. A large
group without the condition can produce many false positives, even at a low false-positive
rate.',question:'Keep the test settings fixed and change prevalence from 1% to 20%. What changes among
positives?',source:'bmj'},
9     dismissal:{title:'Errors are part of the assessment.',focus:'Benefit and harm need
evidence',body:'The model detected wording that may dismiss all screening because errors occur. False
results matter, but this simulator alone cannot establish a real screening test\u2019s overall
benefits, harms, or suitability.',question:'What evidence would you need beyond these counts to judge
a real screening program?',source:'nci'},
10    uncertain:{title:'Let\u2019s use the counts as our anchor.',focus:'The model is unsure',body:'Your
explanation does not match the model\u2019s limited examples clearly enough. Try stating how
prevalence and false positives affect the group of positive results.',question:'Complete this:
\u201cSome people test positive without the condition, so\u2026\u201d',source:'bmj'},
11    personal:{title:'Keep this experiment fictional.',focus:'Personal results need clinical
context',body:'This lab cannot interpret your personal symptoms, test result, or health risk. Its
numbers describe an invented population. Discuss personal results with an appropriate healthcare
professional.',question:'For this exercise, explain the fictional population shown above
instead.',source:'nci'},
12    empty:{title:'Start with one or two sentences.',focus:'Your reasoning matters',body:'Write an
explanation in your own words, then the coach can look for a reasoning pattern.',question:'Why might a
positive result include a person without the condition?',source:'bmj'}
13  };
14  const sources={bmj:{title:'Gigerenzer \xb7 What are natural frequencies? (BMJ,
2011)',url:'https://www.bmj.com/content/343/bmj.d6386'},nci:{title:'National Cancer Institute \xb7
Cancer Screening
Overview',url:'https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq'}};
15  const indexes=new WeakMap();
16  function tokenize(text,model){const stop=new
Set(model.stopWords),words=(text.toLowerCase().replace(/\u2019/g,"").match(/[a-z]+/g)||[]);return
[...words.filter(w=>stop.has(w)),...words.slice(0,-1).map((w,i)=>w+'_'+words[i+1])];}
17  function classify(text,model){
18    if(typeof text!=='string' || text.length>1800)throw new RangeError('Explanation must be text of at
most 1800 characters.');
```

19 if(!text.trim())return {label:'empty',method:'no inference',patterns:[]};

20 if(/(?:\bmys+(:test|result|symptom|diagnosis|blood|scan)|\bhs+(:have|had|got)\s+(?:a\s+)?(?:po
sitive|negative|symptoms|cancer|disease|pain)|\bb(?:diagnose me|should i(?:get|take|stop|start)|do i
have|am i(?:sick|ill)|my chance of))/i.test(text))return {label:'personal',method:'scope
guard',patterns:[]};

21 if(!model)return {label:'unavailable',method:'model unavailable',patterns:[]};

22 if(!indexes.has(model))indexes.set(model,new Map(model.vocabulary.map((v,i)=>[v,i]));)

23 const index=indexes.get(model),feats=tokenize(text,model),known=feats.filter(w=>index.has(w));

24 const scores=model.labels.map(label=>({label,score:model.logPriors[label]+known.reduce((sum,w)=>su
m+model.weights[label][index.get(w)],0)}));

25 const

max=Math.max(...scores.map(s=>s.score)),den=scores.reduce((sum,s)=>sum+Math.exp(s.score-max),0);

26 for(const s of scores)s.vote=Math.exp(s.score-max)/den;

27 scores.sort((a,b)=>b.vote-a.vote);

28 const coverage=known.length/Math.max(1,feats.length),a=model.abstention;

29 const abstain=known.length<a.minKnownFeatures || coverage<a.minCoverage || scores[0].vote<a.minVote || s
cores[0].vote-scores[1].vote<a.minMargin;

30 const first=scores[0],second=scores[1];

31 const patterns=[...new Set(known)].map(word=>({word,diff:model.weights[first.label][index.get(word
)]-model.weights[second.label][index.get(word)]})).filter(x=>x.diff>0).sort((a,b)=>b.diff-a.diff).slic
e(0,4).map(x=>x.word.replace(/_/g,' '));

32 return {label:abstain?'uncertain':first.label,rawLabel:first.label,method:'Multinomial Naive Bayes
,patterns,coverage,vote:first.vote,scores:scores.map(s=>({label:s.label,vote:s.vote})),abstained:abst
ain};

```

33   }
34   function feedback(result,counts){const
item=content[result.label]||content.uncertain,format=root.BaseRateScience?.formatNumber||String;return
  {...item,source:sources[item.source],counts:`In the current fictional experiment:
  ${format(counts.tp)} true positives and ${format(counts.fp)} false positives; ${counts.ppv===null?'the
  positive predictive value is undefined':format(counts.ppv)}% of positive results are expected to be
  true positives'}.`};}
35   root.BaseRateCoach={classify,feedback,tokenize};
36   if(typeof module!=='undefined')module.exports=root.BaseRateCoach;
37 }(typeof window!=='undefined'?window:globalThis);

```

docs/model-card.md

```
1 # BaseRate local coach: model 1.0.0, feedback policy 1.1
2
3 ## Intended use and behavior
4
5 English educational explanations of fictional screening examples. A Multinomial Naive Bayes classifier
  proposes a review topic. It does not verify the explanation, grade understanding or interpret a
  person's test result. Arithmetic is calculated separately in `dist/science.js`.
6
7 The five original labels are `balanced`, `base_rate`, `sensitivity`, `certainty`, and `dismissal`. The
  UI maps them to a positive-group, starting-population, condition-group, uncertainty, or
  benefits/harms topic. Labels, votes, features and coverage remain inspectable as model diagnostics.
  Learners can choose another topic, answer a fixed factual question, inspect counts/reference groups,
  and revise their explanation. Correctness feedback applies only to the selected answer.
8
9 Personal-query and unfamiliar/empty/missing-model guards remain. These are imperfect scope heuristics,
  not a guarantee of detecting every personal question. Missing model data still permits manual topic
  selection and model-independent questions. The app makes no remote inference or analytics calls.
  Explanations are processed and retained in tab memory, without persistent app storage or server
  transmission; notebook exports omit them. Static app/model downloads still contact GitHub Pages, which
  may log requests. Optional connected browser assistants have separate data handling. See [privacy
  scope](privacy.md) and [architecture](architecture.md) for exact data flow and classifier equations.
10
11 ## Training and original evaluation
12
13 `model/dataset.json` contains 100 synthetic English training sentences and 30 separately authored
  synthetic holdout sentences. Both subsets were generated with AI assistance in the same project. They
  are not independently collected learner data. Word and adjacent-word counts, additive smoothing alpha
  0.7, class priors and learned likelihoods are implemented by the standard-library Python trainer and
  browser inference.
14
15 Run `python3 model/train.py` to reproduce the distributed artifacts. Run `python3 model/audit.py` to
  reproduce and compare them in a temporary directory, check the split, and rerun the archived/current
  feedback probes. The audit does not modify the model or evaluation. `--write` regenerates the
  deterministic audit artifact after an intentional change. Node and Python standard libraries suffice.
16
17 Reproduction verified byte-identical `dist/model.js` and `model/evaluation.json`. Original holdout:
  29/30 raw labels correct; uncertainty gates accept 28, of which 27 are correct. These are synthetic
  development results, not clinical accuracy, calibrated confidence or a broad language benchmark.
  Weights, thresholds, data and holdout are unchanged by feedback policy 1.1. All 24 diagnostic probe
  classifications are unchanged. There is no claim of improved classifier accuracy.
18
19 ## Split distinctness and known errors
20
21 The deterministic split audit found zero exact sentence duplicates. Mean nearest-training token
  Jaccard similarity is 0.333; none reaches 0.5. The nearest training sentence has the same expected
  label for 23/30 holdout sentences. These lexical checks do not establish meaningful semantic
  independence: the subsets share authorship, task, claim types and vocabulary. They test phrasing
  variation within a small synthetic task. No holdout example was moved into training, removed or
  relabeled to hide an error.
22
23 The known holdout error is a correct sentence distinguishing sensitivity from probability among
  positives. It receives the sensitivity-confusion label with an uncalibrated vote of about 98.1%. The
  model cannot reliably resolve negation, reported speech, qualifiers or conflicting ideas from lexical
  counts.
24
25 `model/probes.json` adds 24 developer-authored diagnostic cases: 8 correct explanations, 6 mixed
  claims, 6 ambiguous statements and 4 clear misconceptions. They were used to choose this feedback
  change and are development regressions, not a pristine new test set. `model/audit.json` preserves each
  sentence, intent, classification and old/new response.
26
27 Examples of failures:
28
29 - \u201cAll positives are actual cases, without exception.\u201d receives `balanced` (76.4% vote),
  despite its false certainty.
30
31 - A mixed explanation that notes numerous false positives and concludes that the test is completely
  useless receives `balanced` (approximately 100% vote).
32
33 - A correct sensitivity/PPV distinction, a correct denial of certainty, and reported speech can
  receive misconception labels with high votes.
34
35 - Neutral short observations such as sensitivity being 90% or the existence of false positives cannot
  establish the learner's reasoning, yet can pass the model's gates.
36
37 Five of eight correct diagnostic explanations were accepted as a misconception-pattern label. This
  small targeted set deliberately stresses known weaknesses; its ratio is not an estimate of real-world
  error frequency. Uncalibrated model votes measure the model's relative fit, not probability that a
  learner is wrong.
38
39 ## Why feedback changed instead of retraining
40
41 The evidence points to semantic limitations and unsafe endorsement, not an arithmetic defect. Adding
  these same diagnostic sentences to training would invalidate a later claim of generalization on them.
```

Independently collected new data and a genuinely separate evaluation are unavailable.
Deadline-conscious policy 1.1 therefore preserves the auditable model and changes the educational contract:

```
39
40 1. All classified/uncertain responses are explicitly provisional. No inferred label confirms
41 correctness or declares a misconception.
42 2. Learners confirm or override a topic through a visible selection control.
43 3. A factual multiple-choice check uses current fictional counts or a fixed evidence/scope
44 distinction. It tells the learner exactly what the selected answer establishes.
45 4. Changing the experiment clears an answered check and recalculates its question; manually chosen
46 topics persist. New explanation submissions reset the topic suggestion.
47 5. Learners can return directly to revise their unchanged explanation. There is no fabricated
48 generative critique or score.
49
50 Among the 10 flawed/mixed diagnostic cases, the archived response used an explicit praise title for 2;
51 current feedback does so for 0. Every current classified diagnostic response is provisional. This
52 measures feedback wording, not model improvement, comprehension or health benefit. The model can still
53 suggest an irrelevant topic; override and factual checking reduce the consequence without solving the
54 classification limitation.
55
56 ## Learning evidence and remaining limits
57
58 The guided lesson now explains the actual ratio implied by each wrong denominator and requires a
59 corrected positive-group choice before progression. Notebook schema 2 retains first and final
60 denominator choices and number of selections, alongside first numeric predictions. This makes observed
61 correction visible. Selection counts are not equivalent to effort, mastery or learning gain.
62
63 Automated tests cover all five topic questions, correct/wrong answers, unknown choices, invalid rates,
64 no-condition and perfect-specificity edges, case feedback and text-free exports. Actual browser
65 checks assess rendering and interaction. AI-assisted code review and synthetic critiques are
66 development QA. No real learner study, educator review, clinical validation, multilingual evaluation
67 or assistive-technology user study has occurred.
68
69 Future claims should use independently collected explanations, adjudicated ambiguous/mixed cases,
70 untouched evaluation sets, calibrated abstention analysis and transfer/delayed learning measures. The
71 current prototype is intentionally limited to fictional educational use.
72
73 ## Demo interpretation
74
75 The refreshed demo shows feedback policy 1.1 and the unchanged 1.0.0 classifier. The narrated 29/30
76 result is a raw same-project synthetic holdout result; it does not include abstention or establish
77 independent generalization. Accepted-label results are 27 correct of 28 accepted. Selected factual
78 answers are checked separately from inferred topics. See [the demo claim qualifications](demo.md).
```

docs/demo.md

```
1 # Final-release demonstration
2
3 [Watch the demo and download the project files](https://nicholas-afk.github.io/baserate/project.html).
4
5 The 112-second, 16:9 demo uses the entrant's supplied 58.697-second narration without synthesized
  words, rearrangement or speech cuts. The source waveform remains unchanged and private. Level gain and
  a peak limiter are applied to the delivered audio. No music or generated voice is included.
6
7 All application pixels come from genuine browser interactions with the final public app. The edit uses
  cropped screen-state captures, holds and hard cuts, rather than a continuous screen recording. There
  is no rebuilt interface or invented cursor. The walkthrough's estimates (90%, 80%, 50%) are staged
  demonstration inputs, not learner-study observations. All three cases were actually completed and the
  notebook export control was used.
8
9 ## Reading the video
10
11 | Time | Actual product behavior shown |
12 | --- | --- |
13 | 0\u201318.532 s | Population squares and frequency tree; default 9 true + 49.5 false positives;
  9/58.5 = 15.4%. |
14 | 18.532\u201336.718 s | Built-in misconception example, provisional topic, current counts, evidence
  link, separate factual check and revision control. |
15 | 36.718\u201358.697 s | Inspectable model card, synthetic evaluation, retained known error and
  limitations. |
16 | 58.697\u20131:05 s | Prediction, reveal, wrong denominator blocking progression, correction, completed
  numeric notebook and export control. |
17 | 1:05\u20131:12 s | Pinned comparison and prevalence curve; independent versus perfectly repeated second
  results. |
18 | 1:12\u20131:15 s | Public app location and honest project scope. |
19
20 ## Narration qualifications
21
22 - **\u201cOf 30 separate synthetic examples\u201d** means the raw result on the unchanged
  same-project synthetic holdout. The splits contain no exact duplicate sentences but share authorship,
  task and familiar claim patterns. On screen: 28 accepted labels, 27 correct. These are development
  checks, not an independent benchmark or proof of learning.
23 - **Coach feedback** proposes a review topic. It can misread a correct explanation. The factual check
  verifies only the selected option; it does not verify the explanation, diagnose a misconception or
  grade competence.
24 - **\u201cThe words stay in the browser\u201d** describes the application's local inference and tab
  memory. Static app/model/media requests still contact GitHub Pages. A connected browser assistant has
  separate data handling.
25 - **Educational value** is an intended use. No learner study, clinical validation or measured health
  benefit has been completed.
26 - **AI assistance** is extensive and disclosed. Automatic local transcription has an uncertain/clipped
  word near 55 seconds; the edit does not reconstruct it. The complete disclosure appears on screen.
  Short editorial annotations are not presented as verbatim captions.
27
28 ## Format and submission status
29
30 The [official overview](https://univabio.devpost.com/) requests an interactive prototype, a demo
  explaining purpose/features and users interacting, a one-page description and repository/code PDF. No
  numeric video-duration or encoding limit was found in the current overview or
  [rules](https://univabio.devpost.com/rules) when checked on 6 October 2026. This observation is not
  organizer clarification of the conflicting deadlines; planning retains the earlier cutoff.
31
32 [Devpost's video guidance](https://help.devpost.com/article/85-uploading-a-demo-video) accepts
  YouTube, Vimeo or Youku for its embedded video field. The direct GitHub Pages MP4 is publicly
  accessible but is not a supported embed URL. The existing YouTube account is at Create channel, which
  accepts Terms. That pending personal acceptance is the exact blocker to an accepted embedded upload;
  no channel or duplicate account has been created and no unsupported URL is substituted into the field.
33
34 The one-page PDF, complete source PDF, runnable ZIP, updated screenshots and thumbnail accompany the
  demo. The video poster remains 16:9; a separate [1200-by-800 Devpost
  cover](../dist/media/devpost-thumbnail.png) uses the recommended 3:2 ratio and keeps the title
  readable. Its [editable SVG source](assets/devpost-cover.svg) uses the existing project typography and
  palette. The raw voice file and private video-working directory are excluded from the repository. The
  original app model, dataset and classifications are unchanged; the five application test suites and
  deterministic audit were rerun for this materials refresh.
```

docs/architecture.md

```
1 # Architecture and explanation guide
2
3 BaseRate is a static browser application. It has no backend, database, account system, API key or app
  dependencies. GitHub Pages serves `dist/` at `https://nicholas-afk.github.io/baserate/`. Assets and
  download links are relative to this path. The model is downloaded once as ordinary JavaScript;
  inference then runs synchronously on the device.
4
5 ## Modules and data flow
6
7 The deferred scripts in `dist/index.html` execute in this order:
8
9 | Module | Responsibility |
10 | --- | --- |
11 | `science.js` | Pure expected-count math, rounded squares, comparisons, curve and conditional repeat
  testing |
12 | `model.js` | Generated vocabulary, class priors, learned log likelihoods, thresholds and original
  evaluation summary |
13 | `coach.js` | Text features, classification, fixed provisional feedback, factual questions and answer
  checks |
14 | `learning.js` | Three fixed cases, predictions, denominator feedback and schema-2 export data |
15 | `app.js` | Input validation, base experiment views and visible coach state |
16 | `lab-tools.js` | Frequency tree, pinned comparison, curve/table, repeat explorer and settings link |
17 | `learning-ui.js` | Guided lesson progression and requested notebook download |
18
19 Editing a valid rate updates one unrounded calculation. The grid, counts and equation share that
  result. A `baserate:experiment` event supplies copies of settings/counts to the extra lab views.
  Pinning stores a snapshot of A; changing B cannot mutate A. Lesson cases have their own fixed settings
  and do not use the current sliders.
20
21 Submitting an explanation stores it in tab memory and calls the classifier. Its label suggests a
  topic; the learner can override it. A separate factual check calls the math module or uses a fixed
  scope/evidence answer. Changing rates rebuilds that check and clears any old answer status. A new
  explanation resets the suggested topic; loading an example clears the previous check. No generated
  medical facts or text-based correctness score are used.
22
23 ## Screening arithmetic
24
25 For 1,000 people, convert percentages to proportions `p`, `s`, `c`:
26
27 ```text
28 TP = 1000 * p * s          FN = 1000 * p * (1 - s)
29 FP = 1000 * (1 - p) * (1 - c) TN = 1000 * (1 - p) * c
30 PPV (%) = 100 * TP / (TP + FP)
31 ```
32
33 A zero positive denominator yields `null` (undefined), not zero. Only the 1,000 squares are rounded,
  using largest remainders to conserve their total. Tables, PPV, comparisons and answers use unrounded
  expectations; fractional counts describe an expected population rather than literal partial people.
34
35 At 1% prevalence, 90% sensitivity and 95% specificity: TP=9, FP=49.5, FN=1, TN=940.5. Sensitivity
  starts with the 10 condition cases; PPV starts with the 58.5 positives. PPV is 15.3846%, displayed as
  15.4%. At 20% prevalence with the same test it is 81.8182%.
36
37 The curve evaluates 101 prevalence points for the current test and offers the same information in a
  table. A second test is applied only among first-positive people: second true positives = first TP
  times conditional detection, second false positives = first FP times conditional false-positive rate.
  The independent-identical preset assumes independence within condition/non-condition groups; reusing
  90%/5% gives 8.1 TP and 2.475 FP, PPV 76.5957%. Perfectly repeated results use conditional 100%/100%
  and leave PPV at 15.3846%. These are explicit fictional assumptions, not recommendations about actual
  repeat tests.
38
39 ## How the text classifier learns and runs
40
41 `model/train.py` uses 100 synthetic English sentences, 20 per class. The five original labels are
  `balanced`, `certainty`, `sensitivity`, `base_rate`, `dismissal`. They describe predefined patterns in
  synthetic text, not assessed learner competence. The mapping to review topics is inspectable in
  `app.js`.
42
43 1. Lowercase text, normalize curly apostrophes, and find ASCII letter sequences (`[a-z]+`). Numbers
  are not features.
44 2. Keep single words except the published stop-word list. Also keep adjacent-word pairs from the
  original word sequence, including stop words in pairs. Repeated features are counted, rather than
  deduplicated.
45 3. Build a sorted vocabulary from training only. Count each feature within each class.
46 4. Store  $\log((\text{feature count} + 0.7) / (\text{all feature counts in class} + 0.7 * \text{vocabulary size}))$ . This
  additive smoothing gives unseen-in-class vocabulary features a finite weight.
47 5. Store log class priors. With 20 examples per class, each prior is 1/5.
48
49 At inference, unknown features are ignored. Each class score is its log prior plus the learned log
  weights of all known feature occurrences. Subtract the maximum score, exponentiate, and normalize to
```

produce relative votes. This softmax-like normalization is numerically stable, but its votes are ****uncalibrated****: 98% vote does not mean 98% chance of correct reasoning.

50

51 The prototype withholds a topic if fewer than four known feature occurrences remain, known-feature coverage is below 18%, the top vote is below 65%, or the top-minus-second vote is below 20 percentage points. Coverage includes repeated features. The UI shows up to four unique known features whose weights favor the top class over the runner-up; these are associations, not a complete semantic explanation. Empty text, heuristic personal queries and missing-model states are handled before normal inference. Input is limited to 1,800 characters.

52

53 High votes can still be wrong. Negation, quotations and mixed claims defeat this lexical method. The current feedback policy therefore proposes a review topic rather than declaring correctness or a misconception. Selected-answer feedback is deterministic and does not verify the free-text explanation. Missing model data leaves math/practice available and permits manual topic checks.

54

55 **## Reproduction and evidence**

56

57 Python 3 and Node.js standard libraries suffice for training/audit/tests; the app requires neither at runtime. ``python3 model/train.py`` deliberately rewrites ``dist/model.js`` and ``model/evaluation.json``. Prefer ``python3 model/audit.py`` for verification: it trains in a temporary directory and compares bytes without changing the release. ``--write`` intentionally updates the audit artifact.

58

59 The original synthetic holdout has 30 sentences: 29 raw correct labels, 28 accepted by the gates, 27 accepted labels correct. Training/holdout have no exact duplicates but share synthetic authorship and claim patterns. The 24 diagnostic probes are development regressions, not a new independent benchmark. See the [model card](model-card.md) for known failures and [README](../README.md) for the five verification commands. No independent learning benefit, clinical accuracy or real-user generalization has been demonstrated.

60

61 **## Explain it to judges**

62

63 Walk through the 9/58.5 denominator, pin A before raising prevalence, and show why perfect repetition adds no information. Predict before revealing a lesson answer, select a wrong denominator, then correct it. Use a known classifier failure and explain why the topic is provisional; demonstrate a manual override and factual check. Explain that math is exact within invented assumptions while language classification is limited. The [entrant walkthrough](code-walkthrough.md) has a presentation route; it is preparation material, not proof that the entrant understands the code.

docs/privacy.md

```
1 # Data handling and privacy scope
2
3 These statements describe BaseRate's application code, not every browser extension, connected
  assistant, operating system or hosting provider. The application has no backend, accounts, analytics,
  third-party scripts/fonts, remote inference endpoint or patient dataset.
4
5 | Action/data | What the app does | Boundary |
6 | --- | --- | --- |
7 | Load the lab | Requests HTML, CSS and seven JavaScript files, including trained model weights, from
  GitHub Pages | Hosting requests expose ordinary connection/request information to the host; caching
  may reuse files |
8 | Load the project page | Requests its HTML/CSS and demo video metadata/media (`preload="metadata`);
  playback requests more media as needed | GitHub Pages hosting requests; no external video player is
  embedded |
9 | Edit fictional rates | Calculates and renders in the current tab | No request is initiated by the
  rate controls |
10 | Submit a coach explanation | Classifies locally and retains the text in JavaScript/DOM memory for
  revision | No app request sends the text; reload clears it. This is transient processing, not a claim
  of zero memory use |
11 | Complete the lesson | Retains first numeric predictions and first/final denominator choices in tab
  memory | No app persistence; reload resets progress |
12 | Export notebook | Creates a JSON Blob and saves a file on explicit request | Schema 2 includes fixed
  case settings, predictions, answer gaps, selected denominator choices and attempt counts; coach text
  is excluded. The downloaded file persists wherever the user/browser saves it |
13 | Copy settings link | Writes a URL to the clipboard, or displays it if copying fails | Contains only
  validated `p`, `se`, `sp` fictional rates and `#lab`; any existing query is removed. If
  shared/visited, query rates reach the hosting provider |
14 | Follow references/source links | Navigates to the external destination | Destination privacy
  policies and the browser's referrer policy apply; links use `noopener`, not universal referrer
  suppression |
15 | Use optional WebMCP | Registers two local tools when the browser supports them: experiment update
  and explanation review | A connected assistant can supply text/rates and receive a label/count
  summary. Its conversation, page access and data retention are governed separately; local app inference
  does not imply the assistant's work is private |
16
17 The app uses no `localStorage`, `sessionStorage`, IndexedDB, cookie-writing API or service worker. It
  makes no `fetch`, XMLHttpRequest, WebSocket or beacon calls. Static scripts/styles and video requests
  still occur. This is a source-level statement, not a claim that the browser/host never uses cookies,
  cache or telemetry. The app does not promise offline availability, secure file deletion or protection
  against a malicious extension. Do not enter real personal health information; the personal-query guard
  is imperfect.
18
19 ## Verification scope
20
21 Publication QA compares served asset/download bytes with the release, audits source data flows,
  inspects browser-observed resources before/after a synthetic explanation and checks an actual notebook
  download. Browser resource inventory is not a packet capture of every browser or extension request.
  Automated credential-signature and filename scans cover reachable public history and source archives;
  they do not prove absence of every possible secret. Specific measured release results are recorded in
  the entrant's continuation ledger rather than presented as a privacy certification.
22
23 ## Public materials
24
25 The public repository intentionally includes the app, synthetic data, trainer, audit, tests,
  educational documentation, PDFs, source ZIP and final competition demo. Raw narration files, private
  hosting metadata, personal contact details, credentials, unrelated work and disposable QA pages are
  excluded. The demo uses entrant-authorized narration; the raw recording is not distributed. See
  [attribution](attribution.md) for media rights.
```

docs/attribution.md

```
1 # Attribution, authorship and rights
2
3 BaseRate's application code, interface, educational copy, synthetic examples, training/audit code and
  submission materials were created with extensive AI assistance at the entrant's direction. Automated
  tests and AI-assisted code review are development evidence. No human learner study, educator
  endorsement, medical expertise or measured learning/health effect is claimed. The entrant must
  understand and explain the implementation under the competition's AI-assistance condition.
4
5 ## Content and design sources
6
7 - [Hoffrage et al., 2015, Frontiers in Psychology](https://pmc.ncbi.nlm.nih.gov/articles/PMC4604268/):
  natural-frequency representations informed the population/counting approach. The paper does not
  evaluate BaseRate.
8 - [Gigerenzer, BMJ 2011, What are natural frequencies?](https://www.bmj.com/content/343/bmj.d6386):
  reference groups and interpretation of frequencies.
9 - [National Cancer Institute, Cancer Screening
  Overview](https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq): benefits,
  harms and screening context, not validation of this fictional simulator.
10 - [Nielsen Norman Group, visual design
  principles](https://www.nngroup.com/articles/principles-visual-design/): hierarchy, contrast and
  grouping informed design decisions.
11 - [GOV.UK Design System, Details](https://design-system.service.gov.uk/components/details/):
  progressive disclosure informed advanced methods sections.
12 - Existing educational tools are compared, with their original links, in [research.md](research.md).
  The calculator and visual Bayes explanation are established ideas; the project does not claim to
  invent them.
13
14 These are linked references, not bundled papers or endorsements. Original prose summarizes concepts.
  No external font, stock-image library, icon package or third-party JavaScript is loaded. The interface
  uses Georgia and system font fallbacks, CSS patterns, native controls and an inline original letter
  favicon. Screenshots show the actual application.
15
16 ## License and media
17
18 [MIT](../LICENSE) covers original application source, documentation and synthetic dataset. Linked
  sources retain their own rights. The competition demo includes the entrant's supplied narration over
  actual app states and a visual feature tour. Public distribution of that final demo is authorized for
  this project; the raw voice recording is excluded. The MIT source license does not grant rights to
  reuse the entrant's voice or impersonate them.
19
20 The refreshed 112-second demo shows the final coaching and denominator-feedback behavior. It preserves
  the original supplied narration, with level gain and peak limiting only. The visual edit uses actual
  screen-state captures and condensed interactions, not a continuous recording. Staged estimates are
  labelled demonstration inputs. No participant identity, biography or outcomes are invented. See [demo
  methods and qualifications](demo.md).
```

docs/research.md

```
1 # Competition and product evidence
2
3 Checked 6 October 2026, Hong Kong time. This is a source review and developer assessment, not user
  research or an estimate of judging results.
4
5 ## Official requirements and timing
6
7 - [Official overview](https://univabio.devpost.com/): students aged 13+, individual or team
  participation; companies/professional organizations excluded; standard country/territory exceptions
  mentioned without a list. Health education is an eligible application of AI. AI assistants are
  encouraged, but entrants must understand and explain their code. Required package: interactive
  prototype, purpose/features/interaction demo video, one-page description PDF and GitHub/code PDF. We
  supply both repository and code PDF to resolve the wording conservatively. Sponsor prizes are
  described as non-cash benefits.
8 - [Official rules](https://univabio.devpost.com/rules): five criteria, each worth five points: Idea &
  Innovation, Implementation, Health Impact & Rigor, Design & Usability, Presentation. The rules body
  states an October 7, 2026, 12 AM CST end; the header states October 13, 11:45 PM EDT. No timezone
  expansion accompanies CST.
9 - [Official schedule](https://univabio.devpost.com/details/dates): submission cutoff October 13, 2026,
  11:45 PM EDT; judging October 14-21; winners October 22.
  [Updates](https://univabio.devpost.com/updates) and the [discussion
  index](https://univabio.devpost.com/forum_topics) contained no visible announcement reconciling the
  two cutoffs at inspection.
10
11 The platform schedule supports the later date but does not expressly replace the earlier rules-body
  date. Plan against October 7 midnight, conservatively treating ambiguous CST as China Standard Time
  (Hong Kong time) rather than assuming the later US-Central interpretation. Internal freeze: October 6
  at 18:00 HKT, leaving six hours before the earliest plausible cutoff. This interpretation is a
  planning assumption, not organizer clarification. No organizer message has been sent. Do not defer a
  working package for speculative extra features.
12
13 The entrant confirmed student status and age eligibility. Country/residency exception eligibility has
  not been verified. AI assistance is disclosed throughout the package. A code walkthrough is supplied;
  the entrant's understanding has not been assessed. The entry already displayed Submitted, 5/5 before
  this pass. Its supported embedded-video field is empty; a direct demo on the materials page does not
  establish completion of that requirement.
14
15 ## Existing tools and the originality boundary
16
17 [Seeing Theory's Bayesian Inference
  chapter](https://seeing-theory.brown.edu/bayesian-inference/index.html) already teaches a
  rare-condition medical-test example with adjustable probabilities, patient simulation and posterior
  counts. [Tool Corner's Bayes Medical Test
  Visualiser](https://toolcorner.net/labs/bayes-medical-test-visualiser) documents sliders/numeric
  fields, population icons, four outcomes, PPV/NPV and presets. These are primary creator pages. This
  comparison inspects their published features; it is not a controlled usability comparison or proof
  that no other equivalent product exists.
18
19 BaseRate's probability calculator, icon population and natural-frequency explanation are established
  approaches. Prediction and scaffolding are also established teaching methods. Its plausible
  contribution is connecting first predictions, denominator correction, a retained notebook, editable
  comparisons, dependence assumptions and inspectable local AI topic suggestions in one small,
  reproducible static lab. This integration may be useful; novelty and effectiveness remain to be
  judged.
20
21 ## Rubric assessment and chosen work
22
23 | Criterion | Strength | Weakness and response |
24 | --- | --- | --- |
25 | Idea & Innovation | Connected learning workflow; explicit conditional repeat testing; private local
  inference. | Basic calculator overlaps existing tools. Emphasize the integrated correction/assumption
  workflow; do not claim a unique calculator or invented pedagogy. |
26 | Implementation | Working static application; separate pure arithmetic; reproducible trainer and
  inference parity. | Synthetic language evaluation is narrow and semantic failures are significant.
  Preserve original weights and evaluation; add a reproducible stress audit and provisional,
  learner-controlled feedback. |
27 | Health Impact & Rigor | Makes reference groups and screening tradeoffs visible; fictional scope and
  assumptions are explicit. | No educator review, learner study, clinical validation or measured
  benefit. Improve factual feedback and record observable correction, while keeping all outcome claims
  limited. |
28 | Design & Usability | Linked representations, keyboard controls, responsive notebook layout and
  equivalent data tables. | Earlier lessons allowed progression after a wrong denominator, and AI
  wording could endorse flawed reasoning. Require denominator correction, explain the selected group's
  actual ratio, retain the first choice, and let learners override the coach's topic. |
29 | Presentation | Real prototype, entrant voice, repository and downloadable PDFs/source. | The
  refreshed 112-second final-app demo shows feedback policy 1.1; its supported Devpost embed remains
  pending channel Terms acceptance; supported Devpost video URL is absent. Disclose the recording's
  version, update written materials, and preserve the explicit channel-Terms approval blocker. |
30
31 ## Evidence boundaries and next validation
```

32

33 Automated arithmetic tests establish behavior for specified inputs. Browser inspection establishes that the inspected controls/states function in the inspected environment. Synthetic examples and an AI code review establish development findings, not learner reactions. None measures comprehension, retention, confidence calibration, accessibility with assistive-technology users or health outcomes.

34

35 The next meaningful study would use independently collected explanations and unfamiliar transfer questions, preserve first answers, compare a suitable control condition, and measure delayed comprehension as well as immediate correction. Selecting the correct option after feedback is practice completion; it is not demonstrated learning gain. No participant study has been conducted.

docs/code-walkthrough.md

```
1 # Entrant code walkthrough
2
3 UnivaBio encourages AI assistance while requiring entrants to understand and explain their code. This
  guide supports preparation; it does not certify that the entrant understands the implementation.
4
5 ## Explain the result first
6
7 In 1,000 fictional people with 1% prevalence, 10 have the condition. A 90% sensitive test detects 9.
  Of 990 people without the condition, 5% test positive: 49.5 expected false positives. The relevant
  denominator is all positive results, 58.5, so PPV is  $9/58.5 = 15.4\%$ . Sensitivity starts with the
  condition group; PPV starts with the positive group.
8
9 ## Find the implementation
10
11 - `dist/science.js`: validate finite rates; compute TP/FN/FP/TN; divide TP by TP+FP; return undefined
  for an empty denominator. Preserve fractional expectations. Largest-remainder rounding supplies
  exactly 1,000 illustrative squares without changing the math.
12 - `dist/app.js`: synchronize inputs and representations; render coach feedback. Model suggestions and
  mathematical questions are separate. Current experiment changes recalculate a displayed factual check
  and remove its previous answer status.
13 - `dist/lab-tools.js`: immutable pinned experiment, prevalence curve/equivalent table, and conditional
  second test. Independence is assumed within each condition group, not inferred from seeing two
  positives. Perfectly repeated results use conditional rates 100%/100%, so PPV does not change.
14 - `dist/learning.js` and `learning-ui.js`: three fixed cases; save the first estimate; explain wrong
  reference groups; require correction; export numeric/selected choices only. The lesson is separate
  from exploratory slider settings.
15 - `model/train.py`, `dist/model.js`, `dist/coach.js`: train synthetic word/bigram likelihoods; score
  labels using sums of log probabilities; convert relative scores to uncalibrated votes; apply
  coverage/separation and scope guards. Fixed feedback is chosen locally. This is not a remote language
  model.
16 - `model/audit.py`: independently rerun training in a temporary directory, compare exact bytes,
  measure lexical split overlap, and compare old/new responses on diagnostic probes.
17
18 ## Demonstrate the limitation honestly
19
20 Enter "All positives are actual cases, without exception." The classifier can suggest the
  positive-group topic even though the explanation is wrong. The UI does not endorse it. Choose the
  factual positive-group check: 9 of 58.5 positive results are actual cases in the default example. The
  learner can override the topic and revise their own sentence. Explain why the model's 76.4% vote is
  not a probability of correct reasoning.
21
22 Run the five Node suites and `python3 model/audit.py`. Explain what they establish (specified
  mathematical and interface behavior, deterministic reproduction) and what they do not (real learner
  benefit, clinical validity or broad semantic accuracy). Be ready to trace one input from validation
  through pure calculation to display, and one explanation from tokenization to provisional feedback.
```

docs/design.md

```
1 # BaseRate learning-lab upgrade \u2014 5 October 2026
2
3 The entrant has delegated design, execution, public GitHub hosting, and continued improvement. The
  audience is students and general readers learning to interpret fictional screening statistics. Success
  means a usable learning journey, inspectable mathematical reasoning, useful AI feedback, and credible
  competition evidence. No claim about winning, clinical accuracy, or measured learning benefit is
  supported.
4
5 ## Design direction
6
7 Use the visual language of a field notebook: paper-colored canvas, ink text, Georgia headings, compact
  system sans-serif controls, ruled divisions and tabular numbers. Avoid gradients, decorative badges,
  shadowed card grids, and stock medical illustrations. Color encodes outcomes consistently; labels and
  patterns also distinguish outcomes. The primary task starts near the top. Preserve keyboard use,
  narrow-screen reflow, visible focus, and reduced-motion preferences.
8
9 The direction draws on NN/g's scale, hierarchy, contrast and proximity principles
  (https://www.nngroup.com/articles/principles-visual-design/) and GOV.UK's advice to disclose optional
  detail without hiding essential information
  (https://design-system.service.gov.uk/components/details/). These are design guidance, not evidence
  that a specific aesthetic is less AI-generated.
10
11 ## Product
12
13 1. Explore: preserve the accurate 1,000-person simulation, add number inputs alongside sliders,
  population and frequency-tree views, a visible denominator equation, and links containing only
  fictional rate settings.
14
15 2. Compare: pin an immutable experiment A and manipulate experiment B. Show both denominators,
  expected counts and percentage-point differences. A continuous SVG prevalence curve uses the current
  test's sensitivity/specificity; a table provides equivalent values.
16
17 3. Repeat testing: demonstrate a second test among first-positive people using explicitly
  conditional detection and false-positive rates. An independent-test preset uses 90% and 5%; a
  perfectly repeated-result preset uses 100% and 100%, yielding no new information. Keep assumptions and
  the statement that this is not a real testing recommendation beside the controls.
18
19 4. Practice: three fictional cases (1/90/95, 20/90/95, 1/90/99). Predict a percentage before
  revealing counts, then identify the denominator. Retain first guesses and arithmetic error in memory
  for this tab only, offer a JSON learning-notebook export, and provide deterministic explanations. This
  is practice, not a validated assessment.
20
21 5. Teach back: preserve the trained 100-example NB model and its unchanged 30-example evaluation.
  Expose model alternatives, coverage, influential features and uncalibrated votes on demand. Connect
  model feedback to a relevant next exercise. Text remains transient and local. Never silently
  reinterpret the synthetic evaluation as learner outcomes.
22
23 ## Interfaces and safety
24
25 `science.js` owns all arithmetic: calculate, roundedCounts, compare, prevalenceCurve, sequential.
26 `learning.js` owns immutable case content and numeric grading; `learning-ui.js` owns the practice
  state. `lab-tools.js` owns comparisons, curve, sequence, settings links and notebook export. Existing
  `app.js` owns the base experiment and AI coach and dispatches `baserate:experiment` with validated
  settings/counts.
27
28 Rates passed to math are finite percentages 0\u2013100; population must be positive. UI prevalence is
  0.1\u2013350%, detection/clearance 1\u2013100%. Sequence conditional rates permit 0\u2013100%. Null
  proportions are shown as undefined. Diagrams round only for display, all calculations use raw expected
  counts. Learner explanations are never included in links or exported notebooks. No analytics, remote
  inference or data storage is added.
29
30 ## Deliverables
31
32 Publish a clean public source repository `Nicholas-afk/baserate`, deploy `dist` on a `gh-pages`
  branch, and verify the resulting live URL. Replace the prior synthetic voice with the supplied
  recording without cloning the voice or inventing spoken content. Capture the new working interface and
  retine the video to the actual recording. Update the one-page description, complete-code PDF, source
  ZIP and Devpost draft. Leave final prize submission for the required user handoff.
```

docs/plan.md

```
1 # BaseRate Upgrade Implementation Plan
2
3 > **For agentic workers:** REQUIRED SUB-SKILL: Use superpowers:executing-plans to implement this plan
  task-by-task. Steps use checkbox syntax for tracking.
4
5 **Goal:** Deepen the educational prototype, redesign its interface, use the entrant's voice and
  publish on the requested GitHub account.
6
7 **Architecture:** Static, dependency-free browser application. Pure arithmetic and learning modules
  feed separate UI controllers, with a visible experiment-change event.
8
9 **Tech Stack:** HTML/CSS/JavaScript, Node assertions, standard-library Python trainer, HyperFrames,
  local WhisperX, ReportLab, GitHub Pages.
10
11 **Spec:** docs/design.md
12
13 ## Global Constraints
14
15 - All rates and scenarios are fictional; no personal risk or test recommendation.
16 - Learner text is transient and local, excluded from links and notebook export.
17 - Preserve the existing 100/30 synthetic model evaluation and disclose limitations.
18 - Handle undefined denominators, finite rate validation, keyboard access and 390px screens.
19 - Public hosting and source on Nicholas-afk are explicitly requested; final prize entry requires
  handoff.
20
21 ## Review Focus
22
23 - Empty/invalid URL rates must not reset or corrupt the app.
24 - Pinning A must remain immutable while B changes.
25 - Repeat testing must use conditional rates rather than silently assume independence.
26 - Changing cases and restarting must clear stale results and guesses.
27 - Missing model, undefined proportions, and imported links must have readable fallback behavior.
28
29 ### Task 1: Mathematical extensions
30
31 **Files:** Modify dist/science.js; create tests/extensions.cjs.
32 **Interfaces:** compare(a,b) consumes three-rate objects and returns both counts and deltas;
  prevalenceCurve(sensitivity,specificity) returns 101 {prevalence,ppv} points from 0 to 50;
  sequential(first,conditionalDetection,conditionalFalsePositive,population=1000) returns first counts,
  both-positive counts, PPV and rejected true/false positives.
33
34 - [x] Write tests for rare/common deltas, curve monotonicity, undefined denominators, independent and
  repeated outcomes, invalid rates and conservation.
35 - [x] Run tests and verify failure on missing exports.
36 - [x] Implement pure functions and run old plus new tests to PASS.
37 - [x] Commit and record evidence.
38
39 ### Task 2: Guided practice
40
41 **Files:** Create dist/learning.js, dist/learning-ui.js, tests/learning.cjs.
42 **Interfaces:** BaseRateLearning.cases contains three settings/prompt/explanation entries;
  gradePrediction(caseId,guess) returns numeric error and counts; notebook(answers) returns only
  settings, numeric guesses/results and denominator answers.
43
44 - [x] Test exact case calculations, 0/100 and invalid guesses, no learner-text export, and unknown
  cases.
45 - [x] Verify failure, implement pure module, then PASS.
46 - [x] Implement predict/reveal/denominator/next/restart flow in session memory.
47 - [x] Commit and record evidence after the core tests pass.
48
49 ### Task 3: Interface and connected lab tools
50
51 **Files:** Replace dist/index.html and dist/style.css; modify dist/app.js; create dist/lab-tools.js;
  update project.html.
52 **Interfaces:** app.js dispatches baserate:experiment with {settings,counts}; lab-tools consumes it
  and science; BaseRateLesson exposes notebook()/restart().
53
54 - [x] Implement the notebook visual system, numeric controls, accessible tree/table, pin/compare,
  curve, conditional second test, settings links and model diagnostics.
55 - [x] Run old/new suites, browser-check controls, practice transitions, link validation, snapshot
  pinning, undefined sequence and mobile overflow.
56 - [x] Inspect desktop and mobile screenshots; commit and record evidence.
57
58 ### Task 4: Publication and truthful competition package
59
60 **Files:** README/SUBMISSION, PDFs/ZIP, video composition and voice assets, clean public repository,
  competition-state.json.
61
62 - [x] Transcribe the provided voice, verify spoken claims and synchronize actual captures to it.
63 - [x] Run video validation, inspect midpoint/cut snapshots, render and verify duration/audio.
```

64 - [x] Regenerate the complete source package and one-page PDF, inspect both.

65 - [x] Create/push requested public GitHub repository, enable Pages and verify live resources.

66 - [x] Update saved Devpost links, story and preview using browser UI; preserve final submission handoff.

67 - [x] Complete fresh code review under executing-plans, fix material findings with regression tests, and record final results.

68

69 Historical publication receipts (before feedback policy 1.1): GitHub main 200bc13; Pages gh-pages 04b81c7e237d0d097cb9f2af339c8d0780a8b642. Live app, project page and four downloads return HTTP 200 and match source bytes. Four test suites pass on the clean public source. 84-second H264/AAC demo uses supplied voice; integrated loudness -16.7 LUFS, true peak -1.54 dBTP. Description 1 page, complete code 101 pages, source ZIP 21 files, visually inspected. Devpost was already Submitted (5/5) when opened; upgraded story, links, cover, tags and gallery saved without invoking final Submit. Embedded-video field remains empty pending action-time YouTube terms approval.

70

71 Current revision: provisional and overridable coach topics, five factual self-checks, explanation revision, denominator correction before progression, notebook schema 2, and a reproducible semantic/split audit. Original classifier weights and evaluation are preserved. See docs/research.md and docs/model-card.md for the current evidence and deadline policy. Five test suites pass; the subsequent AI review finding about stale example feedback is fixed with an actual-handler regression.

dist/model.js

```
1 window.BaseRateModel = {
2   "version": "1.0.0",
3   "algorithm": "Multinomial Naive Bayes",
4   "labels": [
5     "balanced",
6     "certainty",
7     "sensitivity",
8     "base_rate",
9     "dismissal"
10  ],
11  "vocabulary": [
12    "a_certainty",
13    "a_clue",
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20    "a_guaranteed",
21    "a_hundred",
22    "a_lot",
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24    "a_ninety",
25    "a_perfect",
26    "a_positive",
27    "a_rare",
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36    "absolute_guarantee",
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40    "accurate_test",
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43    "actual_disease",
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45    "after_a",
46    "after_testing",
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49    "alarms_means",
50    "all",
51    "all_groups",
52    "all_positive",
53    "all_positives",
54    "all_screening",
55    "altogether",
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81  "any_cases",
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94  "are_no",
95  "are_possible",
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98  "are_true",
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107 "be_false",
108 "be_free",
109 "be_ill",
110 "be_interpreted",
111 "be_left",
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113 "be_much",
114 "be_perfect",
115 "be_read",
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221 "difference",
222 "different",
223 "different_group",
224 "different_populations",
225 "different_predictive",
226 "directly",
227 "directly_answers",
228 "disease",
229 "disease_chance",
230 "disease_completely",
231 "disease_if",
232 "disease_in",
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234 "dismiss",
235 "dismiss_screening",
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237 "divided_by",
238 "do",
239 "do_not",
240 "do_with",
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242 "does_not",
243 "effect",
244 "effect_on",
245 "equal",
246 "equal_to",
247 "errors",
248 "errors_show",

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275 "five",
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305 "guarantees",
306 "guarantees_there",
307 "happen",
308 "harmful",
309 "has_a",
310 "has_no",
311 "has_nothing",
312 "has_the",
313 "has_to",
314 "has_zero",
315 "have_the",
316 "healthy",
317 "help",
318 "help_anybody",
319 "high",
320 "high_most",
321 "how",
322 "how_common",
323 "how_likely",
324 "how_many",
325 "how_widespread",
326 "hundred",
327 "hundred_percent",
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329 "i_would",
330 "identical",
331 "if_any",
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```

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```

417 "matters_because",
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475 "no_difference",
476 "no_effect",
477 "no_errors",
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479 "no_one",
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```

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 510 "of_positives",
 511 "of_prevalence",
 512 "of_sensitivity",
 513 "of_ten",
 514 "of_the",
 515 "of_time",
 516 "of_true",
 517 "on_how",
 518 "on_positive",
 519 "on_prevalence",
 520 "on_the",
 521 "one",
 522 "one_case",
 523 "one_should",
 524 "only",
 525 "only_need",
 526 "only_on",
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 529 "or_common",
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 542 "people_start",
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external, multilingual, clinical, or educational-outcome validation."
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README.md

```
1 # BaseRate
2
3 A field guide to screening uncertainty for UnivaBio 2026. Count a fictional population, compare
  assumptions, predict the answer, then explain your reasoning to a small on-device AI coach.
4
5 **[Open the learning lab](https://nicholas-afk.github.io/baserate/)** \xb7 **[Demo and project
  materials](https://nicholas-afk.github.io/baserate/project.html)**
6
7 ## Run locally
8
9 Clone the public repository and serve its `dist/` folder:
10
11 ```sh
12 git clone https://github.com/Nicholas-afk/baserate.git
13 cd baserate
14 python3 -m http.server 8000 --directory dist
15 ```
16
17 Open http://localhost:8000. No app package installation, API key or account is needed. Python 3 runs
  the server/trainer; Node.js runs the verification suites. The app itself needs only a modern
  JavaScript-enabled browser. Use HTTP locally or HTTPS on Pages; clipboard support depends on browser
  permissions and a secure context. A displayed link provides a fallback if copying fails.
18
19 `main` contains the source, tests, documentation and four submission downloads in `dist/downloads/`.
  The existing `gh-pages` branch contains the contents of `dist/` at its root, with `.nojekyll`. GitHub
  Pages is configured for that branch and serves `*/baserate/`, not the account root. All in-app
  assets, navigation and downloads use relative paths. To update this deployment, test `main`, update
  the source package/code PDF when source changes, commit the source, then copy the contents of `dist/`
  to the existing `gh-pages` branch and push it. Do not publish repository tooling, credentials or a raw
  voice recording. Confirm the Pages build and the actual served files and interactions after
  publishing; a successful build alone is insufficient. No bundler or environment-variable substitution
  is required.
20
21 The runnable source ZIP excludes the four submission downloads to avoid recursively packaging itself.
  Serve its `dist/` folder to use the app; obtain the demo and PDFs from the public materials page. The
  main repository includes those downloads. Raw narration and private video authoring files are
  excluded.
22
23 Read the [architecture and worked classifier explanation](docs/architecture.md), [data
  handling](docs/privacy.md) and [entrant walkthrough](docs/code-walkthrough.md) before presenting the
  project.
24
25 ## Working features
26
27 - Three fictional test presets, live sliders and editable numeric rates.
28 - A 1,000-person square grid, positive-only focus and natural-frequency tree using the same underlying
  counts.
29 - Exact expected outcomes, explicit denominator and positive predictive value (PPV).
30 - Pin experiment A while varying B; compare outcomes and PPV changes in percentage points.
31 - A prevalence curve using the current test, with an equivalent accessible table.
32 - A conditional second-test explorer: independent identical tests, perfectly repeated results or
  explicit conditional rates. Assumptions are visible; a repeated result can add no information.
33 - Three predict/reveal/denominator cases requiring correction before progression; first numeric
  guesses and first/final denominator choices retained in a numeric notebook. No coach explanation
  enters the export.
34 - Local AI topic suggestions with learner override, five factual self-checks and an
  explanation-revision step; inspect influential features, coverage and uncalibrated votes.
35 - Conservative uncertainty handling, personal-health query guard and readable missing-model fallback.
36 - Keyboard controls, labelled inputs, patterned outcomes, responsive layout, printable worksheet and
  validated share links containing only fictional rates.
37 - Optional feature-detected WebMCP controls that operate on the same visible state.
38
39 ## Reproduce and verify
40
41 Training uses only Python's standard library. `python3 model/train.py` reproduces the distributed
  model and evaluation. Tests use only Node's standard library:
42
43 ```sh
44 node tests/verify.cjs
45 node tests/extensions.cjs
46 node tests/learning.cjs
47 node tests/display.cjs
48 node tests/coaching.cjs
49 python3 model/audit.py
50 ```
51
52 The checks cover 336 base arithmetic/rounding combinations, 2,525 prevalence-curve points, 405
  conditional second-test combinations, all three lesson equations, invalid inputs, immutable experiment
  snapshots, undefined denominators, count conservation, precise tiny-count display, text-free exports
  and missing-model feedback. Python/browser inference parity is checked on all 30 holdout examples.
```

Manual browser checks covered lesson progression/restart, rate/link validation, comparison pinning, conditional presets, local feedback and a 390-pixel layout without horizontal overflow.

53
54 ## AI, data and limitations
55
56 A Multinomial Naive Bayes model uses learned word and adjacent-word features, additive smoothing (alpha 0.7), and five educational reasoning labels. The original synthetic dataset contains 100 English training explanations and 30 separately authored synthetic holdout examples from the same project; no patient data are used.
57
58 The raw synthetic holdout result is **29/30 correct**. Uncertainty checks accept 28/30 explanations, with 27/28 correct accepted labels. One correct distinction between sensitivity and PPV is misread. See `model/evaluation.json` for complete predictions and confusion matrix. These are small synthetic development checks, not an external benchmark, clinical validation, calibrated confidence or evidence of learning or health benefit. No holdout sentences were moved into training to hide the known error.
59
60 The model's votes are uncalibrated and may fail on negation, mixed reasoning, unfamiliar language or wording. The classifier proposes a topic; it cannot verify an explanation. Feedback policy 1.1 makes all inferred topics provisional, permits override, and checks only a selected factual answer. A 24-case diagnostic audit found semantic failures and reduced explicit praise of flawed/mixed explanations from 2/10 to 0/10 through feedback wording; raw classifications are unchanged. This is not an accuracy or learning-benefit improvement. Its fixed responses are inspectable in `dist/coach.js`. Screening arithmetic runs separately from the model. Conditional second-test rates apply only among first-positive people; reusing unconditional rates requires the displayed independence assumption.
61
62 Only squares are rounded with largest remainders, preserving 1,000 squares. Mathematical expectations and PPV use unrounded values. Display precision adapts for small nonzero counts so a defined denominator never appears as 0/0.
63
64 ## Privacy
65
66 Loading the app downloads HTML, CSS, JavaScript and the trained model from GitHub Pages. The project page also requests the demo video's metadata/media. GitHub may retain ordinary request information. The application has no analytics, external scripts/fonts or remote inference calls.
67
68 Coach explanations and lesson answers stay in tab memory; the app does not persist them or send them to a server. Reload resets them. An explicit notebook export saves numeric predictions, case settings and first/final denominator choices to a file; it excludes coach explanations. A copied settings link contains only three validated fictional rates. External reference links contact their destination when followed. Optional WebMCP exposes experiment controls and coach review to a connected browser assistant; that assistant has its own conversation/data handling. These claims describe the app, not every browser extension or hosting provider. See the [full privacy scope](docs/privacy.md).
69
70 All rates and cases are fictional. The app does not diagnose, estimate a person's risk, interpret personal results, select a real test or recommend treatment. The prototype is English only. No independent learner study, clinical validation or measured health impact has been completed.
71
72 See the [full model card](docs/model-card.md), [deterministic audit](model/audit.json), [competition and product review](docs/research.md), and [entrant code walkthrough](docs/code-walkthrough.md). No exact train/holdout duplicates were found, but lexical disjointness does not make their shared synthetic authorship semantically independent. Publication and responsive browser checks are development QA, not a physical-device, screen-reader or learner study. The refreshed 112-second demo shows the final release. Its staged inputs are demonstration data, not a learner study; see [demo methods and claim qualifications](docs/demo.md).
73
74 ## Research and design
75
76 Natural frequencies and explicit reference groups informed the grid/tree and denominator-first teaching. Hierarchy, restrained typography, spacing, meaningful labels and progressive disclosure informed the editorial interface. Sources support the underlying content and design choices, not this prototype's effectiveness:
77
78 - [Hoffrage et al., 2015, original natural-frequency study](https://pmc.ncbi.nlm.nih.gov/articles/PMC4604268/)
79 - [Gigerenzer, BMJ 2011, What are natural frequencies?](https://www.bmj.com/content/343/bmj.d6386)
80 - [National Cancer Institute, Cancer Screening Overview](https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq)
81 - [Nielsen Norman Group, visual design principles](https://www.nngroup.com/articles/principles-visual-design/)
82 - [GOV.UK Design System, Details component](https://design-system.service.gov.uk/components/details/)
83
84 ## Authorship and next validation
85
86 Code, synthetic data, educational copy, interface and submission materials were created with extensive AI assistance at the entrant's direction. The demo uses the entrant's unchanged supplied recording over edited screen-state captures of the actual final app, followed by a silent, annotated walkthrough of prediction, correction, export, comparison and repeat-test assumptions. It is not a continuous screen recording. No medical expertise, personal biography, clinical research or learner-testing results are claimed. The entrant must understand the implementation and be ready to explain it to judges.
87
88 Next steps: independently collected learner explanations, health-educator review, evaluated multilingual support, accessibility testing with users and a properly designed comprehension study.

```
89 These are proposed work, not completed outcomes.  
90 ## License  
91  
92 MIT for original application source, documentation and synthetic dataset. Linked research retains its  
own rights. The entrant-provided narration is used for this competition demo; the source license does  
not grant rights to impersonate the entrant or reuse their voice. See [attribution and media  
scope](docs/attribution.md).
```

SUBMISSION.md

```
1  ## Inspiration
2
3  A test that detects 90% of cases can still produce a positive group in which most results are false
  positives. The denominator explains the apparent contradiction. BaseRate turns that distinction into a
  learning experiment: predict, count, change an assumption and explain it back.
4
5  The audience is students, health-literacy learners and educators. The intended contribution to human
  health is understanding screening uncertainty. The project does not diagnose or estimate personal
  risk. [Natural-frequency research](https://pmc.ncbi.nlm.nih.gov/articles/PMC4604268/) informed its
  representations, and the [National Cancer
  Institute](https://www.cancer.gov/about-cancer/screening/patient-screening-overview-pdq) informs its
  screening context. These sources do not validate this prototype.
6
7  ## What it does
8
9  The working lab connects several views of the same fictional population. Change prevalence,
  sensitivity and specificity with sliders or numeric inputs. A 1,000-person grid, positive-group focus,
  frequency tree, four expected outcome counts and PPV equation update together. Only the squares are
  rounded.
10
11 Pin experiment A and vary B to compare the positive groups. A prevalence curve, with an equivalent
  table, shows why the same test changes meaning across populations. In the default case, 1% prevalence,
  90% sensitivity and 95% specificity give 9 true positives and 49.5 false positives:  $9/58.5 = 15.4\%$ .
  At 20% prevalence, the unchanged test gives 81.8%.
12
13 The second-test explorer applies explicit conditional rates only to first-positive people. Independent
  identical tests yield 76.6% in the default fictional example. A perfectly repeated result leaves PPV
  unchanged. The assumptions are visible so the tool does not imply that two positives automatically
  provide independent evidence.
14
15 A three-case lesson asks for an estimate before revealing counts, then asks which group belongs in the
  denominator. Wrong choices receive the actual ratio for their chosen reference group and require
  correction before progression. First numeric guesses and first/final denominator choices stay in tab
  memory; learners can export a numeric notebook. A demonstration estimate is explicitly labelled as an
  example, not learner-study data.
16
17 The AI coach reads an explanation locally and proposes a provisional review topic. Learners can
  override it, answer one of five factual checks using current counts or an evidence distinction, and
  revise their explanation. Feedback checks only the selected answer; it does not declare an inferred
  misconception or endorse the explanation. Learners can inspect influential words, feature coverage,
  label alternatives and uncalibrated model votes, then follow a relevant practice link. The coach
  suggests an interpretation rather than grading competence.
18
19 ## How it was built
20
21 A static HTML/CSS/JavaScript app runs without dependencies, accounts or API keys. Pure arithmetic and
  learning modules feed separate interface controllers. A dependency-free Python trainer learns a
  Multinomial Naive Bayes classifier from 100 synthetic English explanations. Word and adjacent-word
  features distinguish contextual reasoning and four common misconception patterns.
22
23 Inference stays in the browser. Coverage and vote separation can trigger uncertainty; personal-health
  queries are redirected away from the fictional experiment. The app has no analytics or remote
  inference calls. Explanations stay in tab memory, without persistent app storage or server
  transmission; exports exclude them. Loading static app/model files and demo media still contacts
  GitHub Pages, which may retain ordinary request logs. Connected browser assistants have separate data
  handling.
24
25 The interface uses a restrained field-notebook style, clear hierarchy, readable labels, patterned
  outcomes and keyboard controls. Essential explanations stay visible; advanced assumptions and model
  details use progressive disclosure. The refreshed 112-second demo uses the entrant's unchanged
  supplied narration over actual final-application screen-state captures, with edited holds and cuts. It
  shows the local coach, staged prediction, correction, notebook export, pinned comparison and explicit
  repeat-test assumptions. The estimates are demonstration data, not a learner study; this is not a
  continuous screen recording. [Watch the final demo and access the
  deliverables](https://nicholas-afk.github.io/baserate/project.html).
26
27 ## Challenges and what was learned
28
29 Exact expectations can be fractional while the grid must contain exactly 1,000 squares.
  Largest-remainder rounding keeps that illustration consistent; calculations, comparisons and practice
  answers retain raw values. Adaptive display precision preserves tiny nonzero denominators.
30
31 Repeated testing exposed a second conceptual trap: multiplying test rates without explaining
  dependence. The conditional explorer makes the selected population and assumption explicit, including
  a repeat that adds no information.
32
33 A small text model can sound overly certain or misread a correct explanation. The app exposes
  uncertainty and failure modes. One holdout sentence correctly distinguishing sensitivity from
  predictive value was misread as confusion. The project retains the known error and complete evaluation
  instead of moving the sentence into training. A 24-case development stress audit also exposed
```

correct, ambiguous and mixed explanations with high but incorrect votes. Among ten flawed/mixed probes, the previous wording praised two; provisional feedback now praises zero. All classifications remain unchanged, so this is a measured feedback-policy correction, not improved classifier accuracy.

```
34
35 ## Validation
36
37 Five reproducible test suites and a deterministic model audit pass. They cover 336 base
  arithmetic/rounding combinations, 2,525 prevalence-curve points, 405 conditional repeat-test
  combinations, all lesson equations, immutable snapshots, invalid inputs, zero denominators, tiny-count
  display, exports excluding learner text and missing-model feedback. All 30 holdout examples match
  between Python and browser inference.
38
39 Browser checks verified comparison pinning, independent/repeated presets, three-case progression and
  restart, numeric-input errors and recovery, validated settings links, local model diagnostics and the
  actual app at 390 pixels without horizontal overflow. A separate code review was completed and its
  display/recovery findings corrected.
40
41 The synthetic holdout from the same project produced 29/30 correct raw classifications. Uncertainty
  checks accepted 28 labels, 27 correct. These are development checks, not a benchmark, clinical
  accuracy claim, evidence of learning gain or health outcome. No independent learner study or clinical
  validation has been completed. No exact split duplicates were found; shared authorship and claim types
  limit semantic independence. The reproducible audit, full model card and entrant code walkthrough are
  in the repository.
42
43 ## AI assistance and honest limits
44
45 The project, interface, synthetic dataset, implementation and submission materials were created with
  extensive AI assistance at the entrant's direction. The entrant supplied the narration and must
  understand and explain the code. No clinical research, medical expertise or user testing is claimed.
46
47 All scenarios are fictional. The English-only classifier can fail on negation, mixed ideas, unfamiliar
  phrasing and other languages. Model votes are uncalibrated. The app does not interpret personal
  results, diagnose, recommend real tests or give treatment advice. The AI neither computes the
  screening probability nor generates medical facts.
48
49 ## What's next
50
51 Independent learner explanations, health-educator review, accessibility testing with users, evaluated
  multilingual support and a properly designed comprehension study. These are proposed validation steps,
  not completed outcomes.
```

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```
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2
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```

docs/assets/devpost-cover.svg

```
1 <svg xmlns="http://www.w3.org/2000/svg" width="1200" height="800" viewBox="0 0 1200 800">
2 <rect width="1200" height="800" fill="#242c29"/>
3 <path d="M100 128H1100" stroke="#819487" stroke-width="2"/>
4 <g fill="#efefe5" text-anchor="middle">
5 <text x="600" y="220" font-family="monospace" font-size="25" letter-spacing="3">AI SCREENING LITERACY
  LAB</text>
6 <text x="600" y="405" font-family="Georgia,serif" font-size="164">BaseRate</text>
7 <text x="600" y="497" font-family="Georgia,serif" font-size="49">Predict. Count. Explain.</text>
8 <text x="600" y="659" font-family="monospace" font-size="23" letter-spacing="2">UNIVABIO 2026</text>
9 </g>
10 <path d="M100 696H1100" stroke="#819487" stroke-width="2"/>
11 </svg>
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